Modeling Soil–Water–Atmosphere Interactions for Irrigation Scheduling and Design

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Abstract

Irrigation decisions depend on a coupled soil–water–atmosphere system in which pores, roots, surface cover and weather exchange water continuously, yet the software that advises farmers and designers seldom shows which variables it represents, which relation produced a number, or the range in which that relation holds. This article models the system in an open, offline framework and makes it explorable through seventeen result views. The modelled column has five compartments (atmosphere, soil surface and cover, soil matrix and structure, soil water, and crop), six water fluxes that close the water balance together with the storage change, and twelve controls (soil, tillage, consolidation rate, crop, mulch strength and capacity, rainfall, warming, irrigation method, trigger, sensor depth and nitrogen). Its variables range from porosity, retention, capillary pore emptying and consolidation of tilled pores, through two-stage soil evaporation, interception and mulch, to canopy cover, rooting depth, root water uptake and the water potential from soil to air, and on to drainage, nitrate leaching, the fate of conserved water and seven feedback loops. Three computing layers stand behind the views: a meta-model of a 10 190-run simulation ensemble, a daily 40-layer model, and closed-form surface, drip and sprinkler design modules. Every relation carries a provenance label (literature, data-driven or engineered) and each layer is tested against an independent reference. Evaluated for ten soils, the system variables show that a common -300 cm trigger uses 24 to 84 % of the plant-available water, that stage-1 evaporation lasts 1.1 to 3.0 days and longest in sand, and that drainage ranges from 28 to 164 mm yr^{-1} under the same rainfall. The evaluation also exposes two limitations: the leaf water potential is set by a conductivity floor rather than by the soil, and the drainage share of a mulch saving is understated in coarse soils because two experiments cover one soil. A reference implementation of the textbook surface procedure reproduces published worked examples (median error 0.08 %), the shipped surface layer reproduces 13 published advance times within 0.5 %, the meta-model

preserves the ranking of the trigger, sensor-depth and rainfall options tested (Spearman 0.64–0.97) with a diagnosed baseline offset, and the daily model conserves mass to 0.1 %. Verification exposed three defects in the first release, in the sprinkler range coefficient, the drip iteration count and the surface layer's treatment of steady intake during advance; all three are corrected in release 1.1.0. No field validation is claimed; the framework supplies a traceable map of the coupled system, of where each output can be relied on, and of the experiments that would validate it.

Keywords. soil–water–atmosphere modelling; soil–plant–atmosphere continuum; soil evaporation and mulch; irrigation decision support; simulation meta-model; software verification; irrigation scheduling and design; open science

List of symbols

Table 1: List of symbols, abbreviations and units used in the text.

Symbol	Meaning	Unit
$ET,\ I,\ D$	annual evapotranspiration, irrigation and drainage at 1 m	mm yr^{-1}
ρ	Spearman rank correlation	$-$
R^2, nRMSE	coefficient of determination; RMSE as a percentage of the held-out mean	$-$, %
$h^*,\ f^*$	critical suction and soil-independent depletion at which irrigation is triggered ($f^* = 0.45$)	cm, $-$
T_{req}	required infiltration (intake opportunity) time	min
$Q,\ Q_{\max}$	furrow inflow and its maximum non-erosive value	m^3 min^{-1}
$L,\ t_L,\ t_{L/2}$	furrow length; advance times to L and $L/2$	m, min
$x = p\,t^r$	power-law advance curve	m
E_a	application efficiency	fraction or %
$h_f,\ H_0$	friction loss along a drip lateral; inlet head	m
E_u	drip emission uniformity	$-$
$h,\ \theta,\ S_e$	suction head, water content, effective saturation	cm, $-$, $-$
$\alpha,\ n,\ m$	van Genuchten shape parameters	cm^{-1}, $-$, $-$
$f,\ f_T$	depletion fraction; its value at the trigger	$-$
$E,\ T$	soil evaporation, transpiration	mm yr^{-1}
P	precipitation	mm yr^{-1}
$N_L,\ Y$	nitrogen leaching, grain yield	kg ha^{-1}, Mg ha^{-1}
$\varphi,\ c,\ \gamma_P,\ \Delta T$	mulch vapour-flux factor, mulch interception capacity, rainfall scale, warming	$-$, mm, $-$, K
γ_y	year factor of the rainfall generator	$-$
$E_{\text{oth}},\ \Delta S$	other evaporation ($ET - T - E$), storage change	mm yr^{-1}
$\theta_s = \phi,\ \theta_r$	saturated (porosity) and residual water content	$-$
K_r	Mualem relative hydraulic conductivity	$-$
$r_c,\ \sigma,\ \gamma_c$	capillary radius of the largest water-filled pore, surface tension, contact angle	cm, N m^{-1}, $-$

Symbol	Meaning	Unit
p, p_0, p_c, τ_c	relative pore space, its value after tillage and when consolidated, consolidation rate	$-$, $-$, $-$, d^{-1}
Δ_{till}, Δ_{cons}	tillage and consolidation effects on drainage	$\mathrm{mm\,yr}^{-1}$
w_i, s_i, f_i, W_{max}	layer water, relative content, depletion; layer capacity	mm, $-$, $-$, mm
N_y	growing-season days on which the sensor is beyond the trigger	d
$e_{0,b}$, e_0	bare-soil and mulched evaporative demand	$\mathrm{mm\,d}^{-1}$
Q_1, $Q(t)$, t_1, τ_E, ℓ	stage-1 storage, cumulative evaporation, stage-1 duration, stage-2 time scale, dry-layer depth	mm, mm, d, d, mm
$\mathcal{I}$	regime index (evaporation against drainage)	$-$
RH, ψ_{air}	relative humidity, water potential of the air	$-$, MPa
$C(d)$	canopy cover	$-$
R_{max}, R_d, β_i	maximum and current rooting depth, root density of layer i	cm, cm, $-$
T_d, q_i^T, a_i	daily transpiration demand, layer uptake, stress factor	$\mathrm{mm\,d}^{-1}$, $\mathrm{mm\,d}^{-1}$, $-$
z_{50}, z_{90}	depths holding 50 and 90 % of root water uptake	cm
N_r	nitrogen rate (per cent of standard)	%
$\psi_{\mathrm{RZ}}, \psi_{\mathrm{root}}, \psi_{\mathrm{stem}}, \psi_{\mathrm{leaf}}$	water potential of root zone, root surface, stem and leaf	MPa
η_I, η_D, η_T, η_{rest}	shares of saved evaporation going to less irrigation, more drainage, more transpiration, storage	$-$
ΔE, ΔI, ΔD, ΔT_{tr}	changes of soil evaporation, irrigation, drainage and transpiration under mulch	$\mathrm{mm\,yr}^{-1}$
κ_N	nitrate cost of drainage	$\mathrm{kg\,N\,ha}^{-1}\,\mathrm{mm}^{-1}$
Z_n	net irrigation depth	mm
k_t	transpiration calibration factor	$-$
K_s, ψ_f, $\Delta\theta$	saturated conductivity, wetting-front suction, moisture deficit (surface layer)	$\mathrm{mm\,h}^{-1}$, mm, $-$
k, A	Kostiakov coefficient and exponent	$\mathrm{m\,min}^{-A}$, $-$
σ_y, σ_z, r	surface-flow shape factors, advance exponent	$-$
E_s, DP, RO	storage efficiency, deep percolation, runoff	$-$
c_v, x_e	emitter coefficient of variation, emitter exponent	$-$, $-$

Symbol	Meaning	Unit
$q,\ d,\ h_n,\ R_w$	sprinkler discharge, nozzle diameter, pressure head, wetted radius	L min^{-1}, mm, m, m
CU	Christiansen uniformity coefficient of a sprinkler pattern	fraction or %
$\rho_1,\ \rho_2$	coefficients of the power-law furrow geometry of Walker and Skogerboe (1987) ($A = \rho_1 y^{\rho_2}$)	–
Δx_e	emitter spacing along a drip lateral	m
SWAFL	the software framework presented here (SWAFL Desktop)	–
Engine 1, Engine 2	annual meta-model; daily bucket model	–

1. Introduction

Irrigation consumes most of the world's freshwater withdrawals, and improving its management is widely seen as a route to water security (Fereres and Soriano, 2007; Geerts and Raes, 2009; Playán and Mateos, 2006). Between a body of soil-water science and a farmer's decision about when, how much and by which method to irrigate lies a layer of software: schedulers, sensor dashboards, design calculators and crop-water models (Datta and Taghvaeian, 2023; García et al., 2020; Jones, 2004). That layer is where science is either translated faithfully or lost. Three weaknesses recur. First, *opacity*: a tool reports a number, but the user cannot tell which equation produced it, whether the equation is from the literature or is an in-house heuristic, and what data it was fitted to. Second, *unverified fidelity*: tools are tested on cases chosen by their authors, and the test results are seldom published in a form that a third party can re-run (Hutton et al., 2016; Wilkinson et al., 2016). Third, *conflation of purposes*: a screening emulator that is rank-faithful is sometimes presented, and used, as an absolute predictor (Bennett et al., 2013; Moriasi et al., 2007; Oreskes et al., 1994).

Process-based crop and soil models such as Daisy (Abrahamsen and Hansen, 2000; Hansen et al., 1991, 2012; Holbak et al., 2026), HYDRUS (Šimůnek et al., 2008, 2016), DSSAT (Jones et al., 2003), APSIM (Keating et al., 2003) and AquaCrop (Steduto et al., 2009) are the scientific backbone, but they are not decision tools for a user who needs an answer in seconds without installing a simulation environment. Emulators (surrogates) bridge the gap (Asher et al., 2015;

Kennedy and O'Hagan, 2001; Razavi et al., 2012; Sacks et al., 1989), but the surrogate literature is concerned mainly with statistical accuracy; the question of *traceability*, that is, whether each output can be tied back to the equations, the sources and the simulation data, receives less attention. Meanwhile classical irrigation-design theory (volume-balance advance and recession in surface irrigation (Elliott and Walker, 1982; Walker and Skogerboe, 1987), lateral hydraulics and emission uniformity in drip systems (Keller and Bliesner, 1990; Keller and Karmeli, 1974; Scaloppi and Allen, 1993), nozzle and pattern analysis in sprinkler systems (Christiansen, 1942; Keller and Bliesner, 1990; Playán et al., 2005)) is published in textbooks as closed-form equations and worked examples. These worked examples are a free and under-used source of verification data.

This article models soil–water–atmosphere interactions for irrigation scheduling and design in SWAFL Desktop, an offline application that integrates (i) a meta-model of a large designed ensemble of process-model simulations, (ii) a daily bucket model for dates and schedules, and (iii) closed-form design layers, under three design principles: every number is equation-traceable; every equation carries a provenance label; and every claim of fidelity is backed by a re-runnable test. The software is then verified against published worked examples, held-out simulations, mass closure and independent numerical solvers, and the tests are used to locate the limits of each layer. The article is therefore both a software description and a verification study, and its emphasis is on what can and cannot be trusted.

Verification is organised around the claims that the software makes. Traceability comes first: every output must lead to a numbered equation and a labelled provenance. The closed-form design layers are then checked against published worked examples, and the annual meta-model against held-out simulations, with rank and absolute fidelity assessed separately. The daily model is tested for mass conservation and for agreement with the annual model. Finally, the accuracy of the numerical solvers is mapped across their stated operating range.

After a summary of the software architecture and design principles (Section 2), the article sets out the coupled system that the framework represents: its compartments, state variables, fluxes and controls, and the relations that link them, from pore structure and retention through evaporation, interception and mulch to roots, transpiration and the water potential from soil to air, drainage, nitrogen and the feedbacks between them (Section 3). It then describes how the numbers are computed (Section 4), evaluates the system variables for ten soils, presents the result views and the sensitivity of the outputs to the controls (Sections 5.1–5.3), and reports the verification

of each computing layer (Sections 5.4–5.8).

This article makes no claim of field validation, of agronomic benefit, or of user-study usability; none of these have been carried out, and no field measurements exist in the project repository to support them. The claim is limited to traceability and to the verification results reported, and to the identification of limitations.

Levels of evidence. Four levels of evidence are kept apart. Computational reproducibility, the regeneration of every reported number from archived code and data, is established. Verification, agreement with published worked examples, held-out simulations and independent numerical solutions, is the subject of this article. Predictive validity against measurements is not claimed. Field transferability, which requires sensor-scheduled plots, catch-can and lateral measurements and a user study, is identified as the next step, and the operating envelopes reported here define the conditions in which those measurements are most informative.

What is new. The application joins three kinds of engine that are normally separate (a meta-model of a 10 190-run Daisy ensemble, a daily 40-layer water balance and closed-form hydraulic modules for surface, drip and sprinkler design) in one offline program. Its distinguishing feature is traceability: a ledger of 134 numbered equations, each with a label of origin (literature, data-driven or engineered), so that every output can be followed to its equation and its source. The design modules reproduce the worked examples of FAO-45 and standard texts within 3 %, the meta-model ranks held-out scenarios with Spearman 0.64 to 0.97, 10^5 scenarios are swept in about 2 s, and one command re-runs every check.

2. Software architecture and design principles

SWAFL Desktop is an Electron application written in JavaScript that runs offline on Windows, macOS and Linux; the installer and source are in the repository. All computation is in the engine and design modules and carries no external service calls; the simulation tables are embedded as JSON. The Manual documents every control, panel and equation (134 numbered equations), and the code reproduces the equations as written. Figure 1 summarises the architecture: an *evidence layer* (archived simulation data), two engines, three design layers, an equation-traceable dashboard and decision outputs.

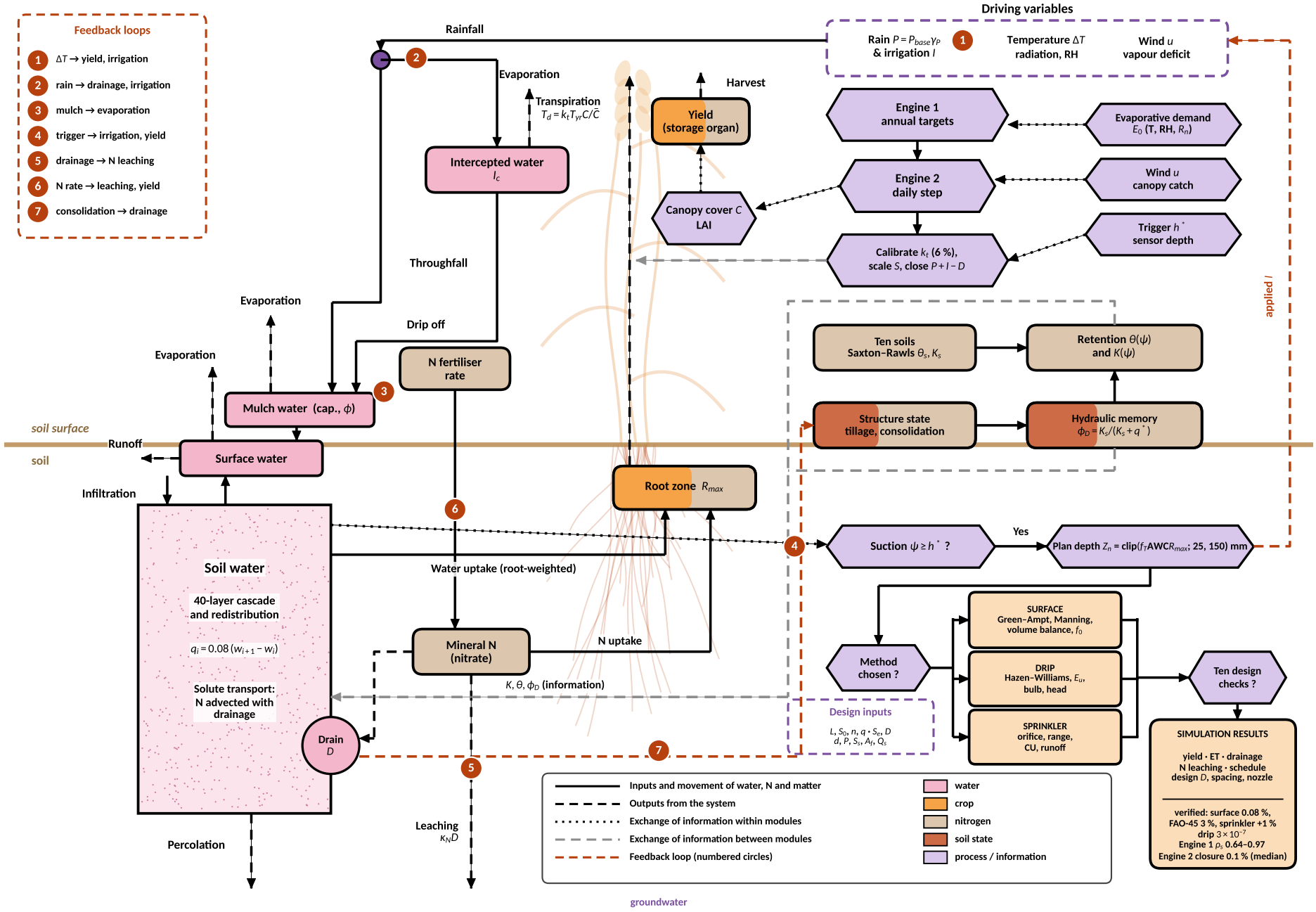


Figure 1: Schematic overview of the main processes in SWAFL Desktop. Boxes indicate storages and solid lines flows; dashed black lines are flows out of the system, dotted lines are flows of information and dashed grey lines exchange of information between modules. Dashed purple boxes with a triangle are the drop-down controls and driving variables of the Properties panel, and the drivers of soil hydraulic properties (SHP). Dashed red lines and numbered circles mark the seven feedback loops. Pink indicates water, orange the crop, light brown nitrogen and rust soil structure and hydraulic state; purple hexagons are processes and information (Engines 1 and 2, trigger, design layers) and peach boxes are design layers and results. The brown line is the soil surface, with groundwater hatched at the base.

Each dashboard tile, chart and arrow is produced by an equation printed in the Manual with a number, and the code comments cite the number. The traceability chain is *tile* → *function* → *equation* → *source*. Appendix A lists the equations used in this article together with their manual numbers, provenance and source.

Each equation is labelled **[Literature]** (taken from a cited source, implemented as written), **[Data-driven]** (a constant or table derived from the Daisy ensemble), or **[Engineered]** (a heuristic or simplification fixed by the developer, e.g. the spray-pattern shapes). The labels allow users and reviewers to see which parts of the software rest on published science and which on judgement. An audit of the first release of the manual found the labels applied inconsistently (some equations labelled, many not); every one of the 134 numbered equations now carries its label (Section 5.8).

Each layer is accompanied by a test whose expected output comes from outside the software: a published worked example, a held-out simulation, a mass balance or an independent numerical solver. The scripts are in the repository (Appendix B).

Engine 1 is an *annual* meta-model: it composes, by multiplication of ratios and addition of differences, 20-year-mean results of Daisy simulations to produce ET, soil evaporation, transpiration, irrigation, yield, drainage and nitrogen leaching for the chosen soil, tillage, mulch, rainfall scale, warming, trigger, sensor depth, method and nitrogen rate. Engine 2 is a *daily* 40-layer bucket model over 20 years of generated weather; it produces dates, depths, the animated field and schedules, and it is calibrated and scaled so that its 20-year means equal Engine 1's. The design layers use hydraulics rather than data; they borrow soil, crop and trigger from the engines.

3. The coupled soil–water–atmosphere system and its variables

This section sets out what the framework represents before it describes how the numbers are computed (Section 4). The modelled system is a field column of 1 m depth, from the air above the canopy to the lower boundary at which drainage is counted, with five compartments: the atmosphere, the soil surface and its cover, the soil matrix and its structure, the soil water, and the crop. The compartments exchange water through six fluxes (precipitation, irrigation, transpiration, soil evaporation, other evaporation and drainage, Eq. (1)), which together with the storage change close the balance, and are driven by twelve controls. Every state variable, flux and control is named in Tables 2 and 3, with the relation that produces it and the provenance label of that relation; the relations themselves follow in Sections 3.3–3.10. Manual equation numbers are given in

brackets so that each relation can be followed into the Manual.

3.1 System boundary and the governing balance

Over any period the water entering the column equals the water leaving it plus the change in storage,

$$P + I = T + E + E_{\text{oth}} + D + \Delta S, \tag{1}$$

where P is precipitation, I irrigation, T transpiration, E soil evaporation, E_{oth} the remaining evaporation (water intercepted by the canopy, evaporation from mulch and litter, and the unattributed term identified in the companion water-balance audit), D the drainage across 1 m (negative values denote net capillary rise) and ΔS the change in profile storage. On the annual scale Eq. (1) is closed by Engine 1 (Eq. (39)); on the daily scale by Engine 2 (Eq. (43)). Nitrate leaching N_L is carried with D, and grain yield Y with T. Table 2 lists the state variables and fluxes of each compartment.

Table 2: State variables and fluxes of the modelled soil–water–atmosphere system, by compartment; the soil matrix and soil structure rows form the soil compartment, and the lower boundary (drainage) and the management decision are listed separately. Source: E1 = Engine 1 (annual), E2 = Engine 2 (daily), D = Daisy output stored in the software, R = relation evaluated by the layer. Provenance: L literature, D data-driven, E engineered.

Compartment	Variable (symbol)	Unit	Relation (manual)	Source; provenance
Atmosphere	rainfall scale γ_P; temperature shift ΔT	−; K	controls (Table 3)	input
	annual precipitation $P = P_{\text{base}}\gamma_P$	mm yr^{-1}	Eq. (39) (5.12)	E1; D
	daily rainfall R, year factor γ_y	mm d^{-1}, −	Eq. (40) (5.15)	E2; E
	bare-soil evaporative demand $e_{0,b}$	mm d^{-1}	Eq. (10) (5.26)	R; E
	relative humidity RH; air water potential ψ_{air}	−; MPa	Eq. (15) (7.5)	R; L
Surface and cover	mulch vapour-flux factor φ; interception capacity c	−; mm	controls	input

Compartment	Variable (symbol)	Unit	Relation (manual)	Source; provenance
	interception per shower I_c; throughfall T_f	mm	Eq. (41) (5.16)	E2; E
	stage-1 storage Q_1; stage-2 time scale τ_E; stage-1 duration t_1	mm; d; d	Eq. (11) (9.2–9.3)	R; E
	cumulative evaporation $Q(t)$; drying-front depth ℓ	mm	Eqs. (11) and (12) (9.3–9.5)	R; E
	annual soil evaporation E; other evaporation E_{oth}	mm yr^{-1}	Eqs. (31) and (39) (5.4, 5.12)	E1; D
	regime index $\mathcal{I}$	–	Eq. (14) (9.8)	R; E
Soil matrix	sand, silt, clay fractions	%	measured texture	D
	porosity $\phi = \theta_s$; residual content θ_r	–	Eq. (2) (8.2)	R; E
	van Genuchten α, n; effective saturation S_e; relative conductivity K_r	cm^{-1}, –	Eqs. (27) and (28) (5.1–5.2)	R; L
	capillary radius of the largest water-filled pore r_c	cm	Eq. (3) (8.4)	R; L
	available-water capacity AWC	mm cm^{-1}	Table 7	E
Soil structure	tillage system; consolidation rate τ_c	–; d^{-1}	controls	input
	relative pore space $p(t)$	–	Eq. (4) (8.1)	R; E
	tillage and consolidation effects on drainage Δ_{till}, Δ_{cons}	mm yr^{-1}	Eq. (5) (8.5)	D
Soil water	layer storage w_i; relative content s_i; depletion f_i (40 layers of 2.5 cm)	mm; –; –	Eq. (6) (Section 5.3)	E2; E
	suction h, water content θ, depletion f	cm, –, –	Eqs. (27) and (29) (5.1, 5.3)	R; L

Compartment	Variable (symbol)	Unit	Relation (manual)	Source; provenance
	sensor suction record, days beyond the trigger N_y	cm; d	Eq. (8) (6.1–6.2)	D
	share of root zone beyond the trigger	–	Eq. (9) (7.6)	D; E
	wetting front; redistribution flux q_i	cm; mm d^{-1}	Eq. (7) (5.17)	E2; E
	storage change ΔS, ΔS_d	mm	Eqs. (39) and (43) (5.12, 5.25)	E1, E2; L
Crop	canopy cover $C(d)$	–	Eq. (16) (5.13–5.14)	E2; E
	maximum and current rooting depth $R_{\max}$, R_d; root density β_i	cm; –	Eqs. (17) to (19) (5.18–5.20)	E2; E
	transpiration demand T_d; layer uptake q_i^T; stress factor a_i	mm d^{-1}	Eq. (42) (5.22–5.23)	E2; E
	depths holding 50 % and 90 % of uptake z_{50}, z_{90}	cm	Eq. (20) (8.7)	R; E
	water potential of root zone, root, stem and leaf $\psi_{\mathrm{RZ}}, \psi_{\mathrm{root}}, \psi_{\mathrm{stem}}, \psi_{\mathrm{leaf}}$	MPa	Eqs. (21) and (22) (7.1–7.4)	R; L, E
	annual transpiration T; yield Y; irrigation efficiency IE	mm yr^{-1}; Mg ha^{-1}; kg m^{-3}	Eqs. (32) and (35) (5.5, 5.8)	E1; D
Lower boundary	drainage D; nitrate leaching N_L	mm yr^{-1}; kg ha^{-1}	Eqs. (36) and (37) (5.9–5.10)	E1; D
	nitrate cost of drainage κ_N	kg ha^{-1} per mm	Eq. (24) (10.2)	R; D
Management	trigger suction h_T; trigger depletion f_T; critical depletion f^*	cm; –; –	Eqs. (25) and (29) (5.3, 6.3)	R; L

Compartment	Variable (symbol)	Unit	Relation (manual)	Source; provenance
	net irrigation depth Z_n; annual irrigation I	mm; mm yr^{-1}	Eqs. (26) and (34) (5.27, 5.7)	R, E1; L, D
	fate of evaporation saved by mulch $\eta_I, \eta_D, \eta_T, \eta_{\text{rest}}$	–	Eq. (23) (9.6–9.7)	R; D

3.2 Driving variables and controls

Every control is an index into the list of values at which simulations exist, so that no displayed value lies between runs (Table 3). Changing a control re-evaluates every view; the last column of Table 3 states which variables respond.

Table 3: The twelve controls of the framework, their values and defaults, and the variables that respond. Run set: the Daisy experiment from which the response is read.

Control	Values	Default	Responding variables (run set)
Soil	ten soils S01–S10	S01	all: (α, n), AWC, baseline vector, trigger anchor; intake parameters of the design layers (E5 baseline)
Tillage	conventional, reduced, no-till	conventional	p_0; D, E, Y (E5)
Consolidation rate τ_c	0.001, 0.0025, 0.005, 0.01, 0.02 d^{-1}	0.001	$p(t)$; D (E5); no effect under no-till
Crop	cereal (simulated); release 1.1.0: 48 library crops (FAO-56 transfer), release 1.0.0: legume, vegetable (scaled)	cereal	calendar, $C(d)$, R_{max}; $Y \times f_{\text{crop}}$
Mulch factor φ	0.1, 0.2, …, 1.0	1.0 (bare)	E, I, D, Y (E7, E14); e_0
Interception capacity c	0.5, 1, 2, 4 mm	1 mm	E (E7, E14)
Rainfall scale γ_P	$\times$0.5, 0.6, …, 1.4	$\times$1.0	all (E11, E14); R, I_c

Control	Values	Default	Responding variables (run set)
Temperature shift ΔT	-1.0, -0.5, …, $+3.5$ K	0	E, T, I, Y (E11); $e_{0,b}$, RH
Irrigation method	surface; overhead; drip at 10–20, 20–30, 40–50, 50–60 cm	surface	I, D, Y (E4); dose placement; β_i near the emitter
Trigger suction h_T	-50 to $-10\,000$ cm (20 values)	-300 cm	I, Y, D (E2); irrigation dates; $R_{\max}$; Z_n
Sensor depth z_s	5, 10, …, 100 cm	20 cm	I, Y (E3); irrigation dates
Nitrogen rate	10 to 150 % of standard (10 values)	100 %	Y, N_L (E13); $R_{\max}$

3.3 Soil matrix: texture, porosity, retention and conductivity

Each soil is described by its measured texture and an effective van Genuchten pair (α, n) (Table 7). Porosity and residual water content follow an engineered texture relation [8.2],

$$\phi = \theta_s = \mathrm{clip}\big(0.36 + 0.0016\,\mathrm{clay} + 0.0009\,\mathrm{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\mathrm{clay}, \quad (2)$$

with clay and silt in per cent, and the water content, effective saturation and relative conductivity follow the van Genuchten–Mualem relations Eqs. (27) and (28) (Mualem, 1976; van Genuchten, 1980) [5.1–5.2, 8.3]. The relation between suction and the pores that hold water is the capillary (Young–Laplace) law [8.4],

$$r_c = \frac{2\sigma\cos\gamma_c}{\rho_w g h} \approx \frac{0.149\ \mathrm{cm}^2}{h}, \quad (3)$$

with surface tension $\sigma = 0.0728\,\mathrm{N\,m^{-1}}$ and contact angle $\gamma_c \approx 0$ (Hillel, 1998): a pore of radius r is water-filled when $r \le r_c(h)$ and empties at $h = 0.149/r$ cm (with r in cm). The pore views draw this rule on a modelled section whose grain classes follow the measured texture; the section illustrates the mechanism and is not a pore-network simulation (Blunt, 2017).

Plant-available water is the water held between field capacity (100 cm) and the wilting point (15 849 cm) (Allen et al., 1998), and the depletion f of Eq. (29) [5.3] expresses the soil-water state as the share of that water already used. The same suction therefore corresponds to different depletions in different soils: -300 cm uses 24 % of the plant-available water of the silty clay loam

and 84 % of that of the sand (Table 8).

3.4 Soil structure, tillage and consolidation

After tillage the loosened topsoil re-settles. The structure views describe the relative pore space p (1 for freshly loosened soil) by a first-order relaxation towards a consolidated value $p_c = 0.30$ [8.1],

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\,(\text{conventional}),\ 0.72\,(\text{reduced}), \qquad p = 0.42\,(\text{no-till}), \tag{4}$$

with t in days since tillage and τ_c the consolidation rate, so that loose pores have a half-life $\ln 2/\tau_c$ of 693 d at 0.001 d^{-1} and 35 d at 0.02 d^{-1}. The exponential form is the first-order consolidation used in the structure sub-model on which Daisy's tillage effect is based (Flanagan and Nearing, 1995); p_0, p_c and the no-till value are presentation constants, and the hydrological effect of tillage is read from the simulations, not from Eq. (4). For each soil the drivers of a change in hydraulic properties are summarised as the tillage effect and the consolidation effect on annual drainage [8.5],

$$\Delta_{\text{till}} = D(\text{conv}, \tau_c^*) - D(\text{no-till}, \tau_c^*), \qquad \Delta_{\text{cons}} = D(\text{till}^*, \tau_{\text{fast}}) - D(\text{till}^*, \tau_{\text{slow}}), \tag{5}$$

with the current settings starred and D taken from the E5 runs; ordinary least-squares lines of each effect on clay content [8.6] show how the response to structure varies across textures (Leuther and Schlüter, 2021; Strudley et al., 2008).

3.5 Soil water in depth and time

In the daily model the profile is 40 layers of $\Delta z = 2.5$ cm. Layer i holds plant-available water w_i between 0 and $W_{\max} = \text{AWC}\,\Delta z$, and its relative content and depletion are

$$s_i = \frac{w_i}{W_{\max}}, \qquad f_i = 1 - s_i, \qquad h_i = \max\{30,\ h(f_i)\}, \tag{6}$$

where $h(f)$ is the inverse of Eq. (29) Eq. (25); the 30 cm floor is engineered. Infiltrating water fills the top layer and cascades downwards; the excess leaving layer 40 is drainage and the deepest wetted layer is the wetting front. Between neighbouring layers a slow redistribution moves water

towards the drier layer [5.17],

$$q_i = 0.08\,(w_{i+1} - w_i), \tag{7}$$

which is displayed as capillary rise when it is upward and larger than $0.03\,W_{\max}$. The tipping-bucket structure conserves water exactly but is not a Richards solution, which is why its drainage is scaled to Engine 1 (Eq. (43)).

The sensor views use the Daisy-simulated daily suction at 20 cm for the 20 seasons, irrigated and rainfed. For year y the number of growing-season days on which the sensor is beyond the trigger, and the share of such days removed by irrigation, are [6.1–6.2]

$$N_y(h_T) = \#\{d \in \text{season of } y:\ h_d > |h_T|\}, \qquad X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \tag{8}$$

over the 183 days from 1 April, and the profile view reports the share of the 0–60 cm root zone that is beyond the trigger [7.6],

$$\text{share} = \frac{\sum_{i:\, z_i \le 60,\ h_i > |h_T|} \Delta z_i}{\sum_{i:\, z_i \le 60} \Delta z_i}. \tag{9}$$

3.6 Atmosphere and the soil surface

Evaporative demand. The bare-soil demand used by the evaporation view combines a season-mean of $4.2\,\text{mm}\,\text{d}^{-1}$ with a seasonal weight, a warming factor of 7 % per kelvin, a cloud factor and the dryness of the year [5.26],

$$\begin{aligned} e_{0,b} = 4.2\,(1 + 0.07\Delta T)\,\frac{0.35 + 0.325\big[1 + \sin\frac{2\pi(\text{doy}-100)}{365}\big]}{0.675} \\ \times\ \text{clip}\big(1 - 0.12(\gamma_P - 1);\, 0.85, 1.12\big)\sqrt{\gamma_y}\ \ \text{mm}\,\text{d}^{-1}, \end{aligned} \tag{10}$$

and under mulch and canopy the demand is $e_0 = e_{0,b}\,\varphi\,(1 - 0.4C)$. The warming factor is close to the Clausius–Clapeyron rate of the saturation vapour pressure; the constants are engineered.

Two-stage drying of the surface. After a wetting the surface loses water at the atmospheric rate while capillary flow keeps it wet (stage 1), and then at a rate limited by vapour diffusion through a growing dry layer (stage 2), with cumulative loss rising roughly as $\sqrt{t}$ (Lehmann et al., 2008;

Or et al., 2013; Philip, 1957; Ritchie, 1972). The evaporation view implements [9.2–9.3]

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad Q(t) = q_0 + \begin{cases} e_0 t, & t \le t_1, \\ (Q_1 - q_0) + 2 e_0 \tau_E \big(\sqrt{1 + (t - t_1)/\tau_E} - 1\big), & t > t_1, \end{cases} \tag{11}$$

whose derivative is the rate $e(t) = e_0\big[1 + (t - t_1)/\tau_E\big]^{-1/2}$ in stage 2. The stage-1 storage is the water held in the top $L_1 = 40\,\text{mm}$ between an initial and a surface-dry suction, $Q_1 = \max\{2,\ [\theta(h_i) - \theta(h_s)]L_1(0.6 + 0.4\gamma_P)\}$ with $h_i = 0.25(1.5 - 0.5\gamma_P)/\alpha$ and $h_s = 4/\alpha$; the stage-2 time scale is $\tau_E = 0.006 + 0.03(\text{clay} + \text{silt})/100\,\text{d}$; and q_0 is the loss already avoided when drip places the water below the surface ($q_0 = (1 - w_f)0.95Q_1$, $w_f = \text{clip}(1 - z_d/50; 0.05, 1)$, zero for surface methods). Because h_i and h_s scale with $1/\alpha$, Q_1 samples the steep part of every retention curve. The dry-layer depth and the suction profile below it are [9.4–9.5]

$$\begin{aligned} \ell(t) &= \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm}, \\ \log_{10} h(z) &= \log_{10} h_s + \big(\log_{10} h_i - \log_{10} h_s\big)\big(1 - e^{-(z-\ell)/\lambda_E}\big), \end{aligned} \tag{12}$$

with $\Delta\theta = \text{clip}(\theta(h_i) - \theta(10^4); 0.05, 0.4)$ and $\lambda_E = L_1/3$; the pores that hold water at each depth follow Eq. (3). All constants of Eqs. (11) and (12) are engineered, and the annual soil evaporation of the framework is the Daisy value of Eq. (31), not this curve.

Interception and mulch. The canopy and litter catch the first I_c mm of each shower (Eq. (41)). A mulch acts through φ on the vapour flux and through c on interception; its effect on the annual fluxes is the difference between Engine 1 evaluated bare and mulched [9.1],

$$\Delta E = E_{1.0} - E_\varphi, \qquad \Delta I = I_{1.0} - I_\varphi, \qquad \Delta D = D_\varphi - D_{1.0}, \qquad \Delta T_{\text{tr}} = T_\varphi - T_{1.0}, \tag{13}$$

and the regime map classifies each combination of mulch and rainfall by the index [9.8]

$$\mathcal{I} = \frac{E - D}{\max\{1, E + D\}}, \tag{14}$$

with evaporation dominating where $\mathcal{I} > 0.05$, drainage where $\mathcal{I} < -0.05$, and the boundary in between.

Air. The water potential of the air follows the thermodynamic relation with relative humidity [7.5] (Nobel, 2009),

$$\psi_{\text{air}} = \frac{\mathcal{R}T_K}{V_w}\ln \text{RH} \approx 135.3\ \ln \text{RH}\ \ \text{MPa}, \qquad \text{RH} = \text{clip}(0.66 - 0.03\Delta T;\ 0.35,\ 0.90), \quad (15)$$

with $\mathcal{R}$ the gas constant, $T_K \approx 293\,\text{K}$ and $V_w = 18 \times 10^{-6}\,\text{m}^3\,\text{mol}^{-1}$; at $\text{RH} = 0.66$ the air is at $-56\,\text{MPa}$.

3.7 The crop: canopy, roots, uptake and the water-potential path

Canopy. The cereal follows the four-year rotation of the simulations (spring barley, winter barley, winter rape, winter wheat). Canopy cover rises logistically, $C(d) = \{1 + \exp[-(d - d_{\text{mid}})/s]\}^{-1}$ with $s = 13\,\text{d}$ for spring crops [5.13], follows a two-phase curve for winter crops [5.14], and declines linearly from the peak to harvest:

$$C \leftarrow C\ \max\Big\{0,\ 1 - \frac{d - d_{\text{pk}}}{d_{\text{harv}} - d_{\text{pk}}}\Big\} \quad (d > d_{\text{pk}}). \quad (16)$$

Roots. The maximum rooting depth responds to texture, trigger and nitrogen [5.18],

$$R_{\max} = \text{clip}\Big(R_c(0.85 + 0.3\,\text{sand})\Big[1 + 0.12\log_{10}\frac{|h_T|}{300}\Big]\Big[1 - 0.18\,\frac{N_r - 100}{100}\Big];\ 25,\ 100\Big)\ \text{cm}, \quad (17)$$

with $R_c = 95\,\text{cm}$ for the cereal, sand as a fraction and N_r the nitrogen rate in per cent, so that drier triggers deepen the roots and high nitrogen rates make them shallower. While the canopy grows, the rooting depth advances towards water [5.19],

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\,\text{clip}\big(\tfrac{C - 0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7(0.35 + \bar{s}_{\text{wet}})\kappa_r\big\}\Big\}, \quad (18)$$

where $\bar{s}_{\text{wet}}$ is the mean relative content of the four layers below the root front and $\kappa_r = 1.3$ when the topsoil is dry; roots die back after harvest and are pruned by drought. Root density decreases exponentially with depth, increases with wetness and, under drip, concentrates around the emitter [5.20],

$$\beta_i \propto \exp\Big(-\frac{z_i}{\lambda_r R_d + 8}\Big)(0.45 + 0.9 s_i)\Big[1 + 3e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip}}, \qquad z_i \le R_d, \qquad \sum_i \beta_i = 1, \quad (19)$$

with $\lambda_r = 0.32$ for the cereal. The demand is divided among layers by availability-weighted root density, with the linear stress factor a_i of Eq. (42) in the spirit of the FAO-56 coefficient (Allen et al., 1998; Feddes et al., 1978). The depth that holds a share u of the annual uptake U_i of layer i is [8.7]

$$z_u = k_u \Delta z, \qquad k_u = \min\Big\{k : \frac{\sum_{i \le k} U_i}{\sum_i U_i} \ge u\Big\}, \qquad u = 0.5,\ 0.9, \tag{20}$$

averaged over the 20 years.

The water-potential path. The continuum from soil to air is a chain of series resistances in the catenary description of van den Honert (1948). The root-zone potential is the root-density-weighted mean of the layer suctions of Eq. (6) [7.1],

$$\psi_{\mathrm{RZ}} = -9.807 \times 10^{-5} \, \frac{\sum_i \beta_i h_i}{\sum_i \beta_i} \ \text{MPa}, \tag{21}$$

and with the day's transpiration E_t (mm d^{-1}) [7.2–7.4]

$$\begin{gathered} R_{\mathrm{soil}} = 0.03 \ \max\{K_r, 0.002\}^{-0.55}, \quad R_{\mathrm{root}} = 0.10, \quad R_{\mathrm{xyl}} = 0.05, \quad R_{\mathrm{leaf}} = 0.08, \\ \psi_{\mathrm{root}} = \psi_{\mathrm{RZ}} - E_t R_{\mathrm{soil}}, \qquad \psi_{\mathrm{stem}} = \psi_{\mathrm{root}} - E_t R_{\mathrm{root}}, \\ \psi_{\mathrm{leaf}} = \mathrm{clip}\big(\psi_{\mathrm{stem}} - E_t(R_{\mathrm{xyl}} + R_{\mathrm{leaf}}); -3.2, 0\big), \end{gathered} \tag{22}$$

with resistances in MPa per mm d^{-1}, K_r the relative conductivity at the root-zone suction (Eq. (28)) and the transpiration terms set to zero when canopy cover is below 0.05; stomata are drawn as closing when $\psi_{\mathrm{leaf}} < -1.6$ MPa. The physical relations are Eq. (15) and the retention and conductivity functions; the resistances are engineered.

3.8 Drainage, nitrogen and the fate of conserved water

Water that a mulch keeps in the soil can reduce irrigation, raise transpiration, drain below the root zone or remain in storage. The fate view divides each millimetre of evaporation saved among these destinations [9.6–9.7],

$$\begin{gathered} \eta_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}}, \quad \eta_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}}, \\ \eta_T = \frac{\max\{\Delta T_{\mathrm{tr}}, 0\}}{\max\{\Delta E, 0.1\}}, \quad \eta_{\mathrm{rest}} = \max\{0, 1 - \eta_I - \eta_D - \eta_T\}, \end{gathered} \tag{23}$$

with the differences of Eq. (13) evaluated at each rainfall scale, which is the annual counterpart of the loss fates analysed in the companion hydraulic-memory study. Nitrate moves with drainage, and the nitrate cost of a millimetre of drainage is the one-step difference [10.2]

$$\kappa_N = \frac{N_L^{(\gamma_P+0.1)} - N_L}{D^{(\gamma_P+0.1)} - D} \quad [\mathrm{kg\,N\,ha^{-1}\,mm^{-1}}]. \tag{24}$$

3.9 Coupling: the seven feedback loops

The compartments act on each other through seven loops, marked by numbered circles in Fig. 1: (1) temperature and the crop, (2) rain and soil water, (3) mulch and evaporation, (4) the sensor trigger and irrigation, (5) drainage and nitrate leaching, (6) nitrogen and roots, and (7) tillage structure and drainage. The loop view quantifies each by moving its control one step on its list and re-evaluating Engine 1; the printed effect is the change of the outputs listed in Table 4, and the drawn arrow width is $w_k = 5 + \mathrm{clip}(\mu_k; 0, 4)$ pixels [10.1] with μ_k the arrow-width input of the table. The effects are one-step differences of simulated results, not partial derivatives, so a step at the end of a list may be smaller than a step in its middle; loops between soil moisture and the atmosphere at regional scale (Entekhabi et al., 1996; Koster et al., 2004; Seneviratne et al., 2010) are outside the column and are not represented.

Table 4: The seven feedback loops: control step, printed effect and arrow-width input μ.

Loop	Control step	Printed effect	μ
1	temperature ± 0.5 K	$\Delta Y\, f_{\mathrm{crop}}$, ΔI	$8\lvert\Delta Y\rvert$
2	rainfall scale ± 0.1	ΔD, ΔI	$\lvert\Delta D\rvert/12$
3	one step stronger mulch	ΔE, ΔI	$\lvert\Delta E\rvert/14$
4	trigger one step drier or wetter	ΔI, $\Delta Y\, f_{\mathrm{crop}}$	$\lvert\Delta I\rvert/8$
5	one more millimetre of drainage	κ_N (Eq. (24))	$\lvert\Delta D\rvert/10$
6	nitrogen ± 15 %	ΔN_L, $\Delta Y\, f_{\mathrm{crop}}$	$\lvert\Delta N_L\rvert/8$
7	consolidation rate one step	ΔD	$\lvert\Delta D\rvert/2$

3.10 From the soil-water state to the irrigation decision

The decision links the state of the soil to the action. A sensor at depth z_s reads the suction of layer $j_s = \mathrm{round}(z_s/\Delta z)$, and irrigation is due when its depletion reaches the trigger depletion $f_T = f(|h_T|)$ of Eq. (29). Because the same suction means different dryness in different soils, the trigger can also be set as a soil-independent depletion f^*, whose suction in each soil is the

inverse [6.3]

$$h(f^*) = \{h : \ f(h) = f^*\}, \tag{25}$$

found by bisection; with $f^* = 0.45$, the depletion beyond which the simulated yield falls by more than 5 %, this gives the soil-specific settings of Table 7 (143 cm for sand to 729 cm for silty clay loam). The net depth that refills the root zone from the trigger back to field capacity is [5.27]

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\ 25,\ 150\big) \text{ mm}, \tag{26}$$

the readily-available-water idea of FAO-56 practice (Allen et al., 1998), which carries the soil, crop and trigger into the surface, drip and sprinkler design layers (Sections 4.4–4.6). The relative yield at the trigger is $Y(h_T)/\max_h Y(h)$ over the E2 runs of the soil [6.5].

4. Materials and methods

4.1 Evidence layer, the simulation ensemble

The evidence layer is a designed ensemble of 10 190 Daisy runs (Hansen et al., 2012; Holbak et al., 2026) in 17 experiments on ten soils (sand to silty clay loam; Ap and Bt horizons) with 20 years of Taastrup weather, described in the companion papers; twelve experiments cover all ten soils and five (E8, E11, E12, E14, E15) the coarse loam only. The experiments used by Engine 1 are summarised in Table 5, and the output variables of every run are listed in Table 6. Daisy solves the Richards equation with van Genuchten–Mualem hydraulics (Mualem, 1976; van Genuchten, 1980) coupled to a structure model for tillage effects (Flanagan and Nearing, 1995). For Engine 1 the effective parameters (α, n) were set so that the soil passes through the anchors of the project's trigger table; for the standard textures they correspond to the class means of Carsel and Parrish (1988) (Table 7).

Table 5: Experiments used by Engine 1, the number of soils they cover and the equation family that consumes them.

Experiment	Varied factors	Soils	Used for
E2	trigger suction	10	trigger correction Eq. (38)
E3	sensor depth	10	sensor-depth correction Eq. (38)
E4	irrigation method (overhead, surface, drip at eight depths from 5–10 to 50–60 cm)	10	method correction Eq. (38)
E5	tillage × structural consolidation	10	baseline vector **b** for Eqs. (31) to (37)
E7	mulch vapour-flux factor × interception capacity	10	soil evaporation $\bar{E}$
E11	warming × evaporation factor × rain	1 (S01)	warming ratios and increments
E13	N rate × water strategy	10	nitrogen correction
E14	mulch × capacity × rain	1 (S01)	mulch ratios and increments

Table 6: Output variables of each Daisy run (20-year means, 1980–1999) that form the evidence layer. The complete run table is in the Data folder of the repository and on Zenodo.

Variable	Unit	Meaning
`rain_mm`, `irrigation_mm`	mm yr^{-1}	precipitation P and irrigation I
`ET_mm`	mm yr^{-1}	actual evapotranspiration
`transpiration_mm`, `soil_evap_mm`, `canopy_evap_mm`	mm yr^{-1}	reported components of ET (T, E, interception)
`drainage_1m_mm`	mm yr^{-1}	flux across 1 m (negative: net capillary rise)
`water_stress_d`	d yr^{-1}	days of crop water stress
`yield_Mg_ha`	Mg ha^{-1}	grain yield Y
`N_fertiliser_kg_ha`, `N_leached_kg_ha`	kg ha^{-1} yr^{-1}	nitrogen applied and leached (N_L)
`IE_kg_m3`, `DPF`, `NLF`	kg m^{-3}, −, −	irrigation efficiency, $D/(P+I)$, N_L/N applied
daily suction at 20 cm; 1994 profiles	cm	sensor record (V1) and profile view (V2), irrigated and rainfed (coarse loam, coarse sand, standard settings); in release 1.1.0 other settings and soils add the difference between two Engine 2 runs (settings minus standard) to the record as depletion

Table 7: Effective van Genuchten parameters used by Engine 1 and the available-water capacity (AWC, mm of water per cm of soil, an engineered texture-class constant), with the critical suction h^* at the soil-independent depletion $f^* = 0.45$.

ID	Soil	α (cm^{-1})	n	AWC	h^* (cm)
S01	Coarse loam	0.0961	1.353	1.5	387
S02	Coarse sand	0.0536	1.600	0.5	262
S03	Sand	0.1450	2.680	0.6	143
S04	Loamy sand	0.1240	2.280	0.9	160
S05	Sandy loam	0.0750	1.890	1.2	195
S06	Loam	0.0360	1.560	1.7	284
S07	Silt loam	0.0200	1.410	2.0	401
S08	Sandy clay loam	0.0590	1.480	1.4	311
S09	Clay loam	0.0190	1.310	1.6	503
S10	Silty clay loam	0.0100	1.230	1.7	729

Retention and relative conductivity follow van Genuchten (1980) and Mualem (1976):

$$S_e(h) = \left[1 + (\alpha|h|)^n\right]^{-m}, \qquad m = 1 - 1/n, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r) S_e(h), \tag{27}$$

$$K_r(S_e) = S_e^{1/2}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2. \tag{28}$$

Plant-available water is defined between field capacity (suction 100 cm) and wilting point (15 849 cm), the FAO convention (Allen et al., 1998):

$$f(h) = \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}, \tag{29}$$

where $f = 0$ is field capacity and $f = 1$ the wilting point. The inverse $h(f)$ is obtained by bisection on $\log_{10}|h| \in [0, 4.6]$ (60 halvings). A trigger suction h_T therefore corresponds to a depletion fraction $f_T = f_{\text{used}}(|h_T|)$.

4.2 Engine 1, the annual meta-model

Let $\mathbf{b} = (E_b, T_b, ET_b, I_b, Y_b, D_b, N_b)$ be the 20-year-mean Daisy results (soil evaporation, transpiration, evapotranspiration, irrigation, yield, drainage, nitrogen leaching) for the chosen soil, tillage and consolidation rate (E5), φ the mulch strength, c the mulch capacity, γ_P the rainfall scale and ΔT the warming. Define mulch lookups $M(\varphi, \gamma_P)$ from E14, warming lookups $W(\Delta T, \gamma_P)$ from

E11, and the shorthand

$$\begin{aligned} \text{now} = M(\varphi, \gamma_P), \quad \text{ref} = M(\varphi, 1), \quad \text{nm} = M(1, \gamma_P), \quad \text{bare} = M(1, 1), \\ \text{tN} = W(\Delta T, \gamma_P), \quad \text{tB} = W(0, \gamma_P), \quad \text{tR} = W(0, 1). \end{aligned} \tag{30}$$

The engine composes ratios (taken as 1 where the denominator is below 10^{-6}), clipped to guard against extrapolation, as

$$E = \bar{E}(\text{soil}, \varphi, c) \left[\frac{\text{now}.E}{\text{ref}.E}\right] \left[\frac{\text{tN}.E}{\text{tB}.E}\right] \left[\frac{E_b}{E_{b,\text{ref}}}\right], \tag{31}$$

$$T = T_b \left[\frac{\text{now}.T}{\text{bare}.T}\right] \text{clip}\left(\frac{\text{tN}.T}{\text{tB}.T}; 0.3, 2.5\right), \tag{32}$$

$$\boldsymbol{ET} = \boldsymbol{ET}_b \left[\frac{\text{now}.\boldsymbol{ET}}{\text{bare}.\boldsymbol{ET}}\right] \text{clip}\left(\frac{\text{tN}.\boldsymbol{ET}}{\text{tB}.\boldsymbol{ET}}; 0.5, 2\right), \tag{33}$$

$$I = I_b \left[\frac{\text{now}.I}{\text{bare}.I}\right] \text{clip}\left(\frac{\text{tN}.I}{\text{tB}.I}; 0.4, 2.5\right), \tag{34}$$

$$Y = Y_b \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right] \left[\frac{\text{now}.Y}{\text{nm}.Y}\right], \tag{35}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D), \tag{36}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N). \tag{37}$$

Here $\bar{E}(\text{soil}, \varphi, c)$ is the linear interpolation in φ of the simulated soil evaporation of E7 at capacity c, closed at $\varphi = 1$ by the bare-soil value, and $E_{b,\text{ref}}$ is the E5 soil evaporation of the soil under conventional tillage at a consolidation rate of 0.005 d^{-1}, so that the last factor transfers the effect of the chosen tillage and consolidation rate. Fluxes are combined multiplicatively and drainage and leaching additively, because ratios of small numbers (drainage near zero) are unstable. Four set-specific corrections then act in turn; for each family $F \in \{$trigger (E2), sensor depth (E3), method (E4), nitrogen (E13; yield and N_L only)$\}$ with key κ and baseline key κ_0 (trigger -300 cm, sensor depth 20 cm, surface method, 100 % nitrogen),

$$\begin{aligned} Q \leftarrow Q \, \text{clip}\left(\frac{r_\kappa.Q}{r_{\kappa_0}.Q}; 0.05, 6\right) \ (Q \in \{E, T, \boldsymbol{ET}, I, Y\}), \\ D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N). \end{aligned} \tag{38}$$

Finally non-negativity, a soil-dependent yield ceiling $Y \le Y_{\max}(\text{soil})$ (the largest simulated yield), rain $P = P_{\text{base}}\gamma_P$ and a closure window are enforced:

$$\begin{aligned} \boldsymbol{ET} \in [P + I - D - 30, \ P + I - D + 15] \ (\text{otherwise clipped; } E, T \text{ scaled equally}), \\ E_{\text{oth}} = \max(0, \boldsymbol{ET} - T - E), \qquad \Delta S = P + I - \boldsymbol{ET} - D, \end{aligned} \tag{39}$$

and irrigation efficiency IE $= 100\,Y/I$ (kg m^{-3}, since 1 mm $=$ 10 m^3 ha^{-1}). The combination of mulch and warming uses the *product* of ratios, which assumes that their effects are independent; no experiment crosses the two factors, so this assumption is untested (Section 6). The meta-model contains no fitted parameters of its own: it is a deterministic lookup/composition over the simulation tables, so its accuracy is limited by (a) the discretisation of the lookups, (b) the validity of the independence assumption and (c) the consistency of the baselines of the different experiments, a point that proved decisive in the verification (Section 5.5).

4.3 Engine 2, the daily bucket model

The soil is $N = 40$ layers of thickness $\Delta z = 2.5$ cm, each holding w_i (mm) up to the plant-available capacity $W_{\max} = \mathrm{AWC}\,\Delta z$, initial value $0.72\,W_{\max}$; the run is 20 years of 365 days. Canopy cover follows the logistic curves of Eq. (16); rainfall is generated with wet-day probability 0.28 and depth

$$R = \min\{70,\ \mu_0\gamma_y(-\ln U)\}, \quad U \sim \mathrm{Uniform}(0,1), \quad \mu_0 = \frac{P}{0.28 \cdot 365}, \tag{40}$$

an exponential distribution capped at 70 mm, seeded by a hash of the soil name so every run is reproducible; the year factor $\gamma_y = \mathrm{clip}\big((m_{\mathrm{med}}/m_y)^{0.3}; 0.65, 1.45\big)$ makes years wetter or drier according to the dryness m_y of the 20-year rainfed suction record. The canopy and litter catch the first I_c mm of every shower,

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\mathrm{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}, \tag{41}$$

where the choice of I_c makes the expected interception over the exponential distribution equal to the annual interception of Engine 1. Water enters the top layer and cascades down (any layer exceeding $W_{\max}$ passes the excess down; the excess leaving layer 40 is drainage), and a slow redistribution moves water toward the drier neighbour (Eq. (7)). Soil evaporation is the annual E distributed seasonally and reduced by canopy shade, taken from the top four layers; transpiration demand is the annual T spread by cover, times a calibration factor k_t, and divided among layers by availability-weighted root density,

$$T_d = k_t\,T_{\mathrm{yr}}\frac{C(d)}{\Sigma_C}, \qquad q_i^T = T_d\frac{\beta_i a_i}{\sum_j \beta_j a_j}, \qquad a_i = \mathrm{clip}\Big(\frac{s_i - 0.15}{0.6}; 0, 1\Big), \tag{42}$$

where β_i is the root density of Eq. (19), $s_i = w_i/W_{\max}$ and a_i is a linear stress function in the spirit of the FAO-56 coefficient (Allen et al., 1998). with Σ_C the sum of the daily canopy cover over a year (so that T_d is in mm d^{-1}). Irrigation of dose $d = \mathrm{clip}(I/10;\, 15,\, 35)$ mm, with I the annual irrigation of Engine 1 (about ten events a year by construction), is applied when canopy cover exceeds 0.2, in the day-of-year window 95–250, the minimum spacing is met, and the sensor-layer depletion reaches f_T. The factor $k_t \in [0.6, 8]$ is calibrated by eight bisection steps on the first eight years so that the mean annual irrigation matches Engine 1's within 6 %. The daily fluxes shown on the dashboard are scaled so that their 20-year means equal Engine 1's:

$$S_P = \frac{P}{\overline{P}_{\mathrm{sim}}},\; S_E = \frac{E}{\overline{E}_{\mathrm{sim}}},\; S_T = \frac{T}{\overline{T}_{\mathrm{sim}}},\; S_D = \frac{D}{\overline{D}_{\mathrm{sim}}}, \qquad \Delta S_d = T_f + I - E - T - D, \tag{43}$$

and ΔS_d closes the daily balance. Drainage and the wetting front are tipping-bucket results, not Richards solutions; that is why they are scaled to Engine 1.

4.4 Surface-irrigation layer

Texture determines water contents by the pedotransfer functions of Saxton and Rawls (2006); the wetting-front suction ψ_f comes from the van Genuchten parameters; and Green–Ampt infiltration (Green and Ampt, 1911) gives cumulative infiltration $F(t)$,

$$F - \psi_f \Delta\theta \ln\!\left(1 + \frac{F}{\psi_f \Delta\theta}\right) = K_s \frac{t}{60} \quad (t \text{ in min}), \qquad \Delta\theta = \max\{0.05,\, 0.95\theta_s - \theta_0\}, \tag{44}$$

with K_s in mm h^{-1} and F, ψ_f in mm, solved by Newton iteration. The Kostiakov intake law (Kostiakov, 1932)

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}) \tag{45}$$

is fitted by least squares on the log–log plot of $F(\tau)/1000$ at 25 geometrically spaced times between 10 and 600 min, and the fit quality R^2 is displayed. The inlet flow area follows Manning's equation (Manning, 1891), $Q = \frac{1}{n} A R_h^{2/3} S^{1/2}$, $R_h = A/P$.

The advance of the stream follows the volume-balance method (Elliott and Walker, 1982; Walker and Skogerboe, 1987):

$$Q\,t = \sigma_y A_0 x + \sigma_z\, k\, t^{A} w x \implies x(t) = \frac{Qt}{\sigma_y A_0 + \sigma_z k t^{A} w}, \qquad \sigma_z = \frac{A + r(1-A) + 1}{(1+A)(1+r)}, \tag{46}$$

with $\sigma_y = 0.8$ the surface-storage shape factor, w the width, and r the exponent of the power-law advance $x = p\,t^r$, obtained iteratively from the times t_L and $t_{L/2}$ at which the front reaches L and $L/2$: $r = \ln 2 / \ln(t_L/t_{L/2})$. The time to infiltrate the required depth Z_n is $t_{st} = (Z_n/k)^{1/A}$, the cut-off time is $t_{co} = t_L + t_{st}$ (for the $E_s = 1$ rule), and the infiltrated depth along the field is $Z(x) = k[t_{co} - t_x(x)]^A$.

By trapezoidal integration of $Z(x)$ over $N = 300$ intervals,

$$\begin{gathered} V_{\text{app}} = Q t_{co}, \quad V_{\text{req}} = Z_n L w, \quad V_{\text{inf}} = w\int_0^L Z\,\mathrm{d}x, \\ V_{\text{st}} = w\int_0^L \min(Z, Z_n)\mathrm{d}x, \quad V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\mathrm{d}x, \quad V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}}, 0), \end{gathered} \tag{47}$$

$$\begin{gathered} E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \\ \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_i |Z_i - \bar{Z}|}{\bar{Z}}\Big)\%, \end{gathered} \tag{48}$$

following Walker and Skogerboe (1987), Burt et al. (1997) and Christiansen (1942); because $V_{\text{app}} = V_{\text{inf}} + V_{\text{ro}}$ and $V_{\text{inf}} = V_{\text{st}} + V_{\text{dp}}$ the identity $E_a + \text{DP} + \text{RO} = 1$ holds exactly and is printed as a built-in closure check.

4.5 Drip-irrigation layer

Friction head loss in the lateral segment feeding emitter j uses the Hazen–Williams equation (Williams and Hazen, 1905), and discharge depends on local head through the power-law emitter equation of Keller and Karmeli (1974):

$$\begin{gathered} h_{f,j} = \frac{10.67\,\Delta x_e\,Q_j^{1.852}}{C^{1.852} D^{4.871}}, \quad Q_j = \sum_{i>j} q_i, \\ q_j = q\Big(\frac{h_j}{H_0}\Big)^{x_e}, \quad h_j = h_{j-1} - h_{f,j-1},\ h_0 = H_0, \end{gathered} \tag{49}$$

with Δx_e the emitter spacing, $C = 140$ and $x_e = 0.5$ (turbulent orifice), 0.6 (tape) or 0.05 (pressure-compensating emitter, with a minimum head of 8 m). Because flows depend on heads and heads on flows, the system is solved by fixed-point iteration from the uniform start $q_j = q$, with under-relaxation after the fourth pass, until the largest relative change in any emitter flow is below 10^{-7} (at most 500 passes); version 1 used four passes. Emission uniformity follows the

design equation

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}}; 0.3, 1\Big\}, \qquad \sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \tag{50}$$

with c_v the manufacturing coefficient of variation, n_e the number of emitters per plant and σ_P the relative standard deviation of hydraulic delivery along the lateral. The wetted bulb is modelled as a half-ellipsoid whose aspect ratio is an engineered function of K_s and the discharge; five representative emitters at the standard-normal quantiles $z \in \{-1.28, -0.52, 0, 0.52, 1.28\}$ deliver $V_i = \max\{0.02\bar{V}, \bar{V}(1 + \sigma_q z_i)\}$, from which deep percolation, evaporation and storage are computed, and the efficiencies satisfy $E_a + \mathrm{DP} + \mathrm{EV} = 1$. The automatic run time is the time at which the driest emitters deliver the need, $t = \mathrm{clip}\{V_{\mathrm{req}}/(\bar{q}E_u); 0.05, 96\}$ h.

4.6 Sprinkler layer

Nozzle discharge follows the orifice (Torricelli) equation with coefficient 0.95 and the wetted radius an empirical range law (Keller and Bliesner, 1990) with an angle function:

$$\begin{aligned} q &= 0.95\,A_n\sqrt{2gh_n} \times 6 \times 10^4\ \mathrm{L\,min^{-1}}, \\ R_w &= c_R\sqrt{d\,h_n}\,f_\alpha(\alpha), \quad f_\alpha = \Big[\frac{\max\{0.05, \sin 3.1\alpha\}}{0.9763}\Big]^{0.45}, \end{aligned} \tag{51}$$

with d in mm, the pressure head h_n in m, $A_n = \pi(d/2000)^2$, c_R a coefficient per sprinkler type, and α the trajectory angle in degrees. The radial pattern shape $\hat{p}(x)$, $x = r/R_e$, is engineered for each type (e.g. $1 - x^{1.3}$ for impact sprinklers), wind elongates the pattern ($R_e(\theta) = R_w\,\mathrm{clip}\{1 + 0.022u\cos\theta; 0.7, 1.4\}$), overlap is the sum over neighbours on the sprinkler grid, and the Christiansen coefficient of uniformity (Christiansen, 1942) is $\mathrm{CU} = 1 - \sum_k |a_k - \bar{a}|/(n\bar{a})$. Wind drift and evaporation loss follow an engineered function of wind, pressure, temperature and trajectory angle; intake and runoff use the Kostiakov law with a ponding-time criterion; and the layout, pipes (Blasius friction (Blasius, 1913), with Christiansen's outlet factor) and a ten-point checklist implement the textbook decision logic (Keller and Bliesner, 1990), including the maximum allowable application rate by soil and slope and the number of laterals operated simultaneously.

4.7 Evaluation of the system variables

The variables of Section 3 were evaluated for each of the ten soils with the shipped engine (`swa_vars.js`), at one reference state: conventional tillage, consolidation rate $0.005\,\mathrm{d}^{-1}$, bare soil ($\varphi = 1$, $c = 1$ mm), unscaled rainfall, no warming, nitrogen 100 %, surface method, common trigger -300 cm, sensor at 20 cm and the cereal rotation (the state of the sensitivity runs, Section 5.3). For each soil the script records the Engine 1 balance, the hydraulic variables, a full 20-year Engine 2 run (rooting depth, uptake by layer, events, doses, wetting depth, k_t), the water-potential path averaged over crop days of day-of-year 120–210 in all years, the two-stage evaporation at the season-mean demand of $4.2\,\mathrm{mm\,d}^{-1}$, the fate of the evaporation saved by a straw mulch ($\varphi = 0.4$, $c = 4$ mm) and the nitrate cost of drainage for a $+0.1$ rainfall step.

4.8 Verification design

Four verification families were run, each with an expected output from outside the software.

1. *Published worked examples.* Surface-irrigation Examples 6.1 (furrow) and 6.4 (border) of Walker and Skogerboe (1987) and a sprinkler design example from the textbook, with the soil parameters, flows, lengths and required depth of the examples, comparing the software's advance times, efficiencies, required times, maximum non-erosive flows and decision logic with the printed results (relative error $= 100\,(\text{software} - \text{book})/\text{book}$).
2. *Held-out simulations.* 1 000 runs of E12 (coarse loam: sensor depth $\times$ trigger $\times$ rain scale) were not used to build Engine 1's lookups; the engine was queried at the same factor levels (the cell is selected by indices of the lookups, so no interpolation occurs). Metrics: R^2, RMSE, bias, Spearman ρ and normalised RMSE (RMSE over the observed mean). "Raw" predictions and "re-anchored" predictions are reported, the latter being those in which the engine's baseline cell (rain scale 1, baseline trigger and depth) is aligned to Daisy's baseline cell, multiplicatively for fluxes and additively for drainage and nitrogen leaching, as a diagnostic of anchor inconsistency.
3. *Mass closure and agreement of the daily and annual models.* The unscaled Engine 2 core was run on 60 random states to measure the closure residual as a percentage of inflow, and its calibrated annual irrigation was compared with Engine 1 over a realistic grid of 10 soils $\times$ 5 triggers $\times$ 2 methods (cereal crop).
4. *Independent numerical solvers.* The drip-lateral solution was compared with an indepen-

dent converged solver (many iterations) over 400 random configurations covering emitter type, spacing, lateral length, diameter, inlet head and coefficient of variation.

In addition, run times were measured in Node.js on a cloud container (single call medians), and the manual was audited for equation count and provenance labels.

4.9 Derivations and error propagation

This section sets out the short derivations behind the closed-form layers, so that each can be audited line by line, and the elementary error analysis that explains which deviations in the verification were to be expected.

Assume a sharp wetting front at depth L_f with suction head ψ_f and a saturated transmission zone of conductivity K_s. Darcy's law across the wetted zone gives the infiltration rate

$$f = \frac{\mathrm{d}F}{\mathrm{d}t} = K_s\Big(1 + \frac{\psi_f \Delta\theta}{F}\Big), \qquad F = L_f \Delta\theta, \tag{52}$$

and separating variables, $\int_0^F \big(1 + \psi_f \Delta\theta / F'\big)^{-1} \mathrm{d}F' = K_s t$, yields Eq. (44) (Green and Ampt, 1911). Two limits are used as sanity checks in the code: for small t the logarithm expands as $F - \psi_f \Delta\theta \ln(1 + F/\psi_f\Delta\theta) \approx F^2/(2\psi_f\Delta\theta)$, so $F \approx \sqrt{2 K_s \psi_f \Delta\theta\, t}$ (sorptivity-dominated, $F \propto t^{1/2}$); for large t, $F \to K_s t$ (gravity-dominated). The Kostiakov exponent A fitted to $F(\tau)$ over 10–600 min is therefore expected to lie between 0.5 and 1 (the fitted exponents have not been tabulated for the ten soils; the interface displays them with the fit R^2). Because Eq. (52) is monotone, the Newton iteration of the code converges from the starting value $F_0 = \sqrt{2 K_s \psi_f \Delta\theta t} + K_s t$.

Let the advance front be at $x(t)$ and the inflow be Q. Conservation of volume over $[0, t]$ states that the inflow equals the volume stored on the surface plus the volume infiltrated:

$$Q t = \sigma_y A_0\, x + \sigma_z\, Z(t)\, w\, x, \tag{53}$$

where A_0 is the flow cross-section at the inlet, σ_y the surface-storage shape factor (the ratio of the mean to the inlet cross-section along the wetted length) and σ_z the sub-surface shape factor. For a power-law advance $x = p t^r$ and Kostiakov intake, the infiltrated volume per unit width is $\int_0^x Z(t - t_x)\,\mathrm{d}x$, which evaluates to $\sigma_z Z(t) x$ with σ_z as in Eq. (46) (Elliott and Walker, 1982; Walker and Skogerboe, 1987). Solving Eq. (53) for x gives Eq. (46), and the exponent r is found from the two measured (or predicted) times $t_{L/2}$ and t_L by $r = \ln 2 / \ln(t_L / t_{L/2})$, since $x \propto t^r$

implies $L/(L/2) = (t_L/t_{L/2})^r$. Because σ_z depends on r and r on the times, the solution is a fixed point, found by iteration; the code reports non-convergence rather than returning a number.

Let $V_{\text{inf}} = V_{\text{st}} + V_{\text{dp}}$, which holds by construction since $\min(Z, Z_n) + \max(Z - Z_n, 0) = Z$ pointwise. If the infiltrated volume does not exceed the applied volume, $V_{\text{ro}} = V_{\text{app}} - V_{\text{inf}} \geq 0$, so

$$E_a + \text{DP} + \text{RO} = \frac{V_{\text{st}} + V_{\text{dp}} + V_{\text{ro}}}{V_{\text{app}}} = \frac{V_{\text{inf}} + V_{\text{app}} - V_{\text{inf}}}{V_{\text{app}}} = 1. \tag{54}$$

The identity fails only when the numerical V_{inf} exceeds V_{app}, which signals a violated volume balance and is flagged in the interface. The same argument gives $E_a + \text{DP} + \text{EV} = 1$ for the drip layer, where evaporation takes the place of runoff.

For n catch-can depths a_k with mean $\bar{a}$, the Christiansen coefficient $\text{CU} = 1 - \sum |a_k - \bar{a}|/(n\bar{a})$ equals one minus the relative mean absolute deviation (Christiansen, 1942). For normally distributed depths the mean absolute deviation is $\sigma\sqrt{2/\pi}$, so $\text{CU} \approx 1 - 0.80\,\sigma/\bar{a}$; this relation allows the coefficient of variation to be inferred from a reported CU and was used to translate the "CU ≥ 85 %" check into a coefficient of variation of at most 0.19. For the drip layer, Eq. (50) of Keller and Karmeli (1974) treats the lowest quartile of emitters; the factor 1.27 is the standardised mean of the lowest quarter of a normal distribution, so that $1-1.27\,c_v$ is the ratio of the low-quarter mean to the mean discharge, and division by $\sqrt{n_e}$ accounts for the averaging of n_e emitters per plant; $n_e = 1$ gives the most conservative estimate.

For a lateral of N emitters at spacing s_e, the flow in the segment that feeds emitter j (counting from the inlet) is $\sum_{i \geq j} q_i$, and the head at emitter j is the inlet head less the accumulated losses. With uniform emitters and the exponent x_e, the total head loss for a pipe of length L with outlet factor F_c is approximated by Christiansen's factor

$$h_f \approx F_c \, \frac{10.67\, L\, Q_0^{1.852}}{C^{1.852} D^{4.871}}, \qquad F_c = \frac{1}{2.852} + \frac{1}{2N} + \frac{\sqrt{1.852 - 1}}{6N^2}, \tag{55}$$

which tends to $1/2.852 = 0.351$ for large N and is the textbook shortcut used as an independent plausibility check on the segment-by-segment iteration. Because $q_j \propto h_j^{x_e}$ with $x_e < 1$, the loop *head* $\rightarrow$ *flow* $\rightarrow$ *loss* $\rightarrow$ *head* is a contraction when the relative loss h_f/H_0 is small, with contraction ratio of the order $x_e\, h_f/H_0$; four iterations then reduce the initial error by a factor of order $(x_e h_f/H_0)^4$, which is of order 10^{-4} (about 2×10^{-4} for $x_e = 0.6$) for $h_f/H_0 \leq 0.2$ and much larger for $h_f/H_0 \geq 0.5$ (where the ratio is of order 0.3). This simple analysis predicts the qualitative result of Table 15 before any numerical test, and it explains why the failures occur only

in poorly designed laterals.

Engine 1 returns products of ratios, Eqs. (31) to (35). If each ratio carries an independent relative error δ_i, the relative error of the product is $\delta \approx \sqrt{\sum \delta_i^2}$; with the three to five factors per flux and ratio errors of a few per cent this gives a few per cent, so random error cannot explain the $+53$ mm bias reported in Section 5.5, which is consistent with a *systematic* baseline offset ($\delta_b = (720 - 673)/673 = 7\,\%$ for ET and $(282 - 228)/228 = 24\,\%$ for irrigation). The additive treatment of drainage and leaching propagates absolute errors, $\sigma_D^2 = \sum \sigma_i^2$, and because the mean drainage is small (35 mm) a modest absolute error is a large relative one, which explains the normalised RMSE of 48.7 % despite $R^2 = 0.82$. Finally, the closure window of Eq. (39) bounds the storage drift to $-15 \le \Delta S \le 30\,\mathrm{mm\,yr^{-1}}$, which over a 20-year simulation could accumulate to a few hundred millimetres in the extreme. The window is not a rare safeguard: in the held-out test of Section 5.5 it is active in 680 of the 1 000 cells (482 at the upper and 198 at the lower bound of ET), so in that experiment Engine 1's ET is mostly set by the closure $P+I-D$ and not by the ET ratios, which is consistent with its weak ET skill ($R^2 = -0.11$ after re-anchoring, $\rho = 0.84$). The user-facing balance reports ΔS, so the clipping is visible.

Rank correlation is invariant to monotone transformations, whereas R^2 and RMSE are not. A systematic multiplicative or additive offset shifts RMSE and R^2 but leaves Spearman's ρ unchanged; this is why the raw irrigation R^2 of 0.18 becomes 0.79 after re-anchoring while ρ is 0.96 in both cases. A screening emulator is judged by whether it orders the options correctly, a budgeting model by whether it reproduces the magnitude, and the verification therefore reports both families of metric (Bennett et al., 2013; Moriasi et al., 2007).

5. Results

5.1 State and fluxes of the coupled system in ten soils

The variables of Section 3 were evaluated for the ten soils at the reference state of Section 4.7. Table 8 gives the soil and surface variables, Table 9 the annual water, nitrogen and yield balance, and Table 10 the crop, water-potential and mulch variables; Fig. 2 shows them together.

Table 8: Soil and surface variables of the ten soils. Texture as sand/silt/clay (%); θ_s, θ_r from Eq. (2); $S_e(100)$ effective saturation at field capacity; $f(300)$ depletion reached at a suction of 300 cm (Eq. (29)); Q_1 stage-1 storage, τ_E stage-2 time scale, t_1 stage-1 duration and $Q(12)$ cumulative evaporation after 12 days at $e_0 = 4.2\,\mathrm{mm\,d^{-1}}$ (Eq. (11)).

ID	Soil	Texture	θ_s	θ_r	$S_e(100)$	$f(300)$	Q_1 (mm)	τ_E (d)	t_1 (d)	$Q(12)$ (mm)
S01	Coarse loam	45/35/20	0.423	0.022	0.444	0.38	6.0	0.022	1.4	9.9
S02	Coarse sand	88/8/4	0.374	0.012	0.356	0.50	7.8	0.010	1.9	10.4
S03	Sand	92/5/3	0.369	0.012	0.011	0.84	12.7	0.008	3.0	15.0
S04	Loamy sand	82/12/6	0.380	0.014	0.040	0.76	11.9	0.011	2.8	14.5
S05	Sandy loam	65/25/10	0.399	0.016	0.165	0.63	10.5	0.017	2.5	13.7
S06	Loam	40/40/20	0.428	0.022	0.466	0.47	8.4	0.024	2.0	12.3
S07	Silt loam	20/65/15	0.443	0.019	0.686	0.37	7.1	0.030	1.7	11.5
S08	Sandy clay loam	55/15/30	0.421	0.028	0.417	0.44	7.4	0.019	1.8	10.9
S09	Clay loam	30/35/35	0.448	0.031	0.753	0.31	5.6	0.027	1.3	9.9
S10	Silty clay loam	10/55/35	0.465	0.031	0.878	0.24	4.6	0.033	1.1	9.4

Table 9: Annual balance of the ten soils from Engine 1 ($\mathrm{mm\,yr^{-1}}$, 20-year means; Eq. (1)): precipitation P, irrigation I, transpiration T, soil evaporation E, other evaporation E_{oth}, drainage D and storage change ΔS; yield Y ($\mathrm{Mg\,ha^{-1}}$), nitrate leaching N_L ($\mathrm{kg\,ha^{-1}\,yr^{-1}}$) and the regime index $\mathcal{I}$ (Eq. (14)).

ID	P	I	T	E	E_{oth}	D	ΔS	Y	N_L	$\mathcal{I}$
S01	464	282	239	175	307	27.6	−1.7	6.63	14.6	+0.73
S02	464	414	237	118	362	163.6	−2.2	6.45	62.5	−0.16
S03	464	278	202	58	408	76.5	−3.3	4.98	85.2	−0.14
S04	464	293	215	74	402	70.1	−4.3	5.46	61.1	+0.03
S05	464	298	230	102	378	57.8	−5.4	6.20	30.8	+0.28
S06	464	302	238	138	344	50.4	−4.6	6.58	10.1	+0.46
S07	464	305	239	159	322	52.1	−3.3	6.62	7.8	+0.51
S08	464	299	237	124	357	50.0	−5.6	6.54	9.0	+0.43
S09	464	306	239	155	326	54.0	−4.9	6.62	4.4	+0.48
S10	464	303	239	161	320	51.7	−5.0	6.62	1.3	+0.51

Table 10: Crop, water-potential and mulch variables of the ten soils. Engine 2 (20 years): maximum rooting depth $R_{\max}$ (Eq. (17)), mean of the yearly deepest roots $\bar{R}_d$, depths holding 50 and 90 % of uptake (Eq. (20)), irrigation events per year, dose d (mm), median wetting depth of an irrigation (cm) and calibration factor k_t. Water-potential path (Eqs. (21) and (22)), means over crop days of day-of-year 120–210: ψ_{RZ}, ψ_{leaf} (MPa) and share of days with $\psi_{\mathrm{leaf}} < -1.6$ MPa. Straw mulch ($\varphi = 0.4$, $c = 4$ mm) against bare soil: evaporation saved ΔE (mm yr^{-1}) and its fate shares (Eq. (23)).

ID	$R_{\max}$ (cm)	$\bar{R}_d$ (cm)	z_{50} (cm)	z_{90} (cm)	Events (yr^{-1})	d (mm)	Front (cm)	k_t	ψ_{RZ} (MPa)	ψ_{leaf} (MPa)	Closed (%)	ΔE (mm)	η_I	η_D	η_T
S01	94	90	19.2	47.0	10.1	28.2	45.0	1.64	−0.049	−2.24	74	127	0.27	0.06	0.14
S02	100	96	24.5	64.8	12.1	35.0	100.0	1.12	−0.039	−2.28	75	102	0.50	0.07	0.14
S03	100	96	27.0	52.4	10.2	27.8	55.0	2.05	−0.020	−2.11	71	50	0.69	0.15	0.30
S04	100	92	20.4	39.5	9.9	29.3	42.5	1.93	−0.024	−2.21	73	60	0.60	0.12	0.27
S05	99	92	17.9	35.6	9.9	29.8	37.5	1.81	−0.033	−2.27	75	77	0.48	0.09	0.22
S06	92	87	16.0	35.8	10.2	30.2	32.5	1.81	−0.044	−2.24	74	100	0.37	0.07	0.18
S07	86	83	15.8	36.6	10.5	30.5	32.5	1.81	−0.056	−2.26	74	117	0.32	0.06	0.15
S08	96	92	19.0	43.2	10.2	29.9	45.0	1.76	−0.045	−2.29	75	91	0.41	0.08	0.20
S09	89	86	19.8	54.2	10.4	30.6	70.0	1.70	−0.047	−2.16	72	114	0.33	0.06	0.16
S10	84	80	18.5	51.1	10.3	30.3	100.0	1.35	−0.043	−1.80	58	119	0.32	0.06	0.15

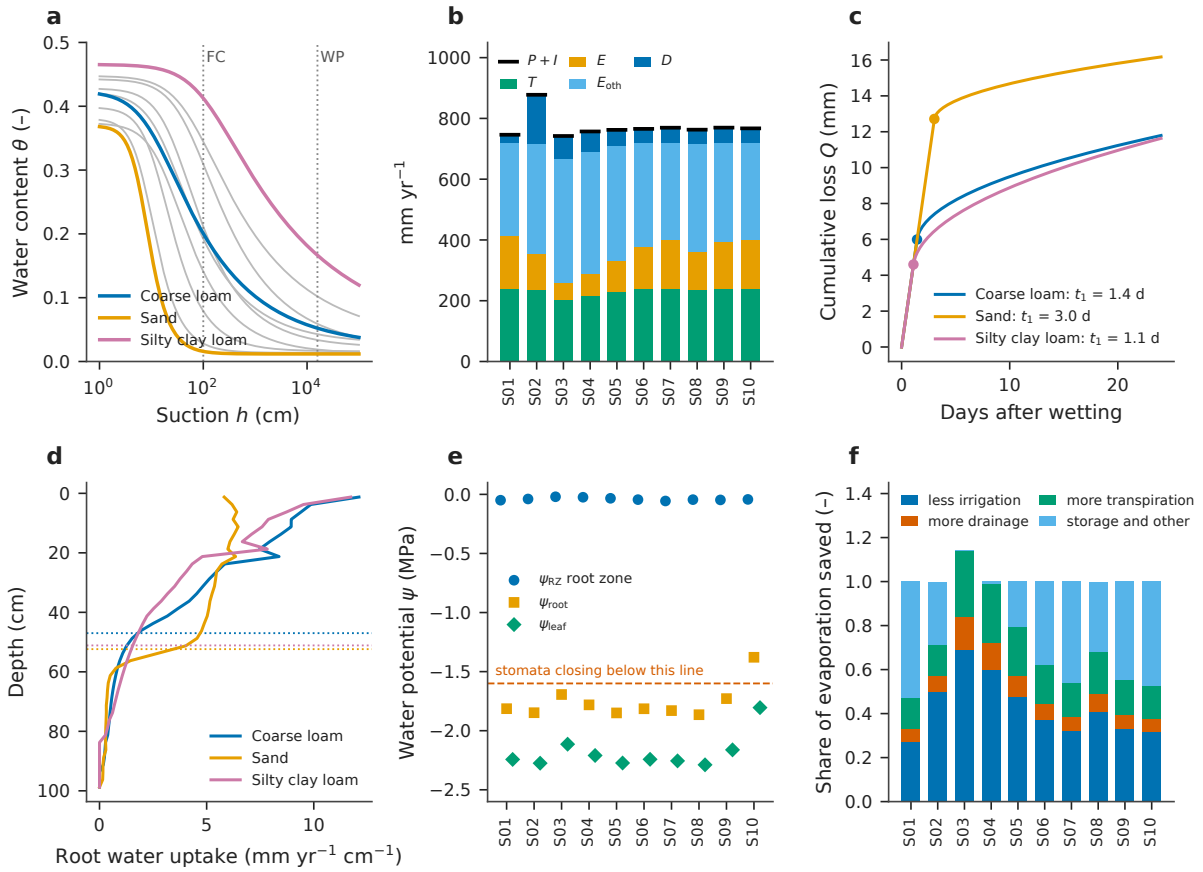


Figure 2: Soil–water–atmosphere variables of the ten soils at the reference state. (a) Retention curves $\theta(h)$ (Eqs. (2) and (27)); sand, coarse loam and silty clay loam highlighted; dotted lines are field capacity (FC, 100 cm) and the wilting point (WP, 15 849 cm). (b) Annual balance: sinks stacked (T, E, E_{oth}, D) against the input $P + I$ (black marks). (c) Cumulative evaporation after a wetting at the season-mean demand (Eq. (11)); dots mark the end of stage 1. (d) Mean annual root water uptake by depth from Engine 2 (unscaled); dotted lines mark z_{90}. (e) Water potential of the root zone, the root surface and the leaf (Eqs. (21) and (22)), crop days of day-of-year 120–210; the dashed line is the stomatal-closure threshold of the view; the air is at -56 MPa (not shown). (f) Fate of the soil evaporation saved by a straw mulch (Eq. (23)); in sand the three positive shares sum to 1.14, so the remainder is zero.

Soil and surface. The soils span the full range of the retention behaviour: at field capacity the sand retains 1 % of its drainable water ($S_e(100) = 0.011$) and the silty clay loam 88 %, and the common trigger of -300 cm uses 24 to 84 % of the plant-available water. The surface variables follow the retention curve. Stage-1 storage ranges from 4.6 mm in the silty clay loam to 12.7 mm in the sand, and stage 1 lasts 1.1 to 3.0 days at the season-mean demand. Stage 1 is therefore longest in the coarse soils, because the window between h_i and h_s scales with $1/\alpha$ and the narrow pore-size distribution of sands releases more water within it; this agrees in direction with the pore-scale analysis of Lehmann et al. (2008), in which coarse media with a narrow pore-size distribution sustain capillary supply to the surface over a larger characteristic length. In stage 2 the order reverses (the longer time scale τ_E of fine soils), so cumulative loss after 12 days ranges

only from 9.4 to 15.0 mm.

Annual balance. With the same rainfall (464 mm yr^{-1}) the soils differ chiefly in how the water leaves (Table 9, Fig. 2b). Transpiration is nearly constant in the loams and clays (237–239 mm) and lower in sand and loamy sand (202 and 215 mm); soil evaporation falls from 175 mm in the coarse loam to 58 mm in the sand; drainage rises from 28 mm in the coarse loam to 164 mm in the coarse sand, which carries the largest irrigation (414 mm) and nitrate leaching of 62 kg ha^{-1}, exceeded only by the sand (85 kg ha^{-1}). The storage change of -1.7 to -5.6 mm yr^{-1} lies inside the closure window of Eq. (39). The regime index separates evaporation-dominated loams and clays ($\mathcal{I} = +0.28$ to $+0.73$) from drainage-dominated coarse sand and sand (-0.16 and -0.14), with the loamy sand on the boundary ($+0.03$); mulch is therefore expected to pay in the former and to feed drainage in the latter, which is the classification of the companion hydraulic-memory study. The term E_{oth} is large, 307–408 mm yr^{-1} or 43–61 % of ET. It is the remainder $ET - T - E$ of the simulations and contains canopy interception together with the evaporation term that the companion audit found unattributed in the Daisy partition; the field view labels it "interception", which overstates what is known about it.

Crop and water-potential path. Maximum rooting depth ranges from 84 cm (silty clay loam) to the 100 cm limit in the sandy soils, and half of the uptake comes from the top 16–27 cm in every soil, with 90 % from the top 36–65 cm (Table 10, Fig. 2d). At the common trigger the calibrated daily model irrigates 9.9–12.1 times a year with doses of 28–35 mm; both follow largely from the dose rule $d = \mathrm{clip}(I/10; 15, 35)$ mm of Engine 2 and the Engine 1 annual irrigation, not from the bucket dynamics. The median wetting depth of an irrigation ranges from 32.5 cm in the loam and silt loam to the 1 m boundary in the coarse sand and silty clay loam. The water-potential path exposes a property of the engineered resistances. The root zone is wet ($\psi_{\mathrm{RZ}} = -0.020$ to -0.056 MPa, irrigated profiles), yet the mean leaf potential is -1.80 to -2.29 MPa and falls below the closure threshold on 58–75 % of mid-season crop days. The cause is the soil resistance: at the root-zone suction the Mualem relative conductivity is below the floor of 0.002 in Eq. (22) on most days (mean $K_r \leq 0.0034$), so R_{soil} sits at its cap of 0.92 MPa per mm d^{-1} and dominates the path in nine of the ten soils; in the silty clay loam the mean K_r (0.0034) is just above the floor, and the leaf potential is correspondingly less negative (-1.80 MPa). The resistances were chosen to give leaf potentials of about -1 to -1.5 MPa for a transpiring cereal; the evaluation shows that they do not, and that the leaf potential is set by the floor and not by the soil. The view is therefore an illustration of the series-resistance concept, and its numbers should not be read as plant water

status.

Mulch. A straw mulch saves 50 (sand) to 127 mm yr^{-1} (coarse loam) of soil evaporation, of which 27–69 % becomes less irrigation and 14–30 % more transpiration (Table 10, Fig. 2f). The extra drainage is 7.3 mm yr^{-1} in every soil, and the nitrate cost of drainage (Eq. (24)) is 0.38 kg N ha^{-1} mm^{-1} in every soil. Both are identical because Engine 1 adds the drainage and leaching increments of mulch and rainfall from experiments E14 and E11, which were run on the coarse loam only (Table 5); the evaporation saving is soil-specific because it is read from E7, which covers ten soils. The consequence is that the drainage share of the saving ($\eta_D = 0.06$–0.15) is much smaller in the coarse soils than the drainage share of 0.61–0.71 that the ten-soil mulch experiment E7 gives directly for the coarse sand, sand, loamy sand and sandy loam in the companion hydraulic-memory study, so the fate partition of the software understates drainage in sandy soils. The one-step nitrate cost likewise differs from the regression slope of 1.06 kg N ha^{-1} mm^{-1} across the water strategies of E13, which is a different quantity (across soils and strategies at fixed nitrogen rate). Both limitations follow from the single-soil design of E11 and E14 and are stated in Section 6.5.

5.2 Result views of the explorer and the evidence behind each

The explorer presents the coupled soil–water–atmosphere system of Section 3 as seventeen result views grouped by question: the field and the soil, irrigation timing, system design, mulch, and the system as a whole. Every view is a rendering of numbered relations in the Manual, and each relation carries its provenance label. Figures 3 to 5 reproduce the result panels only (screenshots of the drawing area, with the interface chrome removed; full-window captures of every view are in Appendix C), and Table 11 gives for every view the physical process shown, the governing relations with their literature origin, and the test in this article that supports it.

Soil and field. The field view closes the annual water balance of the root zone, $P + I = E + T + D + \Delta S$, with the fluxes taken from the Daisy ensemble (Abrahamsen and Hansen, 2000; Hansen et al., 2012) and the crop term referenced to the FAO-56 framework (Allen et al., 1998). The soil-water view shows the water-content and suction profile, converting between the two with the van Genuchten–Mualem retention and conductivity functions (Mualem, 1976; van Genuchten, 1980) that close Richards' equation (Richards, 1931) in widely used solvers (Šimůnek et al., 2008). The pore view shows how tillage and consolidation change the structural porosity that controls

drainage, in line with X-ray evidence on pore networks and their temporal dynamics (Blunt, 2017; Leuther and Schlüter, 2021; Strudley et al., 2008). The profile view displays the hydraulic drivers by texture, derived from pedotransfer functions (Saxton and Rawls, 2006; Schaap et al., 2001) and from the global soil-hydraulic data compilations that now underpin such functions (Vereecken et al., 2016; Weber et al., 2024). The evaporation view follows the two-stage description of soil drying (Philip, 1957; Ritchie, 1972), in which a capillary-supplied first stage gives way to a vapour-diffusion-limited second stage, as established by pore-scale and macroscopic work (Lehmann et al., 2008; Or et al., 2013; Shokri and Or, 2011). The root view distributes uptake over depth with a sink-term formulation (Feddes et al., 1978) scaled to the reference demand (Allen et al., 1998).

Irrigation timing. The two timing views express the trigger as depletion of plant-available water, the quantity used in the FAO-56 management-allowed-depletion concept (Allen et al., 1998), and translate it into a sensor suction at the chosen depth. Sensor-based scheduling is reviewed by Datta and Taghvaeian (2023), and the sensitivity of schedules to deficit strategy by Fereres and Soriano (2007). Collapsing ten soils onto one depletion curve is the result that distinguishes the framework from soil-specific suction thresholds: across the E2 runs of the irrigated soils the pooled R^2 of relative irrigation rises from 0.48 against suction to 0.75 against depletion, and that of relative yield from 0.23 to 0.70, with yield falling beyond $f^* = 0.45$ (Section 3.10).

System design. The surface view reproduces the volume-balance design procedure of Walker (1989) and Elliott and Walker (1982), with Green–Ampt infiltration (Green and Ampt, 1911), Kostiakov intake (Kostiakov, 1932) and Manning flow resistance (Manning, 1891); the results are compared with the worked examples in Tables 12 and 13, and the efficiency definitions follow Burt et al. (1997). The drip view solves the lateral hydraulics with a Hazen–Williams friction law (Williams and Hazen, 1905), uniformity as defined by Keller and Karmeli (1974) and the bulb geometry of Warrick and Yitayew (1988) and Schwartzman and Zur (1986); it is tested against an independent solver in Table 15. The sprinkler view builds the spray pattern, the overlap and the Christiansen coefficient (Christiansen, 1942) from nozzle hydraulics (Keller and Bliesner, 1990), with ballistic and drift losses following Playán et al. (2005) and Dechmi et al. (2003), and is compared with the textbook design checks.

Mulch and the whole system. The mulch views separate the three effects of a surface cover on evaporation, rain interception and soil warming. Their sign and size agree with field and mod-

elling evidence for straw and residue mulches (Balwinder-Singh et al., 2011; Kang et al., 2025; Vuaille et al., 2024; Zheng et al., 2021), and the regime map shows where the saving is recoverable, which is the question treated in the companion study on hydraulic memory. The closing views connect the chain from the soil surface to the atmosphere through the catenary description of water flow from soil to leaf (Nobel, 2009; van den Honert, 1948) and the energy-limited and water-limited regimes of evaporation (Roderick and Farquhar, 2002; Seneviratne et al., 2010), and then follow the feedbacks between soil moisture and the atmosphere (Entekhabi et al., 1996; Koster et al., 2004). No earlier open tool known to the literature reviewed here shows all of these processes under one set of controls with every number traceable to a labelled relation and a stated test.

Table 11: Result views, governing relations and verification. Equation numbers in brackets refer to the Manual; provenance labels are L (literature), D (data-driven), E (engineered).

View	Result shown	Governing relations [manual]; origin	Evidence in this article
Field	annual and daily water balance, yield, leaching	Eqs. (1), (39) and (43) [5.4–5.12, 5.24–5.25] (D, E, L)	Engine 2 closure (median 0.1 %); Engine 1 rank fidelity; Table 9
S1 Soil to sky	water potential from deep soil to air	Eqs. (15), (21) and (22) [7.1–7.5] (L, E)	Table 10: leaf potential set by the conductivity floor
V1 When to irrigate	sensor suction against trigger	Eq. (8) [6.1–6.2] (D)	Engine 2 calibration grid
N1 Irrigation rule	depletion trigger, ten soils	Eqs. (25), (26) and (29) [5.3, 6.3–6.5, 5.27] (L, D)	data collapse of E2: pooled R^2 0.48 $\rightarrow$ 0.75 (irrigation), 0.23 $\rightarrow$ 0.70 (yield)
V2 Water in the soil	pressure-head profile, share beyond trigger	Eqs. (9) and (27) [5.1–5.2, 7.6] (L, E)	Engine 2 closure; agreement with Engine 1
V3 Pores and tillage	pore space, water- and air-filled pores	Eqs. (2) to (4) [8.1–8.4] (E, L)	drainage response of Engine 1 (held-out tests)
V7 Soil hydraulic drivers	tillage and consolidation effects by texture	Eq. (5) [8.5–8.6] (D)	ledger audit of provenance

View	Result shown	Governing relations	Evidence
R1 Roots	rooting depth, uptake by depth	Eqs. (17) to (20) and (42) [5.18–5.23, 8.7] (E)	Engine 2 closure; Table 10
E1 Evaporation	stage-1 and stage-2 evaporation, drying front	Eqs. (10) to (12) [5.26, 9.2–9.5] (E)	Table 8; sensitivity
V4, V6 Mulch	evaporation, irrigation, drainage under mulch	Eqs. (13) and (31) [5.4–5.9, 9.1] (D, E)	Engine 1 held-out tests; sensitivity (Section 5.3)
N2, N3 Saved water, regime map	fate of saved water; dominant loss	Eqs. (14) and (23) [9.6–9.8] (D, E)	Table 10: drainage share understated in coarse soils
V5 How it connects	seven coupled effects	Table 4, Eq. (24) [10.1–10.2] (E, D)	sensitivity analysis
S2 Surface irrigation	advance, intake, efficiencies	Eqs. (44) to (48) [11.4–11.19] (L)	FAO-45 advance times within 0.5 %; textbook procedure: median 0.08 % (Tables 12 and 13); steady-intake term corrected in 1.1.0 (third defect)
S3 Drip irrigation	lateral head loss, bulb, uniformity	Eqs. (49) and (50) [12.3–12.19] (L, E)	independent solver, 3.2×10^{-7} for $h_f/H_0 \leq 0.6$; Christiansen factor to 0.01 % (Table 15)
S4 Sprinklers	nozzle, pattern, overlap, checklist	Eq. (51) [13.1–13.23] (L, E)	diameter within 1 %, nozzle discharge within 1.3 %, decision logic as the textbook

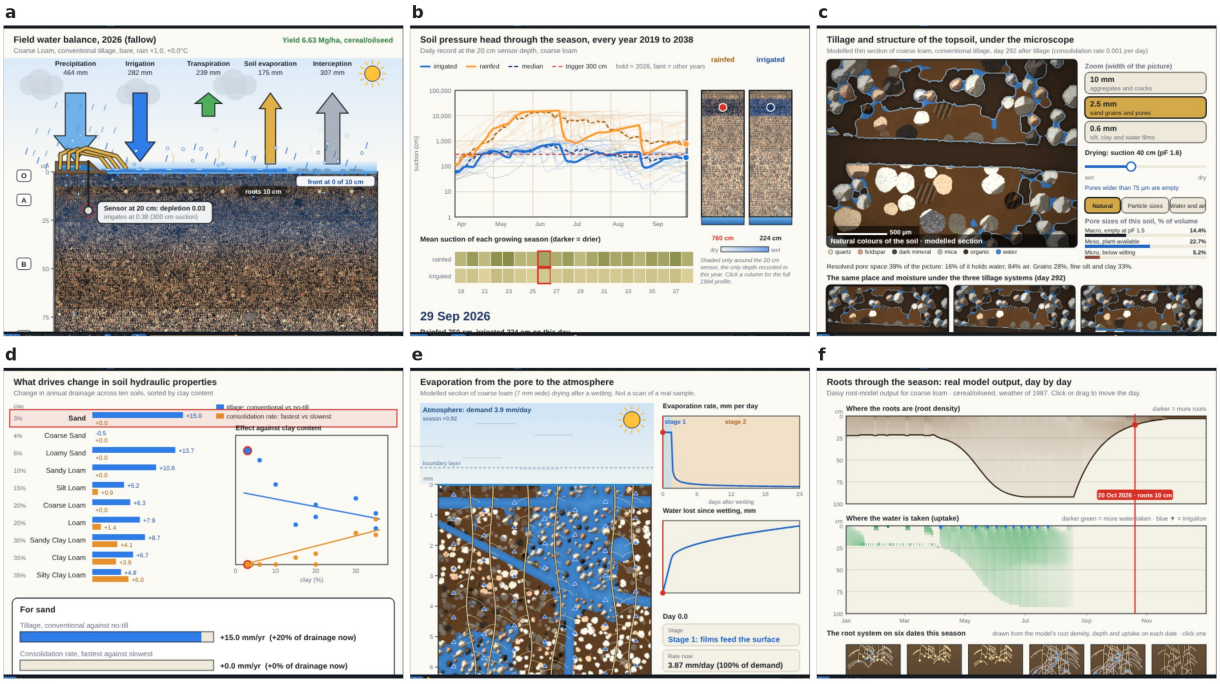


Figure 3: Result views of the explorer for the soil and the field (coarse loam, conventional tillage, cereal rotation). (a) Field water balance of the root zone; (b) soil-water pressure-head profile; (c) pores and tillage; (d) soil hydraulic drivers by texture; (e) evaporation from the pore to the atmosphere; (f) root density and uptake. Governing relations are listed in Table 11.

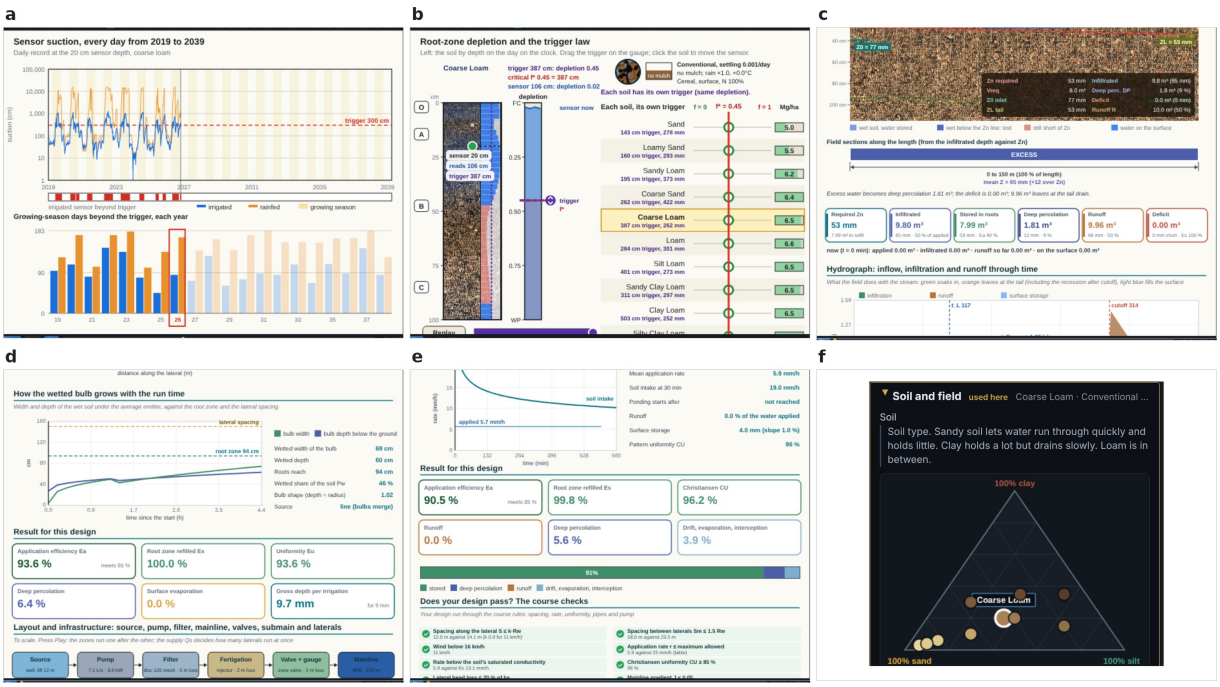


Figure 4: Result views for irrigation timing and design. (a) Sensor and trigger; (b) irrigation rule collapsing ten soils onto one depletion curve; (c) surface irrigation; (d) drip irrigation; (e) sprinklers; (f) soil selection on the texture triangle, which drives every view.

5.3 Sensitivity of the outputs to the input controls

One-at-a-time sensitivity runs were made on the shipped engine from a reference state (soil S01, conventional tillage, consolidation rate 0.005 d^{-1}, bare soil, surface method, trigger -300 cm, sensor depth 20 cm, unscaled rainfall, no warming, nitrogen 100 %), moving one control to the two settings listed in Fig. 6a,b. The reference state gives 282 mm of irrigation and 27.6 mm of drainage per year. Rainfall dominates: scaling rainfall by 0.7 raises irrigation to 430 mm and lowers drainage to 10.6 mm, and scaling by 1.3 lowers irrigation to 189 mm and raises drainage to 87.7 mm. The irrigation method is the next strongest control of irrigation: drip at 20–30 cm lowers it to 201 mm and raises drainage to 31.1 mm, which is the E4 result for this soil (simulated irrigation 162 mm under drip at 20–30 cm against 228 mm under surface irrigation, ratio 0.71), whereas overhead irrigation leaves irrigation and drainage almost unchanged (-0.3 and $+1.1$ mm) but lowers transpiration from 239 to 141 mm, because in E4 the canopy intercepts and evaporates part of the overhead application. The trigger suction follows (335 mm at -150 cm against 208 mm at -1000 cm), then warming (-1 °C: 224 mm; $+2$ °C: 314 mm) and the mulch vapour-flux factor (218–258 mm for factors 0.4–0.7). Sensor depth (315 to 270 mm for 10 to 40 cm) and tillage have smaller effects. Nitrogen rate leaves irrigation and drainage unchanged because its correction (Eq. (38)) acts on yield and nitrogen leaching only, a design property of the annual model. Soil changes the outputs as much as the strongest management controls (Fig. 6c): irrigation ranges from 278 mm (S03) to 414 mm (S02) and drainage from 27.6 mm (S01) to 164 mm (S02).

For the design layers, the same procedure was applied to the sprinkler and drip solvers (Fig. 6d–f). In the sprinkler layer, the operating pressure controls uniformity (CU 0.82 at 200 kPa, 0.96 at 300 kPa and 0.96 at 400 kPa, with the range coefficient corrected in Section 5.4), whereas raising wind speed from 0 to 5 $\mathrm{m\,s}^{-1}$ changes CU by less than 0.01 but reduces application efficiency by 0.02 at 200 kPa (0.75 to 0.73) and by 0.04 at 300 and 400 kPa (0.92 to 0.88 and 0.89 to 0.84); the absence of a wind effect on CU is a limitation of the engineered wind term (Section 5.7). In the drip layer, emission uniformity decreases with lateral length and with smaller diameter or lower inlet head, as expected from friction loss; beyond the design range ($h_f/H_0 > 0.2$, dashed in Fig. 6d,e) the converged solver gives a monotonic E_u except for the longest, narrowest laterals (loss ratios above 0.6, where the head reaches the model's 0.5 m floor and the layer prints its warning); these portions of the curves should not be interpreted as predictions.

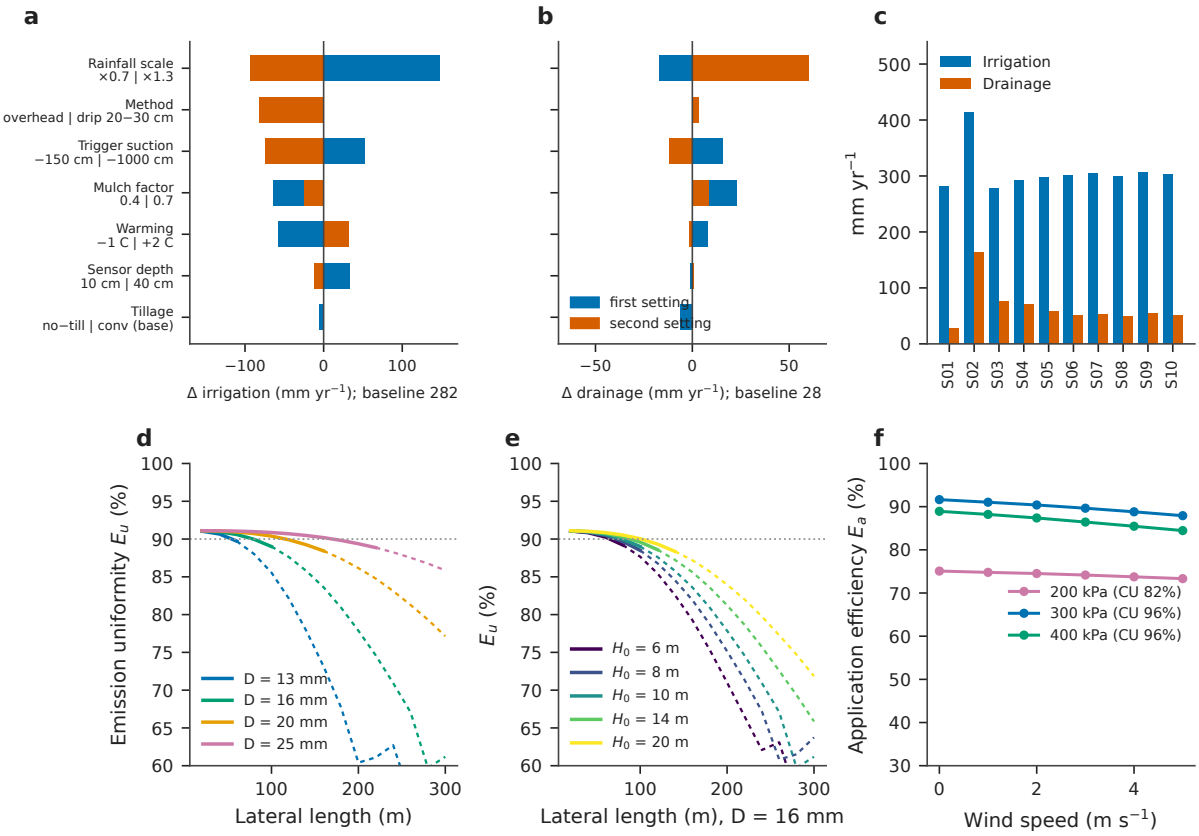


Figure 6: One-at-a-time sensitivity of SWAFL Desktop outputs. (a) Engine 1 annual irrigation and (b) drainage change for each control moved to the two settings listed under its name (blue: first setting; orange: second setting) from the reference state; nitrogen rate, which leaves both unchanged, is not shown. (c) Irrigation and drainage by soil. (d) Drip emission uniformity against lateral length by lateral diameter and (e) by inlet head; dashed where $h_f/H_0 > 0.2$. (f) Sprinkler application efficiency against wind speed at 200, 300 and 400 kPa (CU in the legend).

5.4 Surface irrigation against published worked examples

Examples 6.1 (furrow) and 6.4 (border) of Walker and Skogerboe (1987) were computed with a stand-alone reference implementation of the textbook volume-balance procedure (`lib/wsmodel.js`), which uses the book's power-law description of the furrow section and its intake functions. These two examples therefore verify the procedure and its arithmetic; the surface layer shipped in SWAFL Desktop, which describes the furrow by a trapezoidal section, is compared with Example 6.1 at the end of this subsection and with a second source in Section 5.7.

The example specifies a 200 m furrow of slope $S_0 = 0.008$ and roughness $n = 0.04$, with infiltration $Z = 0.0028\tau^{0.534} + 0.00022\tau$ (first irrigation) and $Z = 0.0021\tau^{0.331} + 0.00015\tau$ (later irrigations), furrow-geometry coefficients $\rho_1 = 0.325$ and $\rho_2 = 2.734$. Table 12 compares 16 quantities. The maximum error is 1.81 % (advance time at $Q = 0.10\,\mathrm{m^3\,min^{-1}}$ for the first

irrigation: 81.65 versus 80.2 min), the median error 0.08 %, and the required infiltration times (378.73 and 868.5 min) agree to better than 0.01 %. For the later irrigation at $Q = 0.10$ the application efficiency recomputed by the software is 33.5 % while the copy of the book used shows 53.0 %, which is inconsistent with the printed trend (the neighbouring values at $Q = 0.075$ and 0.0625 are 44.2 % and 52.5 % and the efficiency increases as the flow falls); this entry was excluded from the comparison and is flagged.

Table 12: Surface-irrigation Example 6.1 (Walker and Skogerboe, 1987): published and computed quantities. T_{req}: required infiltration time (min); t_L: advance time to the end of the furrow (min) at flow Q ($m^3\ min^{-1}$); E_a: application efficiency (%). Error = 100(SWAFL − book)/book.

Irrigation	Quantity	Book	SWAFL	Error (%)
first irrigation	Treq	378.73	378.73	+0.00
first irrigation	tL Q=0.1	80.2	81.65	+1.81
first irrigation	Ea Q=0.1	65.4	65.16	−0.36
first irrigation	tL Q=0.075	198.6	198.67	+0.04
first irrigation	Ea Q=0.075	69.3	69.28	−0.04
first irrigation	tL Q=0.0625	441.6	442.27	+0.15
first irrigation	Ea Q=0.0625	58.5	58.46	−0.06
later irrigation	Treq	868.5	868.52	+0.00
later irrigation	tL Q=0.1	26.2	26.16	−0.16
later irrigation	tL Q=0.075	36	36.05	+0.14
later irrigation	Ea Q=0.075	44.2	44.22	+0.05
later irrigation	tL Q=0.0625	46.3	46.35	+0.11
later irrigation	tL Q=0.05	68.2	68.23	+0.04
later irrigation	Ea Q=0.05	64.1	64.05	−0.08
later irrigation	tL Q=0.0375	143.5	143.64	+0.10
later irrigation	Ea Q=0.0375	79.1	79.04	−0.08

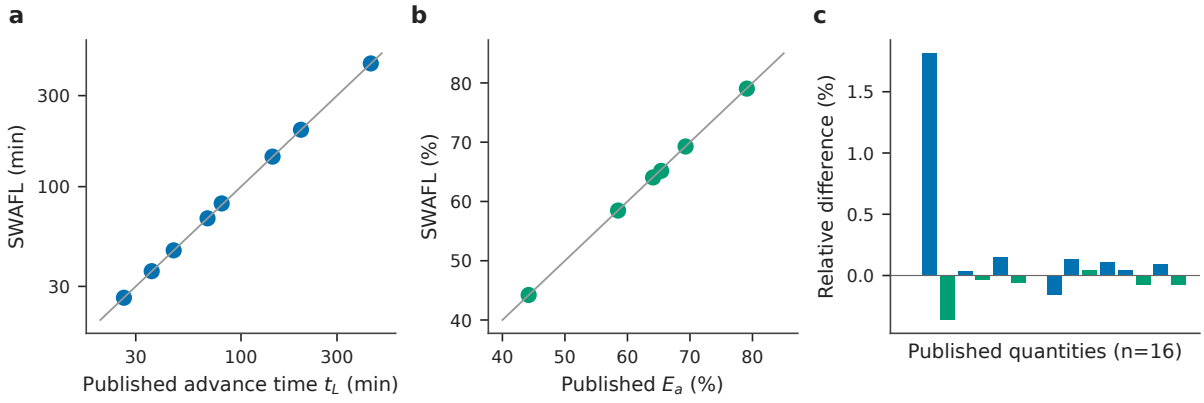


Figure 7: Verification against Example 6.1. (a) Advance time t_L, (b) application efficiency E_a (published versus SWAFL; grey line is 1:1), (c) relative differences of the 16 quantities (blue: advance and required times; green: efficiencies).

For the border-strip example the software gives T_{req} (first irrigation) of 146.40 against

146.4 min, T_{req} (later) 162.23 against 162.8 min (-0.35 %), maximum non-erosive flow $Q_{\max} = 1.883$ against 1.88 $\mathrm{m^3\,min^{-1}m^{-1}}$ ($+0.17$ %), the advance time for $Q = 0.14$ and $L = 360$ m of 158.68 against approximately 158 min ($+0.43$ %) and the inlet area $A_0 = 0.03034$ against 0.0303 $\mathrm{m^2}$ ($+0.13$ %). All are within 0.5 %.

The shipped surface layer and Example 6.1. To test the shipped advance solver itself, Example 6.1 was run through it with the book's intake, length, slope and roughness and with the inlet flow area of the book's furrow geometry, so that only the solver differs from the reference implementation (`v_surface_shipped.js`). In release 1.0.0, where the stream advances quickly, the shipped solver agreed with the book within -1.7 to $+4.0$ % (first irrigation at $Q = 0.10$; later irrigation at $Q = 0.05$–$0.10\,\mathrm{m^3\,min^{-1}}$), but where advance was slow it was too fast: -9.8 % for the later irrigation at $Q = 0.0375$ and -14.7 and -29.0 % for the first irrigation at $Q = 0.075$ and 0.0625. The cause is a single term. In the volume balance the infiltration of the steady-intake component f_0 during advance is divided by $(1 + r)$, with r the advance exponent; the code of release 1.0.0 used a fixed $r = 0.55$, whereas the textbook procedure uses the current, iterated r, which is 0.32–0.45 in the slow cases (and 0.48–0.76 where advance is fast). With r iterated, the shipped solver reproduces the reference implementation to 0.002 % and the book to between -0.2 and $+1.8$ % in all eight cases (`v_surface_shipped.js` runs both versions). This is the third defect found by the verification; it matters only when $f_0 > 0$ and advance is slow. It is corrected in release 1.1.0, which iterates r.

The design problem (180 m × 180 m field, sprinklers spaced 12×15 m, required net depth $Z_n = 42$ mm, application efficiency 0.7 so a gross depth of 60 mm, an irrigation interval of 10 d, 16 h $\mathrm{d^{-1}}$ of operation, farm supply 21 $\mathrm{m^3\,h^{-1}}$, silty loam, 10 % slope) tests decision logic rather than a smooth function. The maximum allowable application rate is 8 mm $\mathrm{h^{-1}}$ and the maximum discharge $q_{\max} = 1.44\,\mathrm{m^3\,h^{-1}}$, as in the book; the layout yields $N = 15$ sprinklers per lateral and $N_m = 12$ lateral positions. With a discharge of 0.675 $\mathrm{m^3\,h^{-1}}$ the book's logic permits one set per day, which would require $N_d = 12$ days for one lateral and is rejected (12 days exceeds the 10-day interval), and the software follows. With $q = 1.35\,\mathrm{m^3\,h^{-1}}$ (two sets per day) one lateral needs $N_d = 6$ days with an inflow of 20.25 $\mathrm{m^3\,h^{-1}}$, which is within the supply; with 2.03 (three sets per day) the supply constraint rejects the option. The software selects $n = 1$ lateral for $q = 1.35$ and $n = 2$ for the lower discharge, as in the textbook. A nozzle of 4.8 mm at a head of 24 m yields 1.343 $\mathrm{m^3\,h^{-1}}$ against the book's 1.36 (-1.3 %). One discrepancy was found in the first version: the wetted diameter predicted for this nozzle was 24.7 m (radius 12.3 m) whereas the book's range

table gives 28.7 m, a deficit of 14 % caused by the range coefficient $c_R = 1.15$ in Eq. (51), which had been chosen rather than taken from the source. The Keller and Bliesner (1990) range relation for impact sprinklers has the coefficient 1.35; with it the software returns a radius of 14.49 m and a wetted diameter of 28.98 m (+1 %), and the coefficient was changed in the shipped code and manual (Table 13.1). The range coefficients of the other three sprinkler types (rotating spray, micro-sprinkler, big gun) remain engineered and have not been verified against published data.

Table 13: Border Example 6.4 (Walker and Skogerboe (1987)) and the sprinkler example: published and computed values.

Quantity	Book	SWAFL	Error (%)
T_{req}, first irrigation (min)	146.4	146.399	−0.00
T_{req}, later irrigation (min)	162.8	162.230	−0.35
$Q_{\max}$ ($m^3\,min^{-1}\,m^{-1}$)	1.88	1.883	+0.17
t_L at $Q = 0.14$, $L = 360$ m (min)	≈158	158.677	+0.43
A_0 (m^2)	0.0303	0.03034	+0.13
Sprinkler: nozzle discharge, 4.8 mm at 24 m ($m^3\,h^{-1}$)	1.36	1.343	−1.3
Sprinkler: wetted diameter (m), version 1	28.7	24.7	−14
Sprinkler: wetted diameter (m), corrected	28.7	28.98	+1.0
Sprinkler: maximum rate $u_{\max}$ ($mm\,h^{-1}$)	8	8	0
Sprinkler: $q_{\max}$ ($m^3\,h^{-1}$)	1.44	1.44	0

5.5 Engine 1 against held-out simulations

Engine 1 was queried at the factor levels of the 1 000 E12 runs (coarse loam, surface method). Without alignment, the engine's baseline differs from Daisy's baseline for this experiment: the engine's baseline (derived from the E5 irrigated set) has ET of 720 and irrigation of 282 mm yr^{-1} whereas the E2/E3/E12 baseline is 673 and 228 mm yr^{-1}, so raw predictions are biased by +53 mm for ET and +71 mm for irrigation (Table 14, Fig. 8). This is a *baseline offset* between experiment families, not a random error: the engine composes ratios taken within one family on a baseline from another family. After aligning the engine's baseline cell with Daisy's (re-anchoring), the normalised RMSE for off-axis cells (all cells except the baseline row and column, which are partly used in the engine's construction) is 6.2 % for ET, 18.9 % for irrigation, 48.7 % for drainage (a small mean, 35 mm), 10.1 % for soil evaporation, 8.7 % for transpiration, 34.6 % for nitrogen leaching and 27.9 % for yield. The R^2 values are 0.79 (irrigation), 0.82 (drainage), 0.71 (soil evaporation), 0.33 (nitrogen leaching), 0.10 (transpiration), −0.11 (ET) and −2.33 (yield); the negative values for ET and yield arise because the variance of these outputs within E12 is small (the coefficient of variation of ET across the cells is 5.9 %), so any constant offset dominates, and

for yield the engine's discrete lookup structure and ceiling produce a stepped response that the simulation does not show (Fig. 8d). Rank correlations are higher than the R^2 values: Spearman ρ is 0.96 for irrigation, 0.97 for soil evaporation, 0.92 for drainage, 0.84 for ET, 0.70 for yield, and 0.64 for transpiration and nitrogen leaching.

Table 14: Engine 1 against 1 000 held-out E12 runs (coarse loam), off-axis cells. Raw: as shipped; re-anchored: baseline aligned. nRMSE: RMSE as % of the observed mean; ρ: Spearman rank correlation.

	Raw			Re-anchored			
Variable	R^2	RMSE	bias	R^2	nRMSE (%)	bias	ρ
ET (mm)	−2.06	69.5	+53.1	−0.11	6.2	+5.5	0.84
Irrigation (mm)	0.18	97.4	+71.4	0.79	18.9	+8.3	0.96
Drainage (mm)	0.72	21.7	+14.6	0.82	48.7	+6.2	0.92
Soil evaporation (mm)	0.71	15.4	−7.9	0.71	10.1	−7.9	0.97
Transpiration (mm)	−1.01	29.3	+21.4	0.10	8.7	+4.0	0.64
Yield (Mg ha^{-1})	−1.82	1.51	−0.79	−2.33	27.9	−1.15	0.70
N leached (kg ha^{-1})	−0.84	18.5	−14.7	0.33	34.6	−0.3	0.64

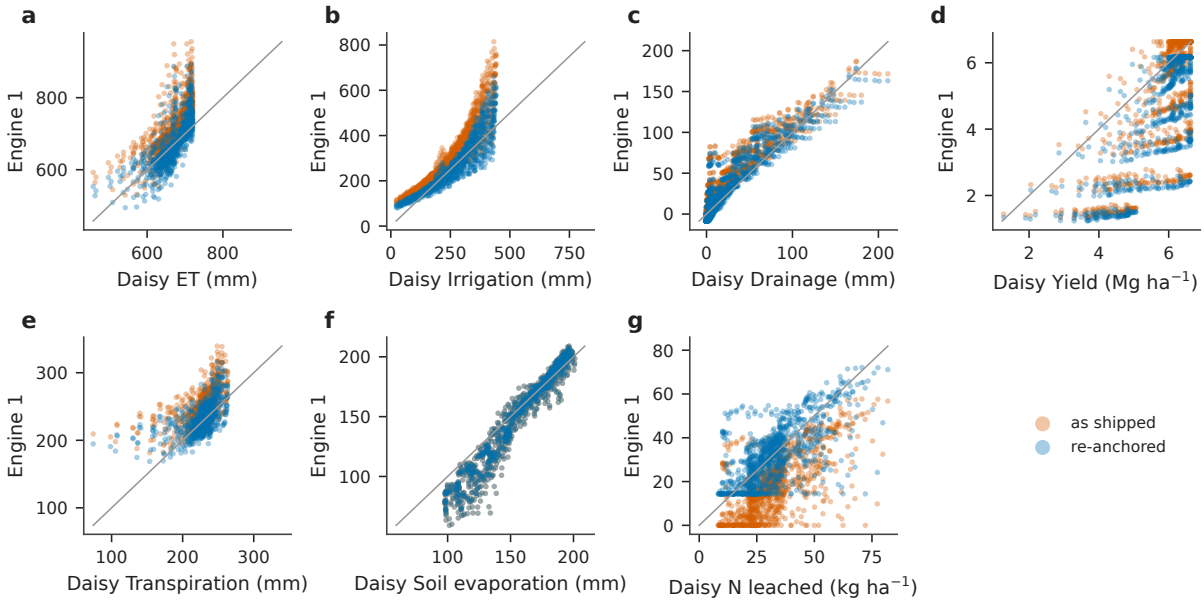


Figure 8: Engine 1 versus Daisy for the 1 000 held-out E12 runs. Orange: as shipped; blue: re-anchored. (a) ET, (b) irrigation, (c) drainage, (d) yield, (e) transpiration, (f) soil evaporation, (g) nitrogen leached. Grey lines are 1:1.

The practical reading is that Engine 1 is a *rank-faithful screening emulator*: it orders scenarios correctly (irrigation $\rho = 0.96$, drainage 0.92, soil evaporation 0.97) and gets the response to trigger, depth and rainfall in the right direction, but its absolute values depend on which experiment family anchors the baseline, and its yield output should not be used quantitatively. The

independence assumption for mulch and warming cannot be tested because no experiment crosses them.

5.6 Engine 2 closure and agreement with Engine 1

The unscaled daily core closes the water balance to within 0.1 % of inflow (median) and at most 0.7 % over 60 random states (Fig. 9a). On a 100-scenario grid the calibrated irrigation closes with residual of median 0.04 % and maximum 0.23 % of inflow.

On a realistic grid of 10 soils $\times$ 5 triggers $\times$ 2 methods (cereal crop), the calibrated mean annual irrigation of Engine 2 agrees with Engine 1 within 10 % in 90 of 100 scenarios (76 within 6 %). The ten disagreements are all in coarse soils at dry triggers: sand (S03) at -500 and -1000 cm (both methods; simulated irrigation 0 versus target 278 mm), loamy sand (S04) at -500 (337 versus 293 mm) and -1000 cm (0 versus 293 mm), and sandy loam (S05) at -1000 cm (8–11 versus 15 mm). The ten cases are of three kinds. In sand at -500 and -1000 cm and in loamy sand at -1000 cm (six cases) the bucket model never reaches the trigger suction, because its layers drain below it, so no irrigation is applied; the Engine 1 value (278 mm for sand at -300 cm and drier triggers in the Daisy set) reflects the simulation's own scheduler behaviour. In loamy sand at -500 cm (two cases) the trigger is reached rarely, the calibration factor rises to $k_t = 3.8$ and the model over-irrigates (337 against 293 mm). In sandy loam at -1000 cm (two cases) the absolute difference is small (8–11 against 15 mm) but exceeds 10 % of the target. An earlier test on randomly drawn states, including legume and vegetable crops, gave much poorer agreement (61 % mean error in irrigation), which is why the realistic grid is reported with its scope: cereal crop, two methods. The unscaled partition (E/T split) and drainage of Engine 2 do not match Engine 1 (Fig. 9c shows unscaled drainage scattering around the 1:1 line by factors of two to five), which is why the displayed series are scaled by Eq. (43); this makes the displayed means correct by construction and the timing within the year *plausible but unverified*.

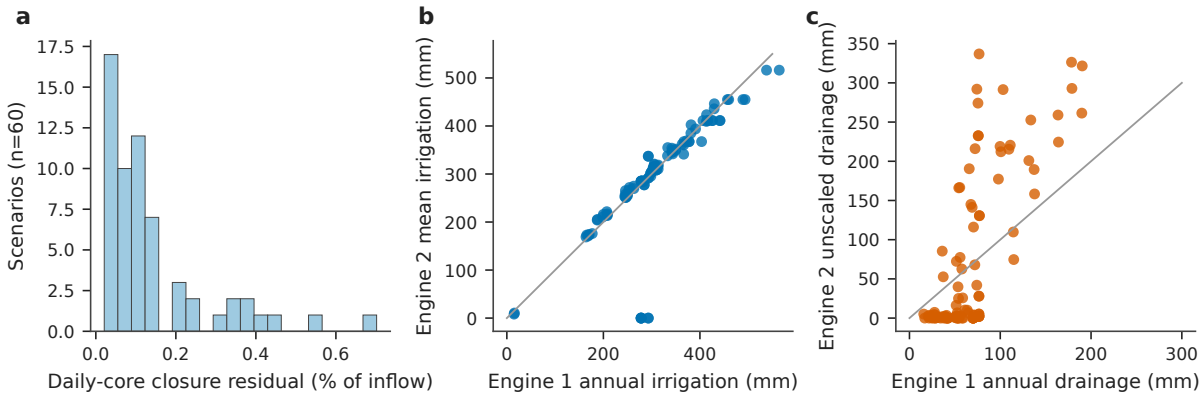


Figure 9: Engine 2. (a) Closure residual of the unscaled daily core over 60 random states. (b) Mean annual irrigation of Engine 2 against Engine 1 for the 100 scenarios of the realistic grid (10 soils × 5 triggers × 2 methods, cereal), including the failures at zero. (c) Unscaled Engine 2 drainage against Engine 1 for the same scenarios.

5.7 Drip, sprinkler and surface layers against the literature

The design layers are not specific to surface irrigation, and each was therefore tested against published relations or examples outside the software.

Drip lateral. Two published results were used. First, the lateral loss along a pipe with N equally spaced outlets equals the full-length Hazen–Williams loss multiplied by the Christiansen reduction factor $F = 1/(m+1) + 1/(2N) + \sqrt{m-1}/(6N^2)$ with $m = 1.852$ (Christiansen, 1942; Keller and Bliesner, 1990). For $N = 5$ to 400 the software's lateral loss agrees with this factor and with the exact discrete sum to 0.01 % or better. Second, the lateral solver was compared with an independent vectorised solver on 400 random configurations (Fig. 10). The first version, which used four iterations, deviated from the independent solution by up to 0.13 % for $h_f/H_0 \le 0.2$, by up to 81 % for $0.2 < h_f/H_0 \le 0.6$ and by up to 89 % above that. The solver was rebuilt to iterate to convergence; it now agrees with the independent solution to a relative error of 3.2×10^{-7} in total head loss for $h_f/H_0 \le 0.6$ (324 configurations), and the warning described below is shown for every configuration with $h_f/H_0 > 0.2$ (156 of 400) and for every configuration in which the iteration does not converge (31 of 400, all with $h_f/H_0 > 0.6$). Above $h_f/H_0 = 0.6$ the maximum difference is 31 % and occurs where the head reaches the 0.5 m floor of the model, a regime in which the lateral is not a usable design. The 20 % limit is the usual design criterion for laterals (Keller and Bliesner, 1990; Keller and Karmeli, 1974); when it is exceeded, the drip layer now prints a red warning stating the loss as a share of the inlet head.

Sprinkler. Four relations were tested: orifice discharge (Torricelli, within 1.3 % of the book value for the textbook nozzle), the range relation (corrected as described above, +1 %), the Christiansen uniformity coefficient and the spacing rule of at most 65 % of the wetted diameter (Keller and Bliesner, 1990). With the corrected radius, a square layout at 65 % spacing gives CU = 85.7 %, at 60 % 92.9 % and at 70 % 78.7 % (Fig. 11b), consistent with the rule that this spacing yields about 85 %. One limitation was found: the engineered wind term changes CU by less than 0.2 percentage points at a wind speed of 3 m s^{-1}, whereas wind is known to degrade uniformity; the wind response of the sprinkler layer should not be used for design.

Surface irrigation, second source. The shipped surface layer was compared with the worked border examples of FAO Irrigation Paper 45 (Walker, 1989), which use the Kostiakov–Lewis intake $Z = k\tau^a + f_0\tau$ and the surface shape factor $\sigma_y = 0.77$ of that source. The shipped layer had no steady-intake term, so f_0 was added to it as an optional parameter (zero by default, so results with the pure Kostiakov law are unchanged). For the 200 m and 100 m open-end borders ($Z_{req} = 8$ cm) the required intake times are 388.5 and 678.9 min, identical to the published values, and the minimum unit flows 0.0226 and 0.032 m^3 min^{-1} m^{-1} from $Q_{\min} = 0.000357\, L\, S_0^{1/2}/n$ agree (0.0226 and 0.0319). Thirteen published advance times, for unit flows of 0.009 to 0.09 m^3 min^{-1} m^{-1}, are reproduced with errors from −0.2 to +0.5 % (mean absolute 0.13 %; Fig. 11a); before the correction of the steady-intake term described above the errors were −2.2 to +2.9 % (mean absolute 1.7 %). The published cut-off and recession times were not compared, because the FAO procedure uses a different cut-off rule from the software's options, so the efficiency tables of that source were not tested.

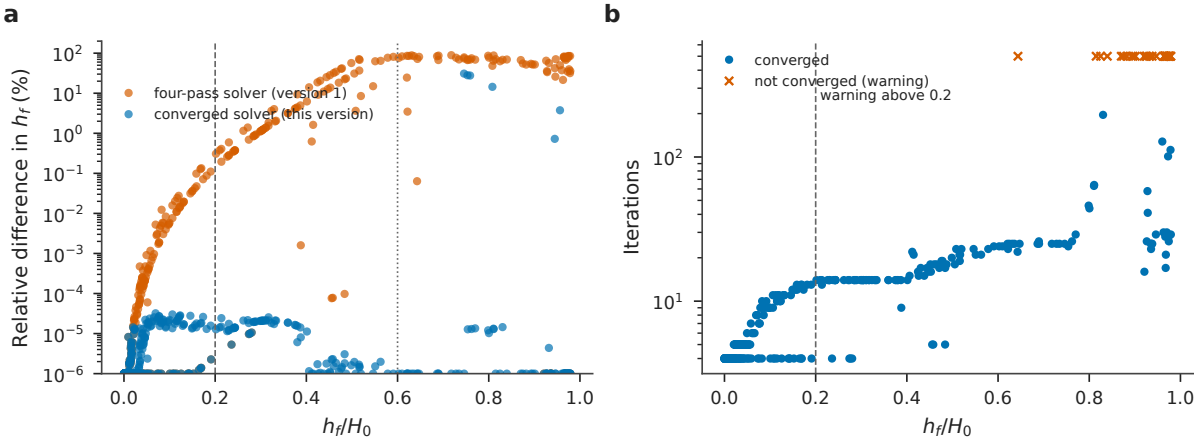


Figure 10: Drip-lateral solver versus an independent converged solver (400 random configurations). (a) Relative difference in total head loss for the four-pass solver of version 1 and the converged solver, against the loss ratio h_f/H_0 (dashed: design limit 0.2; dotted: 0.6); (b) number of iterations of the converged solver; crosses mark configurations that did not converge.

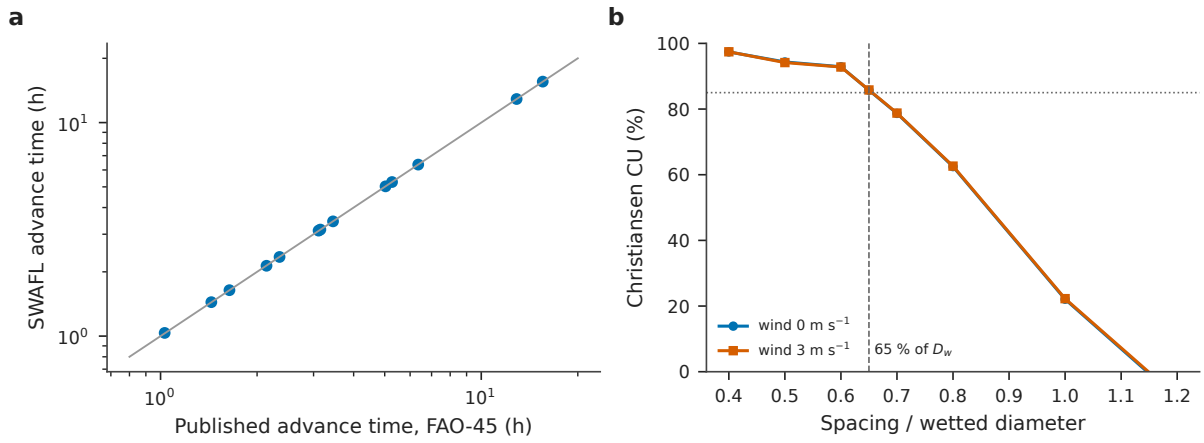


Figure 11: Checks against published sources. (a) Thirteen advance times of the FAO Irrigation Paper 45 border examples (published against SWAFL; grey line 1:1). (b) Christiansen coefficient of uniformity of the impact-sprinkler layer against spacing as a share of the wetted diameter, without wind and at 3 m s^{-1}; dashed: 65 % spacing rule, dotted: CU of 85 %.

Table 15: Drip lateral solver of version 1 (four passes) and the converged solver against an independent solver, by loss ratio h_f/H_0 (400 random configurations). Maximum relative error in h_f and maximum absolute error in E_u.

		Version 1		Converged	
h_f/H_0	Configurations	h_f (%)	E_u	h_f (%)	E_u
≤ 0.2	244	0.13	4.6×10^{-5}	3.2×10^{-5}	2.6×10^{-8}
0.2–0.6	80	81.2	0.133	2.5×10^{-5}	9.2×10^{-8}
> 0.6	76	89.1	0.444	31	0.080

5.8 Run time and documentation audit

Single-call medians in Node.js were 0.021 ms for Engine 1, 0.25 ms for the drip layer, 1.7 ms for the sprinkler layer and 0.14 s for a 20-year daily Engine 2 simulation. Interactive operation (slider movement) is therefore limited by rendering, not by computation, and a design-space sweep of 10^5 Engine 1 evaluations takes about 2 s.

The Manual contains 134 numbered equations (124 carry a tag and ten are rows of aligned groups). Each now carries its provenance label directly beneath it, and a ledger of all 134 is given in the manual's appendices: 38 are **[Literature]**, 18 are **[Data-driven]** and 78 are **[Engineered]**. The labelling was done by reading each equation against its cited source; it records where a relation comes from and does not certify that it is correct. The equations used in this article, including those of the system variables of Section 3, are classified in Appendix A. The engineered ones comprise the root and canopy curves, rainfall and interception generators, wetted-

bulb geometry, spray patterns, wind-drift loss and similar heuristics. An independent reader can therefore see that the sprinkler pattern and drift functions have no external validation, whereas the surface layer relies on published equations.

6. Discussion

6.1 What the system variables add

Representing the whole column, and not only the irrigation decision, changes what a user can ask of the framework. The evaluation for ten soils (Section 5.1) shows the chain from pore structure to the decision in numbers: retention sets how much of the plant-available water a fixed suction consumes (24 to 84 % at -300 cm), which is why the trigger is better expressed as a depletion; the same retention window sets how long the surface stays wet after a wetting, longest in the coarse soils (Lehmann et al., 2008); and under one rainfall the soils divide between evaporation-dominated and drainage-dominated regimes, which decides whether a mulch saving is recovered as less irrigation or lost to drainage. These are the couplings that soil-physics reviews identify as decisive for water use and that a decision tool usually hides (Or et al., 2013; Vereecken et al., 2016).

The same evaluation is a test of the engineered relations that no worked example covers, and it found two defects of representation. The leaf water potential of the soil-to-air view is controlled by the floor on the relative conductivity, so it is nearly the same in every soil and more negative than the resistances were chosen to produce; a conductance scaled by saturated conductivity and root length, as in sink-term models (Feddes et al., 1978), would replace the floor. The fate of a mulch saving inherits the coarse-loam drainage increments of E14, so it understates the drainage share in sandy soils, which the ten-soil mulch experiment E7 resolves directly; composing the mulch drainage response from E7 instead would remove the defect. Both are reported as limitations of the released version (Section 6.5), and both illustrate why the variables of an explanatory view need the same scrutiny as the outputs used for decisions.

6.2 What the evidence supports

Taken together the four verification families support four statements. (i) The closed-form design layers implement the published relations: the textbook surface procedure reproduces a worked

example to a median of 0.08 % and a maximum of 1.8 %, and the shipped surface layer, with the one-term correction of its steady-intake term (release 1.1.0), reproduces 13 FAO-45 advance times within -0.2 to $+0.5$ % and the slow-advance cases of Example 6.1 within -0.2 to $+1.8$ % (before the correction, up to -29 %); the drip lateral matches the Christiansen factor to 0.01 % and an independent solver to 3×10^{-7} within $h_f/H_0 \leq 0.6$; the sprinkler decision logic, nozzle discharge, wetted diameter ($+1$ %) and spacing rule match the textbook. (ii) Engine 1 is a useful screening emulator, with high rank fidelity for irrigation, soil evaporation and drainage, with absolute values that depend on the baseline family; yield and nitrogen estimates are best used for ranking. (iii) Engine 2 conserves mass and reproduces Engine 1's annual irrigation in most realistic scenarios, but not in sandy soils at dry triggers, and its partition is scaled rather than simulated. (iv) Numerical sub-solvers can be accurate in their design range and inaccurate outside it, as the first drip solver was; the failure mode should be flagged to the user, and the revised drip layer does so.

The evidence does not support claims that the software predicts field irrigation requirements, that it improves yields or water productivity, or that it is easier to use than alternatives. The evidence of fidelity is relative to the process model (Daisy) and to textbook examples; Daisy itself has been evaluated in the literature (Hansen et al., 2012; Holbak et al., 2026) but the ten virtual soils here were not compared with field data. A field comparison (lysimeters or sensor-scheduled plots against the software's schedule and design table) and a user study would be needed to make such claims.

Traceability lets a reviewer or a user ask the right questions: which equation produced this number, where did it come from, and how was it tested? It also made the verification possible, since the test for each layer could be written against a named equation. The cost is documentation: the Manual (with a screenshot atlas in which every screen, every state of every control and every part of every scrolled drawing carries numbered arrows and is paired with its equations, and 134 numbered equations) must be kept synchronised with the code, and, as the label audit of the first release showed, this discipline is easily lost. It is recommended that provenance labels be treated as part of the code review, enforced by a check that every numbered equation carries a label and a source, and that a published-example test accompanies every closed-form layer (Jakeman et al., 2006; Refsgaard and Henriksen, 2004).

The annual meta-model avoids fitting any parameter by composing simulated ratios; the advantage is that it cannot produce results outside the simulated envelope without clipping, and

that its behaviour is explainable by the tables. The disadvantage is revealed by the held-out test: ratios from one experiment family applied to the baseline of another transmit the inconsistency between the baselines. The inconsistency here arises because the baseline for E5 (tillage and structure) used the irrigated cereal setting with different trigger and depth from the baseline of E2, E3 and E12. Two remedies are possible: re-anchor all families to a common baseline run that is part of every experiment (a design remedy), or calibrate an anchor offset per family (a statistical remedy). The first is preferable, as it removes the cause.

The combination of mulch and warming multiplies ratios taken from separate experiments. The independence assumption is plausible but the companion hydraulic-memory paper shows that the drainage response to evaporation changes depends on climate (warming and rainfall change the recoverable share of an evaporation saving), which suggests that the mulch effect is not independent of climate for the drainage and irrigation fluxes. A crossed experiment would test this.

Reviews of irrigation decision-support tools identify recurring needs for physics-informed and process-based components, for open and FAIR software, for uncertainty handling, and for shared benchmarks and reproducible releases, and note that accuracy is often demonstrated against calculated or simulated targets and not against operational endpoints (Umutoni and Samadi, 2024; Zhang et al., 2021). The present contribution addresses the needs for open, process-based and reproducible software at the level of the individual equation: provenance labels, a published map from displayed quantity to equation, and verification against references that exist outside the software, including a published design textbook. It does not address sustained multi-season savings or cross-site transfer, which are outside the scope of this verification, and the verification reported here should be read as a necessary step before field validation. The verification suite also clarifies the roles of the three computing components in a decision. The closed-form design layers are suited to design calculations in which the user supplies the field geometry, soil intake and supply, because their outputs are determined by the equations and were reproduced against published values. The annual meta-model is suited to comparison of management options, because it preserves their ordering with a rank correlation of 0.64 to 0.97 across the variables tested, and because each of its outputs is a composition of simulated ratios that can be inspected. The daily model is suited to the timing and size of events within a season and to the visualisation of the field water balance; its seasonal means are tied to the annual meta-model by construction, and its within-season timing is a model result that a field experiment would test.

Equation traceability is a property that can be audited in any decision tool. A reviewer can

select a displayed quantity, locate its equation in the manual, check the equation against its source, and re-run the test that supports it; the procedure requires no access to the developers. The audit of the manual shows where the procedure applies most strongly (the surface and drip layers, whose equations are published), and where the provenance rests on engineered functions (spray patterns, wind drift, wetted-bulb geometry). This distinction is useful to users: a user relying on the surface layer relies on published theory that has been reproduced against its textbook; a user relying on the sprinkler pattern relies on an engineered shape and should confirm it with manufacturer data. The wetted radius, which the first version under-predicted by 14 % for the example nozzle until its coefficient was replaced by the published value, illustrates the point, since the range law is a coefficient on a general expression and a manufacturer's table is the proper reference for a specific nozzle (Keller and Bliesner, 1990; Playán et al., 2005).

Field evaluation is the step that converts verification into validation. A suitable design compares the software's recommended triggers and doses with sensor-scheduled plots on contrasting soils, records applied water, drainage below the root zone and yield, and evaluates the software with the efficiency and bias statistics that are standard for model evaluation (Bennett et al., 2013; Moriasi et al., 2007; Nash and Sutcliffe, 1970). For the design layers, field evaluations of surface advance, drip emission uniformity and sprinkler catch-can uniformity provide the corresponding data (Burt et al., 1997; Playán et al., 2005; Walker and Skogerboe, 1987). The open availability of the code, the manual and the verification scripts allows such an evaluation to be carried out by any group and compared with the results reported here.

Equation-traceable software has teaching and extension uses in addition to its use for decisions. A student can follow a number from the dashboard to the equation, to the source, to the test, and can modify the inputs to see how the number responds. A practitioner can obtain a transparent first estimate before commissioning a field-specific simulation. For both uses the relevant feature is that the numerical behaviour of each layer is documented together with its range of verified operation, so that the tool states what it can be relied on for (Hutton et al., 2016; Wilkinson et al., 2016).

6.3 Operating envelopes established by independent verification

A tool that exposes its equations earns trust only if each layer is tested against something outside itself and the test is reported with the range in which it holds. The verification programme therefore delivers, for every layer, an *operating envelope*: the inputs and the decision purpose

for which agreement with an external reference has been shown, with the mechanism that fixes its edge. Verification separates errors in the implementation of a relation from limitations of the relation and of the reference against which it is judged (Oreskes et al., 1994; Refsgaard and Henriksen, 2004), and a deviation bears on fitness for purpose only relative to that purpose: a bias is immaterial for ranking options and decisive for budgeting water (Bennett et al., 2013; Moriasi et al., 2007). The envelopes below are the substantive result of the work, and they are what allows each layer to be used with a stated confidence.

Engine 1 and a diagnosed anchoring effect with a design remedy. Before alignment Engine 1 over-predicts ET by 53 mm and irrigation by 71 mm yr^{-1} in the held-out experiment (Table 14). The structure of the error identifies its cause. A random error averages out over 1 000 runs; a constant offset does not, and after one baseline cell is aligned the biases fall to $+5.5$ and $+8.3$ mm yr^{-1} while the rank correlations are unchanged (0.84 and 0.96). The offset therefore belongs to the construction, ratios formed within one experiment family and applied to the baseline of another, and not to the ratios. In the literature on computer-model calibration this is a systematic model discrepancy, which Kennedy and O'Hagan (2001) argued should be represented explicitly and not treated as independent noise. Reviews of surrogate modelling likewise stress that a surrogate inherits the properties of the design that trained it, and that the design should cover the region of use (Asher et al., 2015; Razavi et al., 2012; Sacks et al., 1989). The proposed design remedy, one common baseline cell in every experiment family, is the direct expression of that principle. That the offset was invisible in the tables used to build the engine and appeared only in an independent experiment is the usual case for held-out testing over internal consistency checks (Bennett et al., 2013; Oreskes et al., 1994). Until the remedy is built in, absolute Engine 1 values carry the offset, and the supported use is the comparison of options within a family.

Engine 1 and rank fidelity as the decision-relevant skill. After re-anchoring, R^2 is -0.11 for ET and -2.33 for yield while ρ is 0.84 and 0.70 (Table 14). The two statistics answer different questions and are compatible. R^2 here is measured against the mean of the held-out observations; the coefficient of variation of ET across the cells of E12 is only 5.9 %, so an RMSE of 6.2 % of the mean is comparable to the spread to be explained, and a small residual bias then dominates the ratio. Correlation-type measures can be insensitive to additive and proportional errors (Legates and McCabe, 1999), which is why model-evaluation guidance asks for several complementary statistics and not one (Bennett et al., 2013; Moriasi et al., 2007). Reporting rank and absolute metrics separately is therefore not hedging. Rank agreement is the relevant statistic when the

decision is a choice between options, as in screening and factor prioritisation (Saltelli et al., 2008); absolute error is the relevant statistic when a quantity is budgeted. Yield is the exception that the data support: its bias ($-1.15\,\mathrm{Mg\,ha^{-1}}$), the stepped response produced by the engine's lookup structure and the weakest useful rank correlation (0.70) define the role of yield as a ranking output, which is how the software presents it.

Engine 2 and an envelope mapped to sensor practice. The ten scenarios in which Engine 2 and Engine 1 disagree are all coarse soils at triggers of -500 or $-1000\,\mathrm{cm}$ (sand, loamy sand and, at $-1000\,\mathrm{cm}$, sandy loam); in six of them the bucket model never reaches the trigger because its layers drain below it, so no irrigation is applied, and in the other four it reaches the trigger too rarely to calibrate (over-irrigation in loamy sand at $-500\,\mathrm{cm}$, 8–11 against 15 mm in sandy loam at $-1000\,\mathrm{cm}$). These triggers lie outside the range in which sensor-based scheduling is practised in coarse soils. Extension guidance for tensiometers places irrigation triggers for sandy soils at about 20–30 kPa and for silty clay loams at 50–60 kPa, and notes that the instrument functions reliably only to about 80 kPa, beyond which the water column breaks (Alam and Rogers, 1997). A trigger of $-500\,\mathrm{cm}$ ($\approx 49\,\mathrm{kPa}$) is therefore well above the usual range for sand and loamy sand, and $-1000\,\mathrm{cm}$ ($\approx 98\,\mathrm{kPa}$) is beyond the working range of the instrument whose signal the trigger law represents. The disagreement thus marks the boundary of the daily-bucket formulation in a regime that practice avoids, and the zero-irrigation outcome is an explicit failure rather than a silent error, whereas the over-irrigation in loamy sand at $-500\,\mathrm{cm}$ is not flagged and should be. It also reflects that the field-capacity convention of 100 cm suction, adopted from Allen et al. (1998), is a convention whose appropriateness varies with the soil (Assouline and Or, 2014; Twarakavi et al., 2009). The practical consequences are to flag triggers beyond the sensor range for a given texture and to read the Engine 1 value, which is the Daisy scheduler's own result, for those cells. The unscaled partition of Engine 2 does not match Engine 1, and its unscaled drainage scatters around Engine 1 by factors of two to five. Mass closure is a necessary condition and not evidence that the partition is right, since agreement in totals does not confirm the process representation (Oreskes et al., 1994); this is why the displayed series are scaled to Engine 1 and why timing within the year is labelled plausible but unverified.

Drip lateral and a convergence-controlled solver with a declared range. The first solver used four fixed iterations. Its error was at most 0.13 % for $h_f/H_0 \leq 0.2$, but up to 81 % for $0.2 < h_f/H_0 \leq 0.6$ and 89 % above (Table 15). The conventional design criterion limits lateral loss to about 20 % of the operating head (Keller and Bliesner, 1990; Keller and Karmeli, 1974), so the

defect lay where a lateral would normally be redesigned, but a user could still reach that region by changing inputs, and a design tool must report its own range of validity. Iterations of this kind typically converge more slowly as the loss ratio grows, so a fixed number of passes stops short; iterating to convergence removed the defect within $h_f/H_0 \leq 0.6$ (3.2×10^{-7}). The residual 31 % above 0.6 occurs where the head reaches the model's 0.5 m floor, a configuration that is not a usable design and that triggers a warning. The independent solver used here is numerical, which tests the implementation of the iteration. The lateral-hydraulics literature offers analytical and alternative numerical solutions that would add evidence of a different kind (Scaloppi and Allen, 1993; Warrick and Yitayew, 1988).

Sprinkler with a published range law and a calm-condition uniformity bound. The first version gave a wetted diameter of 24.7 m against the textbook 28.7 m (-14 %). The range law is a general expression with an empirical coefficient, and the textbook value follows the coefficient published with it (Keller and Bliesner, 1990); adopting it gives 28.98 m ($+1.0$ %). The error was therefore one of the parameter, not of the structure, and it identifies the reliance that a user can place on the layer: for a specific nozzle the manufacturer's range and pressure table is the proper reference. The wind term is a different case. It changes CU by less than 0.2 percentage points at $3\,\mathrm{m\,s^{-1}}$, whereas field experiments find wind to be the dominant influence on uniformity: in a solid-set maize field wind speed explained a large share of the variation in CU, with CU of about 80 % for single events against 88 % over the season, and yield variability followed that of applied water above $2\,\mathrm{m\,s^{-1}}$ (Dechmi et al., 2003); wind drift and evaporation losses have been quantified in solid sets and moving laterals (Playán et al., 2005). The engineered wind response is therefore too weak, which is why it is marked as not for design: the CU the layer returns is an upper bound for calm conditions.

Surface irrigation and agreement at the precision of the published tables. The residuals of the reference implementation against the textbook (median 0.08 %, maximum 1.81 %) and of the shipped layer against FAO-45 (13 advance times, -0.2 to $+0.5$ % after the correction) have identifiable sources: printed values are rounded or tabulated at limited precision, and the volume-balance method with fixed shape factors is itself an approximation to the advance process (Elliott and Walker, 1982; Walker and Skogerboe, 1987). Their size matters only against the uncertainty of the inputs. Soil variability contributes more to the variability of furrow infiltration within a single field than does intake opportunity time (Childs et al., 1993), so errors of a few per cent in advance time are small beside the uncertainty in the intake parameters that a user must supply.

One printed value, the later-irrigation E_a at $Q = 0.10\,\mathrm{m^3\,min^{-1}}$ (53.0 % in the copy used against 33.5 % computed), was excluded because it breaks the monotone trend of its neighbours (44.2 % and 52.5 % at lower flows). The exclusion is a judgement and is disclosed as such; had the entry been kept, the maximum error would be -36.7 %. The entry should be checked against a physical copy of the source, and the software value is the one consistent with the trend.

Mulch and warming as a specified crossed experiment. Combining mulch and warming multiplies ratios taken from separate experiments. No experiment crosses the two factors, and one-factor-at-a-time designs cannot detect interactions (Saltelli et al., 2008); the companion hydraulic-memory analysis indicates that the drainage response to evaporation change depends on climate. The multiplicative composition is therefore a transparent, falsifiable hypothesis for drainage and irrigation, and a crossed factorial of mulch and warming is the experiment that tests it.

Read together, the envelopes sort the software's outputs by the kind of reliance they support. Closed-form design outputs (surface advance, drip lateral loss, sprinkler discharge and diameter) are reproduced against external references to within a few per cent inside stated ranges, including slow surface advance with steady intake since the correction of the third defect. Screening outputs (Engine 1 irrigation, evaporation and drainage) preserve the ranking of management options. Scheduling outputs (Engine 2) conserve mass to 0.1 % (median) and match the annual irrigation across the realistic part of the trigger range. Each envelope has a mechanism, a stated test and a reported edge, and no edge is hidden inside an aggregate statistic. Documenting where each layer holds is itself a methodological contribution: it converts a decision tool from a black box that is believed into an instrument whose range of use can be checked by a reader.

6.4 End-to-end workflow from question to decision

To make the traceability concrete, one decision is followed through the software, naming at each step the equation that produces the displayed quantity and the test that supports it.

Step 1. Choose the soil and crop setting. The soil sets (α, n), the porosity and retention of Section 3.3 and the baseline vector **b** (Table 7, E5). *Support.* class means of Carsel and Parrish (1988); the baseline is Data-driven.

Step 2. Set the trigger and sensor depth. The trigger suction h_T is converted to a depletion fraction by Eq. (29). *Support.* FAO convention (Allen et al., 1998); the inverse is a monotone bisection.

Step 3. Read the annual consequences. Engine 1 returns E, T, ET, I, Y, D, N_L by Eqs. (31) to (38). *Support.* held-out test (Table 14); the interface should be read as a ranking of options, and absolute yield is not to be used.

Step 4. Inspect the seasonal timing. Engine 2 produces dates and doses with Eqs. (40) to (43). *Support.* mass closure and annual agreement (Section 5.6); timing within the year is plausible but unverified.

Step 5. Select an application method. The surface, drip or sprinkler layer is opened with the soil, depth and trigger carried over. *Support.* Tables 12, 13 and 15.

Step 6. Read the design check. Closure identities (Eq. (54)) are printed, and the drip layer prints a red warning when the lateral loss exceeds 20 % of the inlet head or the iteration does not converge (Section 5.7).

Step 7. Record. The configuration and the equation numbers (manual) allow the decision to be reconstructed by hand.

This chain is deliberately shallow: each displayed number is at most three function calls from its equation, so a reader can reproduce a value with a calculator.

A user comparing two triggers for coarse loam can conclude, with the rank fidelity reported, which uses less irrigation and by roughly how much *relative* to the other; they should not quote the absolute millimetres as a field requirement. A user designing a furrow can use the volume-balance outputs as a textbook calculation. A user designing a sprinkler layout can rely on the corrected wetted radius for the impact type (within 1 % of the textbook nozzle), should check the manufacturer's tables for the other three sprinkler types, and should treat the uniformity coefficient as an upper bound for calm conditions. A user designing a long drip lateral should verify that the head loss does not exceed 20 % of the inlet head.

6.5 Domain of validity and the field-validation programme

1. Validation status. The verification has not yet been extended to field or user data, and uncertainty propagation is not included. It is against published worked examples and relations, which tests that equations are implemented as written; it does not test that they describe a given field. The published examples are few (two for surface irrigation, one for sprinkler

design, relations only for drip), and no field data for drip or sprinkler systems exist in the repository.

2. One climate (Taastrup) and one cereal rotation simulated. Release 1.0.0 scaled the cereal result by fixed legume and vegetable multipliers; release 1.1.0 replaces them by 48 library crops (field crops, vegetables, trees, vines and palms) to which the cereal result is transferred by the ratio of two daily FAO-56 root-zone balances on the same weather years, so the values of these crops are transfers and not Daisy simulations.
3. Engine 1 baseline offset between experiment families; yield is used for ranking.
4. Engine 2 partition and drainage are scaled to Engine 1; irrigation is not matched in sand and loamy sand at dry triggers.
5. Sprinkler range coefficient verified only for the impact type (+1 % in diameter); the coefficients of the rotating-spray, micro-sprinkler and big-gun types, the pattern shapes, the wind term (which barely changes CU) and the drift function are engineered and unverified.
6. Drip laterals with loss above 60 % of the inlet head are not solved reliably (flagged by a warning), and the wetted-bulb geometry is engineered and untested against measured wetting patterns.
7. Single-soil experiments. E11 (warming and rain) and E14 (mulch and rain) cover the coarse loam only, so the drainage and leaching increments of mulch and rainfall, and the nitrate cost of drainage, are the same in every soil; the drainage share of a mulch saving is understated in sandy soils.
8. Water-potential path. The soil resistance is at its cap on most crop days, so the leaf potential is set by the conductivity floor and not by the soil; the soil-to-air view is qualitative.
9. Partition of evaporation. E_{oth} (43–61 % of ET) is the remainder of the simulated partition and includes an unattributed term; the field view labels it interception.
10. Documentation. Provenance labels record origin, not correctness.
11. Surface layer. In release 1.0.0 the steady-intake term of the advance volume balance used a fixed advance exponent of 0.55, which under-predicted slow advance when $f_0 > 0$ (up to -29 % in Example 6.1); release 1.1.0 iterates the exponent, which removes the error. Examples 6.1 and 6.4 verify the textbook procedure through a reference implementation; the shipped layer is verified against FAO-45.
12. Engine 2 over-irrigates without a warning in loamy sand at -500 cm.
13. Atmosphere. The evaporative demand and relative humidity of the explanatory views are engineered (no radiation, vapour-pressure-deficit or wind term), and the available-water

capacity of the daily model is a texture-class constant that is not derived from the van Genuchten retention used for depletion (e.g. sand 0.6 mm cm^{-1}).

14. Textbook Example 6.1: one printed value (later-irrigation E_a at $Q = 0.10$) does not follow the monotone trend of its neighbours in the printed table and was excluded from the comparison.

Process-based models (HYDRUS, DSSAT, APSIM, AquaCrop, Daisy) and decision platforms emphasise physical or biological realism and require setup and data; SWAFL emphasises immediacy and transparency at the cost of fidelity. A reasonable workflow is to use a process model for site-specific prediction and an equation-traceable emulator for scenario screening, teaching and design checks, and to use verification suites such as the one presented here to document the boundary between the two (Asher et al., 2015; Razavi et al., 2012). The method of verifying against textbook worked examples is generic and could be applied to other design software.

Recommendations for decision-support software. (1) Label provenance per equation. (2) Test every closed-form layer against a published worked example and publish the test. (3) Test every emulator against held-out simulations and report rank and absolute metrics separately. (4) Check solver convergence and warn users when the design is outside the validated range. (5) Publish failure cases. (6) Version the data and the code together, with an archive DOI (Wilkinson et al., 2016).

7. Conclusions

SWAFL Desktop models the coupled soil–water–atmosphere system of a field column, with five compartments, six water fluxes and the storage change, twelve controls and the relations that link pore structure, retention, evaporation, mulch, roots, the water-potential path, drainage, nitrogen and seven feedback loops to the irrigation decision, and it shows that an offline decision tool can be built in which every output is tied to a numbered equation, a provenance label and a re-runnable check. Evaluated for ten soils, the variables show the coupling quantitatively: a common -300 cm trigger uses 24–84 % of the plant-available water, stage-1 evaporation lasts 1.1–3.0 days, and drainage ranges from 28 to 164 mm yr^{-1} under the same rainfall; the same evaluation found that the leaf water potential is set by a conductivity floor and that the drainage share of a mulch saving is understated in coarse soils. The checks show that the textbook surface procedure and

the shipped sprinkler layer reproduce published worked examples and that the shipped surface layer reproduces published advance times (surface: median error 0.08 %, maximum 1.81 % for the reference implementation against one source and -0.2 to $+0.5$ % for the shipped layer against a second; drip lateral loss to 0.01 % of the Christiansen factor; sprinkler wetted diameter $+1$ %), that its annual meta-model is a rank-faithful but anchor-sensitive screening emulator (Spearman 0.64–0.97; off-axis nRMSE 6.2 % for ET and 18.9 % for irrigation after re-anchoring), that its daily model conserves mass and reproduces the annual irrigation in 90 % of realistic scenarios but not in sand at dry triggers, and that its drip solver, now iterated to convergence, is reliable up to a head loss of 60 % of the inlet head and warns beyond 20 %. Verification exposed three defects: the sprinkler range coefficient and the drip iteration count, which were corrected in the first released code, and a fixed advance exponent in the surface layer's steady-intake term, which under-predicted slow advance (up to -29 % in Example 6.1) and is removed in release 1.1.0 by iterating the exponent; the provenance labelling is now complete for all 134 equations. The baseline offset, the unverified range coefficients of three sprinkler types, the wind term, the single-soil mulch and warming increments and the conductivity floor of the water-potential path remain open. Field validation remains the necessary next step and is not claimed here.

Release 1.1.0 and companion papers

This paper describes the engines and design modules. Release 1.1.0 adds a graphical Daisy workbench (Paper 4), the bridge from Daisy output to the decision layers (Paper 5) and, in the whole suite (Paper 6), a library of 48 crops, deterministic phenotyping, a field sketch, Sentinel-2 indices, zone-by-zone variable-rate irrigation and field reports. The working date of the application is now the date of the toolbar, which moved the Engine 2 closure maximum of Fig. 9a from 0.78 to 0.70 % of inflow (the median, 0.1 %, is unchanged).

Software, data and code availability

Software. SWAFL Desktop (version 1.1.0) is open source; the Software source is in the project repository (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Software) and the program for Windows is SWAFL.exe (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loo

ps/releases/latest/download/SWAFL.exe).

Manual. The Manual (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/raw/main/Software/v1.1.0/SWAFL_Desktop/docs/manual_v1.1.0/SWAFL_Manual_v1.1.0.pdf) documents every control, view and equation and labels the provenance of all 134 numbered equations; it can be reviewed or downloaded as a PDF.

Data. The complete Daisy ensemble outputs (all 10 190 runs, raw and derived) are archived on Zenodo, doi: 10.5281/zenodo.23043626. The raw run table, and a linked list of every data file used, are in the Data folder of the repository (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Data).

Code. The verification scripts (Node.js with Playwright, driving the shipped engine) and the analysis and figure scripts (Python) are in the Code folder (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Manuscripts). The folder `Manuscripts/reproducibility` is a self-contained package: pinned software versions (Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, statsmodels 0.15.0, Matplotlib 3.11.2, Pillow 12.3.0, Node.js 22.22.0, Playwright 1.56.0), fixed random seeds in every stochastic step, SHA-256 checksums, and one command (`run_all.sh`) that regenerates all results and figures and then compares every regenerated value with the archived result and with the headline numbers printed in the manuscript. Each table and figure names the script that produced it (Appendix B). Daisy itself was not re-run for this package, no field measurements are used, and small intermediate arrays and the interface screenshots are archived rather than regenerated; these limits are listed in `README_REPRODUCE.md`.

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N.H.: software, verification, writing.

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A. Equation ledger

Table 16: Equations of this article, their manual numbers, provenance and source.

Eq.	Manual	Content	Provenance	Source
Eq. (27)	5.1	van Genuchten retention	Literature	van Genuchten (1980)
Eq. (28)	5.2	Mualem relative conductivity	Literature	Mualem (1976)
Eq. (29)	5.3	depletion fraction between 100 and 15 849 cm	Literature	Allen et al. (1998)
Eq. (31)–Eq. (37)	5.4–5.10	ratio/lookup composition for E, T, ET, I, Y, D, N_L	Data-driven	Daisy ensemble
Eq. (38)	5.11	set-specific corrections	Data-driven	E2, E3, E4, E13
Eq. (39)	5.12	non-negativity, ceiling, closure window	Engineered	–
Eq. (40)	5.15	rainfall generator	Engineered	–

Eq.	Manual	Content	Provenance	Source
Eq. (41)	5.16	interception	Engineered	–
Eq. (42)	5.22–5.23	transpiration and stress	Engineered (Literature form)	Allen et al. (1998)
Eq. (43)	5.24–5.25	scaling and daily closure	Engineered	–
–	11.1–11.2	Saxton–Rawls pedotransfer	Literature	Saxton and Rawls (2006)
–	11.3	wetting-front suction	Engineered	–
Eq. (44)	11.4–11.5	Green–Ampt	Literature	Green and Ampt (1911)
Eq. (45)	11.6–11.7	Kostiakov fit	Literature	Kostiakov (1932)
–	11.8	Manning	Literature	Manning (1891)
Eq. (46)	11.9–11.11	volume balance	Literature	Elliott and Walker (1982); Walker and Skogerboe (1987)
–	11.12–11.13	alternative advance laws	Engineered	course notes
Eqs. (47) and (48)	11.15–11.19	depths, volumes, efficiencies, CU	Literature	Burt et al. (1997); Christiansen (1942); Walker and Skogerboe (1987)
–	11.20–11.21	hydrograph and recession	Engineered	–
Eq. (49)	12.3–12.5	Hazen–Williams, emitter equation	Literature	Keller and Karmeli (1974); Williams and Hazen (1905)
Eq. (50)	12.7–12.8	emission uniformity	Literature	Keller and Karmeli (1974)

Eq.	Manual	Content	Provenance	Source
–	12.9–12.15	wetted bulb and losses	Engineered	–
–	12.16–12.19	emitter spread and efficiencies	Engineered	–
Eq. (51)	13.2–13.3	nozzle, wetted radius	Literature / Engineered	Keller and Bliesner (1990)
–	13.4–13.7	pattern, wind, overlap, CU	Engineered	Christiansen (1942)
–	13.8	wind drift and evaporation	Engineered	–
–	13.9–13.12	intake, ponding, efficiencies	Engineered	–
–	13.13–13.23	checklist, pipes, pump	Literature	Blasius (1913); Keller and Bliesner (1990)
System variables (Section 3)				
Eq. (1)	5.12, 5.25	water-balance identity	Literature	–
Eq. (16)	5.13–5.14	canopy cover	Engineered	–
Eq. (7)	5.17	layer redistribution	Engineered	–
Eqs. (17) to (19)	5.18–5.20	rooting depth and root density	Engineered	–
Eq. (10)	5.26	evaporative demand under warming	Engineered	–
Eq. (26)	5.27	net irrigation depth	Literature (engineered clip)	Allen et al. (1998)
Eq. (8)	6.1–6.2	days beyond the trigger	Data-driven	Daisy daily output
Eq. (25)	6.3–6.4	inverse depletion; net depth per soil	Literature	Allen et al. (1998); van Genuchten (1980)

Eq.	Manual	Content	Provenance	Source
–	6.5	relative yield at the trigger	Data-driven	E2
Eqs. (6) and (21)	7.1	layer suction, root-zone potential	Literature (engineered floor)	van Genuchten (1980)
Eq. (22)	7.2–7.4	series resistances, root, stem, leaf potential	Engineered	van den Honert (1948)
Eq. (15)	7.5	air water potential	Literature	Nobel (2009)
Eq. (9)	7.6	share of root zone beyond the trigger	Engineered	–
Eq. (4)	8.1	pore space and consolidation	Engineered	Flanagan and Nearing (1995)
Eq. (2)	8.2	porosity and residual content	Engineered	–
–	8.3	retention $\theta(h)$	Literature	van Genuchten (1980)
Eq. (3)	8.4	capillary radius	Literature	Hillel (1998)
Eq. (5)	8.5–8.6	tillage and consolidation effects; regression on clay	Data-driven	E5
Eq. (20)	8.7	depth holding a share of uptake	Engineered	–
Eq. (13)	9.1	mulch effects on the fluxes	Data-driven	E7, E11, E14
Eq. (11)	9.2–9.3	two-stage evaporation	Engineered	Philip (1957); Ritchie (1972)
Eq. (12)	9.4–9.5	drying front and suction profile	Engineered	–
Eq. (23)	9.6–9.7	fate of saved evaporation	Data-driven	E7, E11, E14
Eq. (14)	9.8	regime index	Engineered	–
Table 4	10.1	feedback-loop arrow width	Engineered	–
Eq. (24)	10.2	nitrate cost of drainage	Data-driven	E11

B. Verification tests and reproducibility map

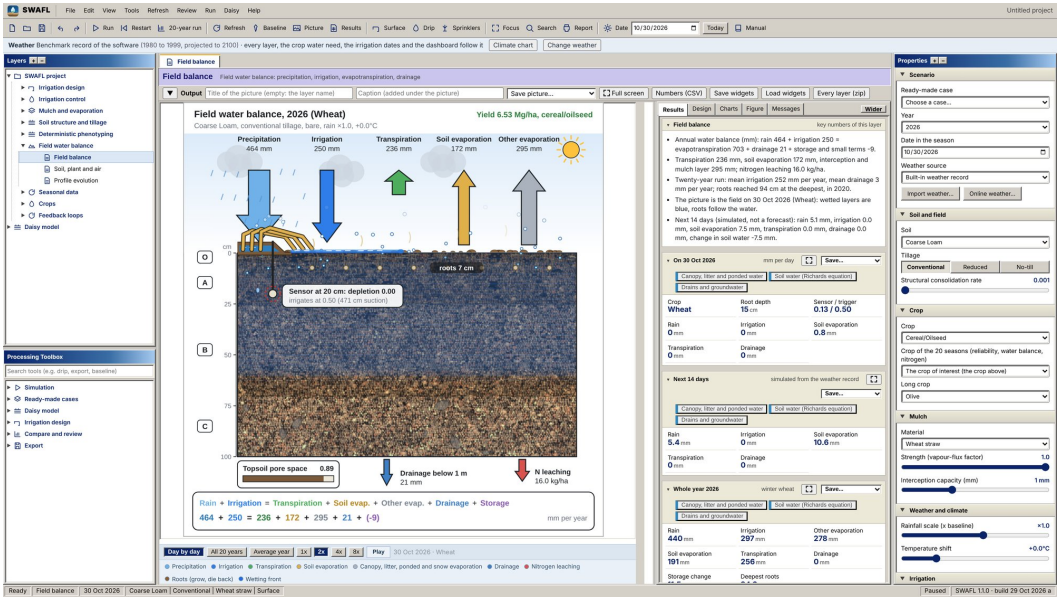
Table 17: Test and result to script map. Scripts are in `Manuscripts/reproducibility/` (`p3_js/` for the Node.js scripts that drive the shipped engine, `analysis/` for Python); outputs are archived in `Manuscripts/results/`.

Test or result	What is compared or computed	Script → output
System variables of ten soils (Tables 8 to 10, Fig. 2)	Engine 1, Engine 2, water-potential path, evaporation, mulch fate	`swa_vars.js` → `p3_swa_vars.json`; `p3_swa_fig.py`
Furrow Example 6.1 (Table 12)	T_{req}, t_L, E_a against the book (reference implementation `lib/wsmodel.js`)	`v_surface.js` → `p3_surface.json`, `p3_ws61.csv`
Example 6.1 through the shipped surface layer	t_L as shipped and with the iterated advance exponent	`v_surface_shipped.js` → `p3_surface_shipped.json`
Border Example 6.4 (Table 13)	T_{req}, $Q_{\max}$, t_L, A_0	`v_border.js` → `p3_border.json`
FAO-45 border examples (Fig. 11a)	13 advance times	`run_fao.js` → `fao_out.json`
Sprinkler example and spacing (Table 13, Fig. 11b)	decision logic, nozzle, diameter, CU	`v_sprinkler.js`, `run_spr.js` → `p3_sprinkler.json`, `spr_out.json`
Engine 1 held-out (Table 14, Fig. 8)	1 000 E12 runs	`make_e12.py`, `e1_heldout.js` → `e12.json`, `e12_pred.json`
Engine 2 closure and calibration (Fig. 9)	60 random states; 100 scenarios	`e2_closure.js`, `e2_grid.js` → `p3_e2_closure.json`, `p3_e2_grid.json`
Drip lateral solver (Table 15, Fig. 10)	independent solver, 400 configurations; Christiansen factor	`v_drip.js`, `run_drip.js` → `p3_drip_swafl.json`, `drip_out.json`
Sensitivity (Fig. 6)	one-at-a-time runs	`e1_sens.js`, `l_sens.js` → `p3_e1_sens.json`, `p3_l_sens.json`
Run time	median call time	`bench.js` → `p3_bench.json`
Depletion collapse of E2 (pooled R^2, Table 11)	cubic fits against suction and depletion	`Results/08_Scripts/novel.py` (figure N1 in `Results/00_Novel_Contributions`)
Figures		`p3_figs.py`, `p3_lit_figs.py`, `p3_sens_fig.py`, `p3_swa_fig.py`, `p3_fig01_workflow_v9.py`
All steps, checks and limits		`run_all.sh`, `compare_results.py`, `audit_numbers.py`, `README_REPRODUCE.md`

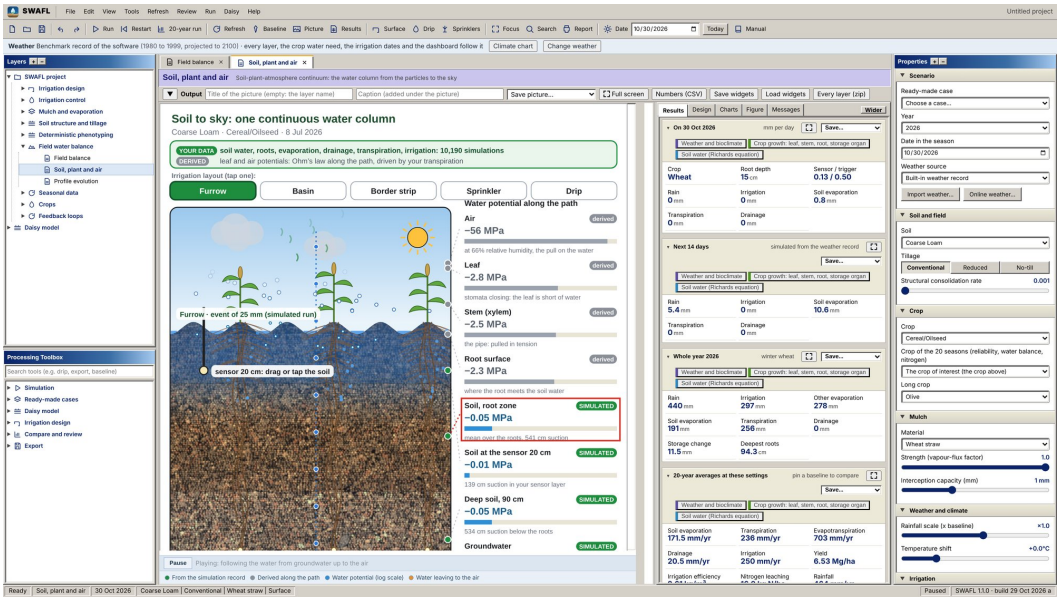
C. Interface screenshots of every view

All interface screenshots of the project are collected here, grouped as the result views of Section 5.2. Each is a full-window capture of SWAFL Desktop (version 1.1.0, window of

2000×1125 points at twice the pixel density); the full-resolution files (4000×2250 px) are in `Manuscripts/figures/shots/` of the repository, and every screen is documented, with numbered arrows on each element, in the Manual.

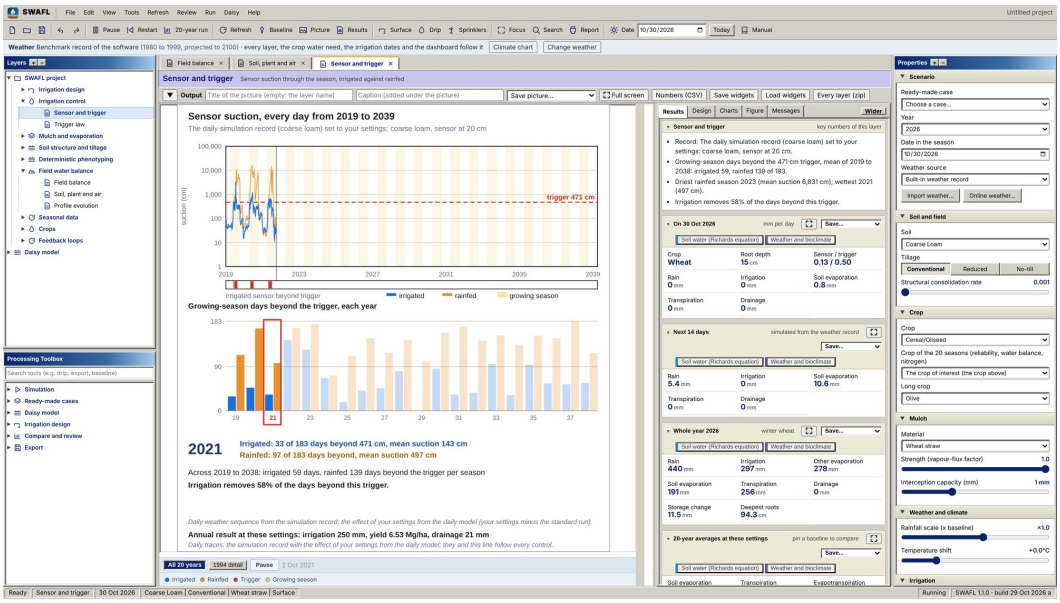


(a) Field balance, the home view: daily fluxes on the field clock, the soil profile with sensor, roots and wetting front, and the closed budget (Eq. (1)).

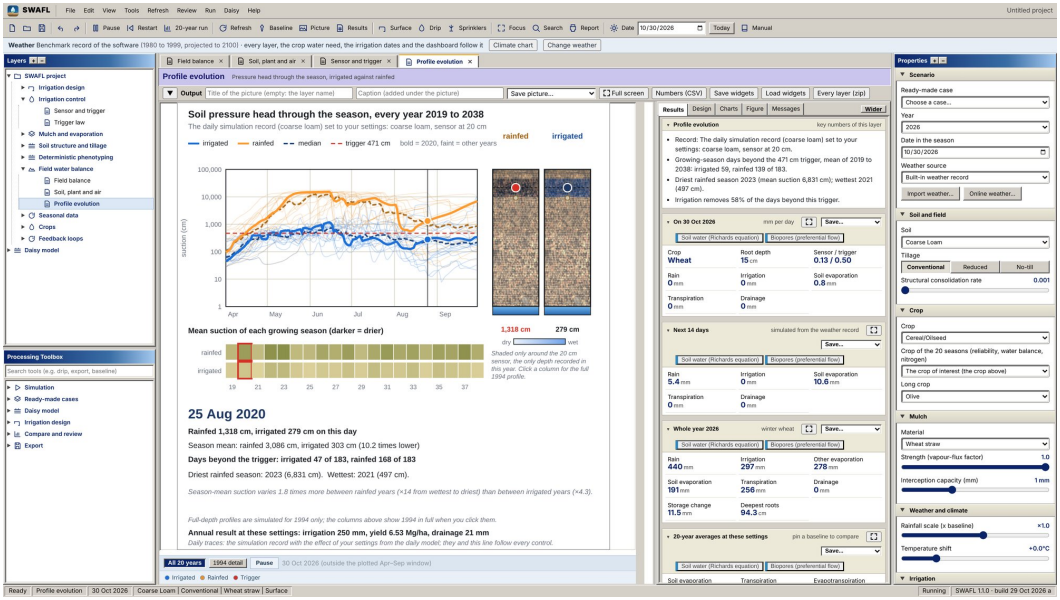


(b) Soil, plant and air: layouts of the five irrigation methods and the water potential along the path from the deep soil to the air (Eqs. (15), (21) and (22)).

Figure 12: SWAFL Desktop: field and the soil–plant–air path.

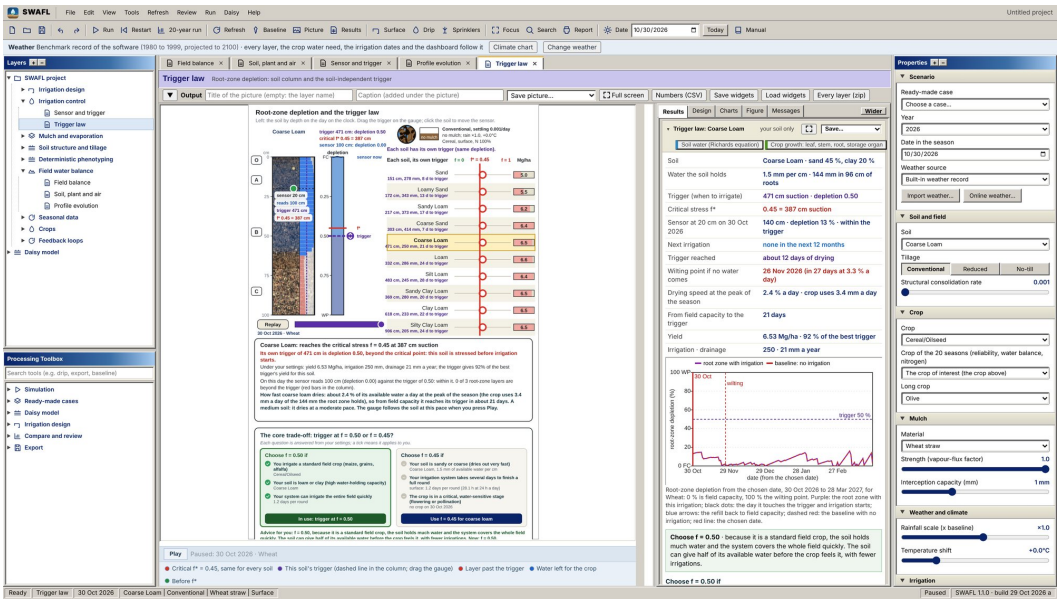


(a) Sensor and trigger: 20 years of daily suction at the sensor, irrigated and rainfed, against the trigger, and the days beyond the trigger per year (Eq. (8)).



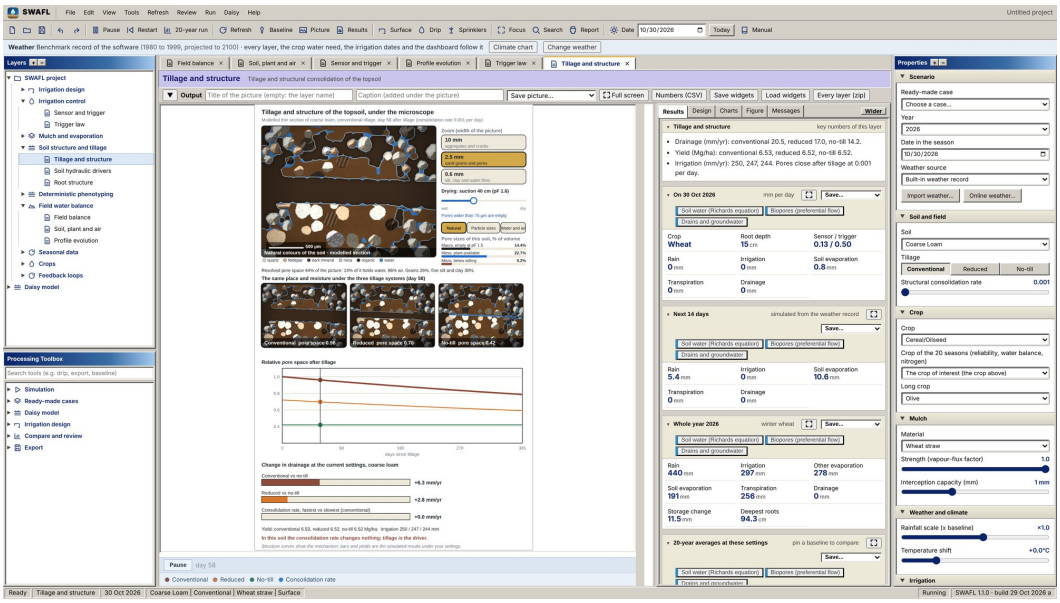
(b) Profile evolution: pressure-head profile through the season, irrigated and rainfed, and the share of the root zone beyond the trigger (Eq. (9)).

Figure 13: SWAFL Desktop: soil water and irrigation timing.

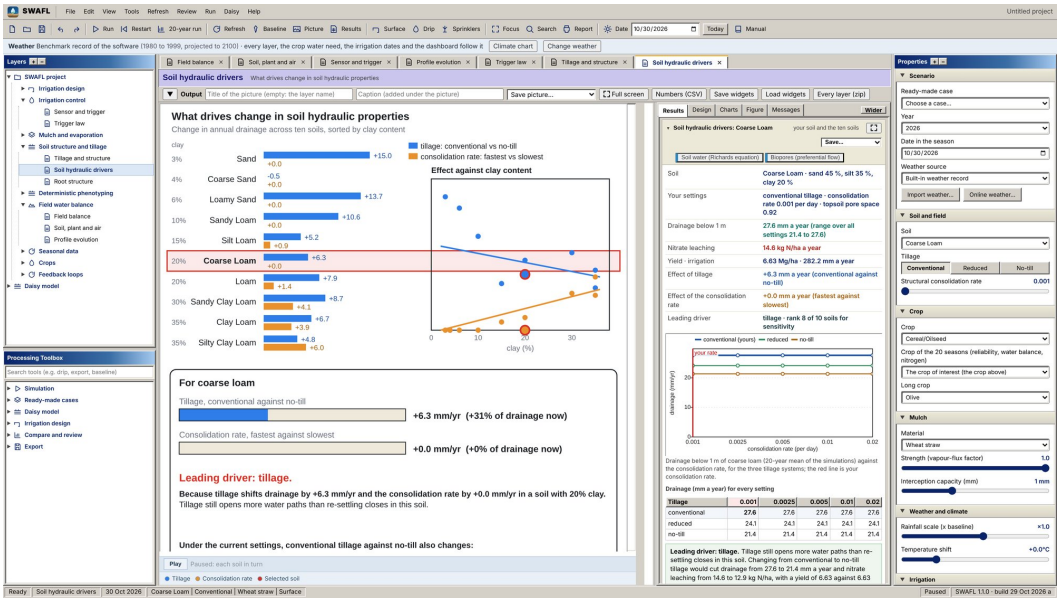


(a) Trigger law: ten soils on one depletion axis with the critical depletion f^* and each soil's own trigger suction (Eqs. (25), (26) and (29)).

Figure 14: SWAFL Desktop: trigger law.

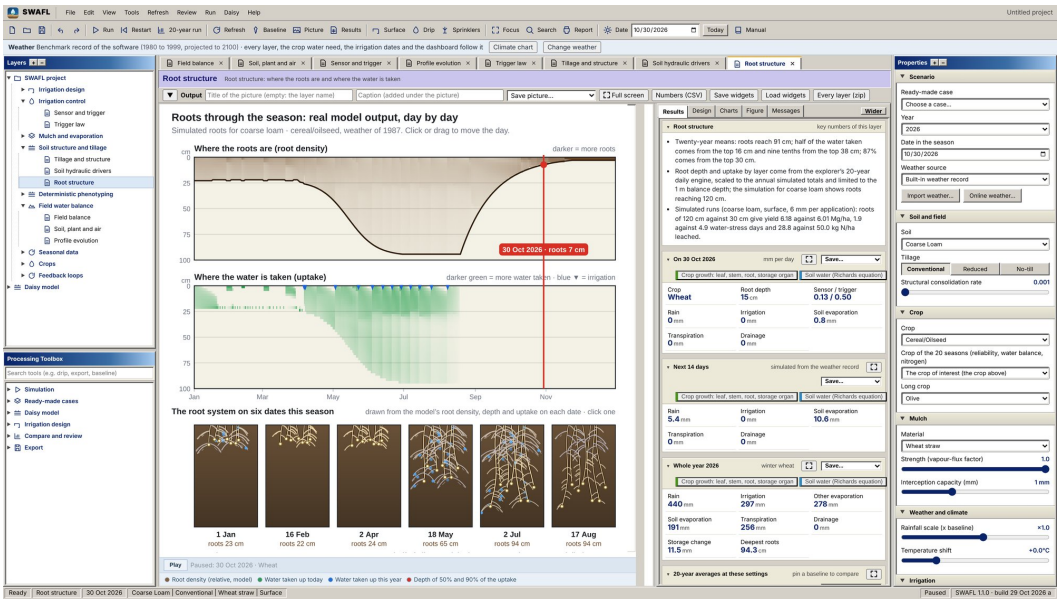


(a) Tillage and structure: modelled thin section of the topsoil with water- and air-filled pores at the chosen suction (Eq. (3)), relative pore space against time (Eq. (4)) and the drainage consequences of tillage. The section renders the model's pore classes; it is not imaged soil.



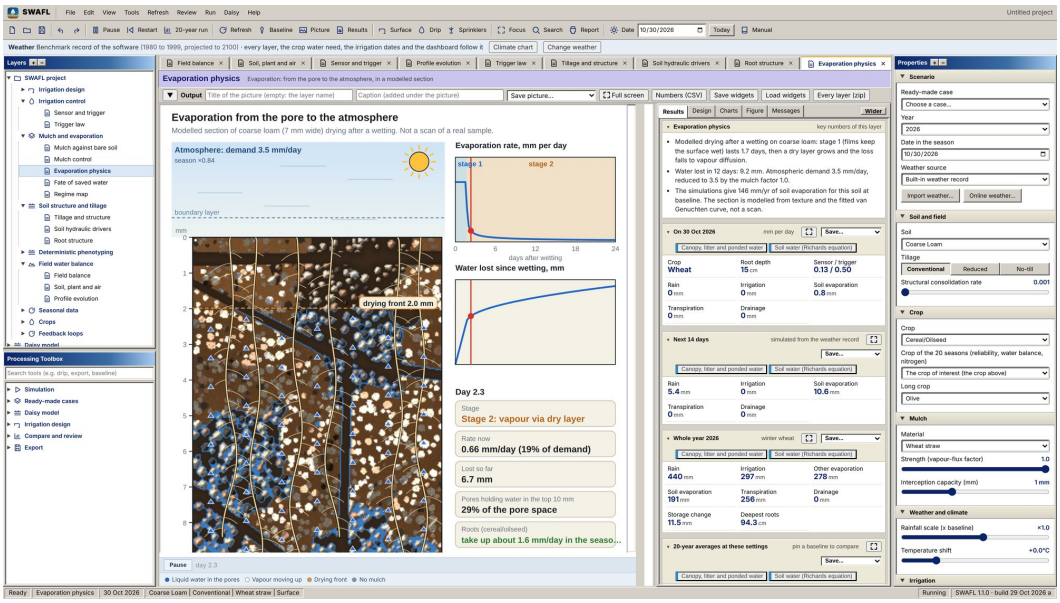
(b) Soil hydraulic drivers: tillage and consolidation effects on drainage for the ten soils and their regressions on clay content (Eq. (5)).

Figure 15: SWAFL Desktop: soil structure and hydraulic drivers.

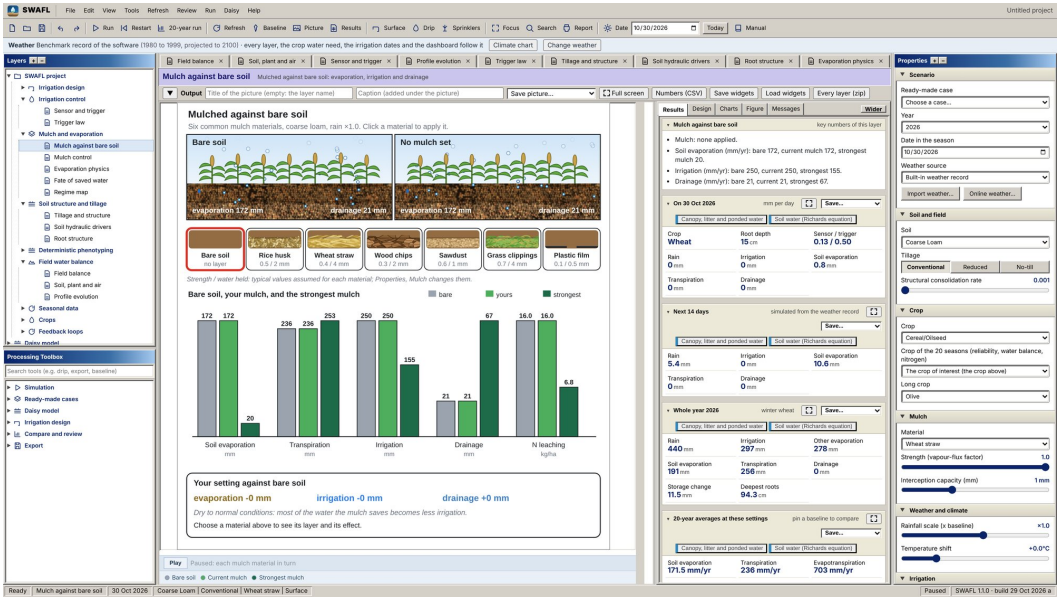


(a) Root structure: rooting depth and the depths holding 50 and 90 % of uptake in the ten soils (Eqs. (17) to (20)), with the E15 rooting-depth heat maps.

Figure 16: SWAFL Desktop: roots.

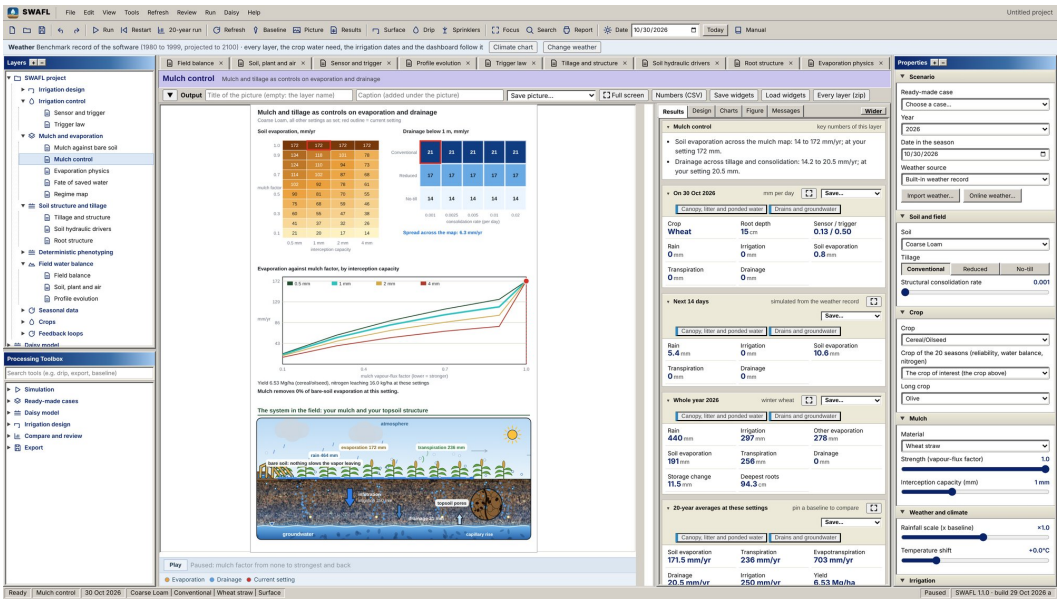


(a) Evaporation physics: drying of the topsoil after a wetting, stage-1 and stage-2 evaporation and the drying front (Eqs. (11) and (12)).

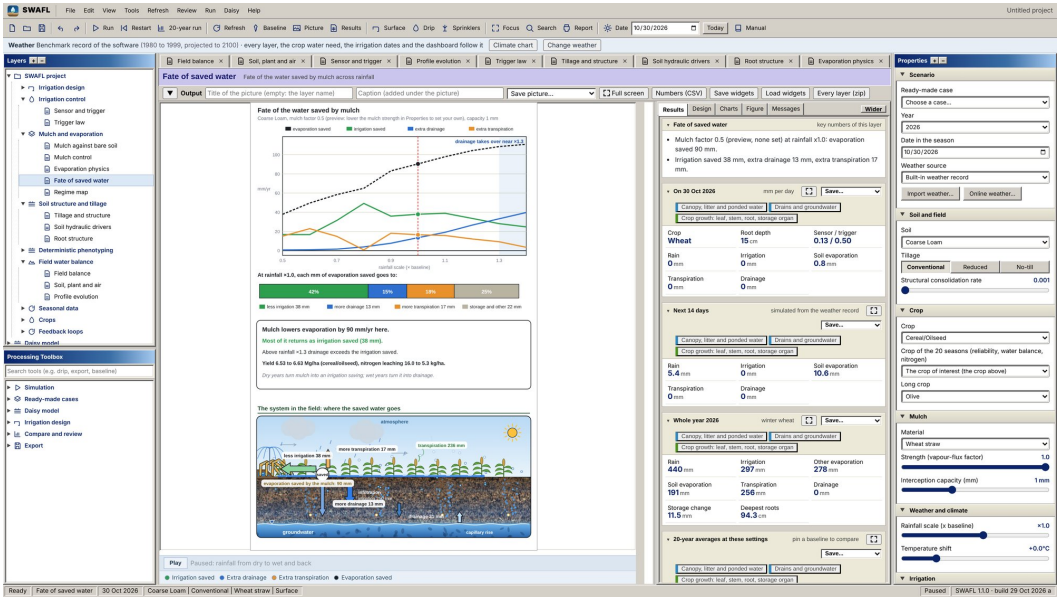


(b) Mulch against bare soil: evaporation, irrigation and drainage under bare soil, the chosen mulch and plastic film (Eq. (13)).

Figure 17: SWAFL Desktop: evaporation and mulch.

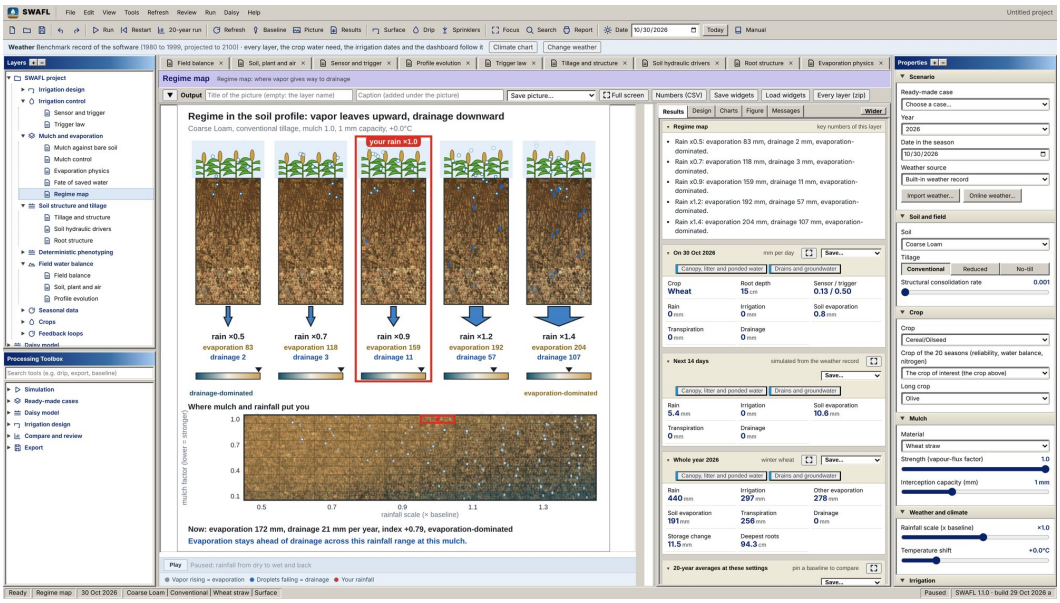


(a) Mulch control: soil evaporation by mulch factor and interception capacity, and drainage by tillage and consolidation rate.

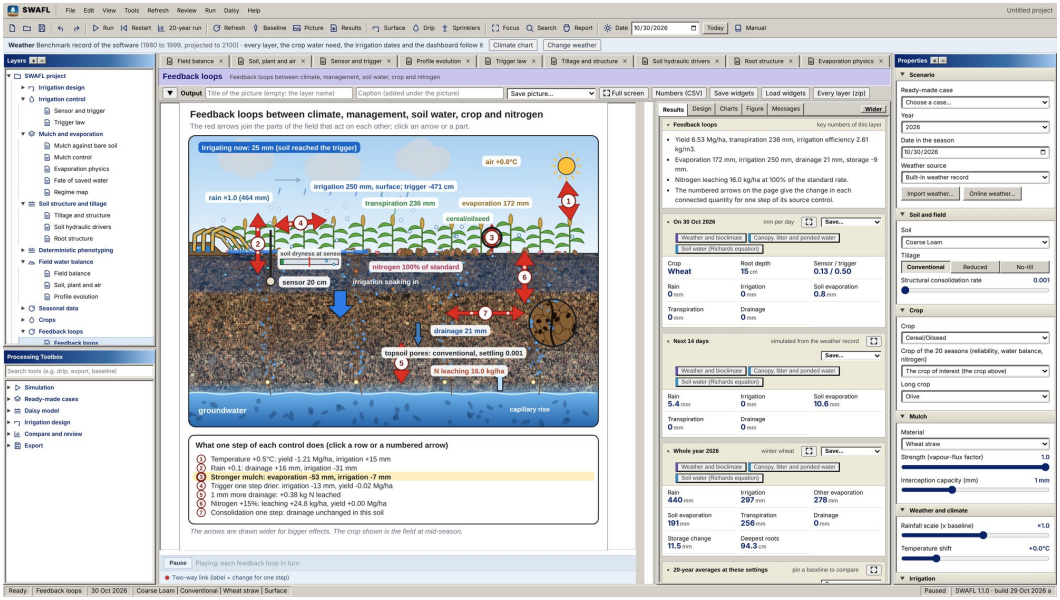


(b) Fate of saved water: evaporation saved, irrigation saved, extra drainage and extra transpiration against the rainfall scale (Eq. (23)).

Figure 18: SWAFL Desktop: mulch control and the fate of saved water.

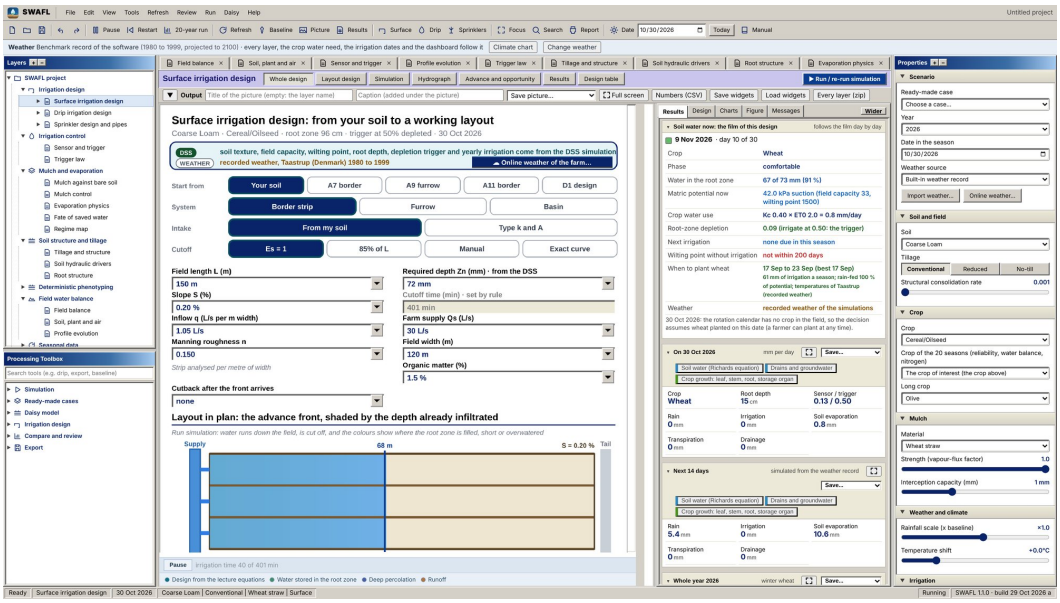


(a) Regime map: mulch factor against rainfall scale coloured by the dominant loss (Eq. (14)).

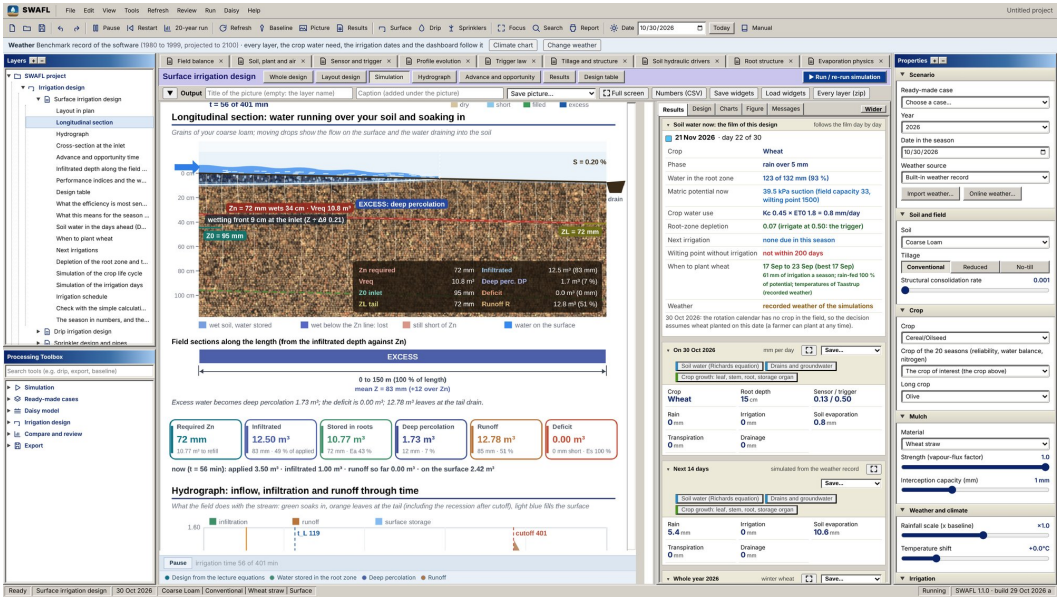


(b) Feedback loops: the seven loops with arrow widths scaled by the effect of one control step (Table 4).

Figure 19: SWAFL Desktop: regime map and feedback loops.

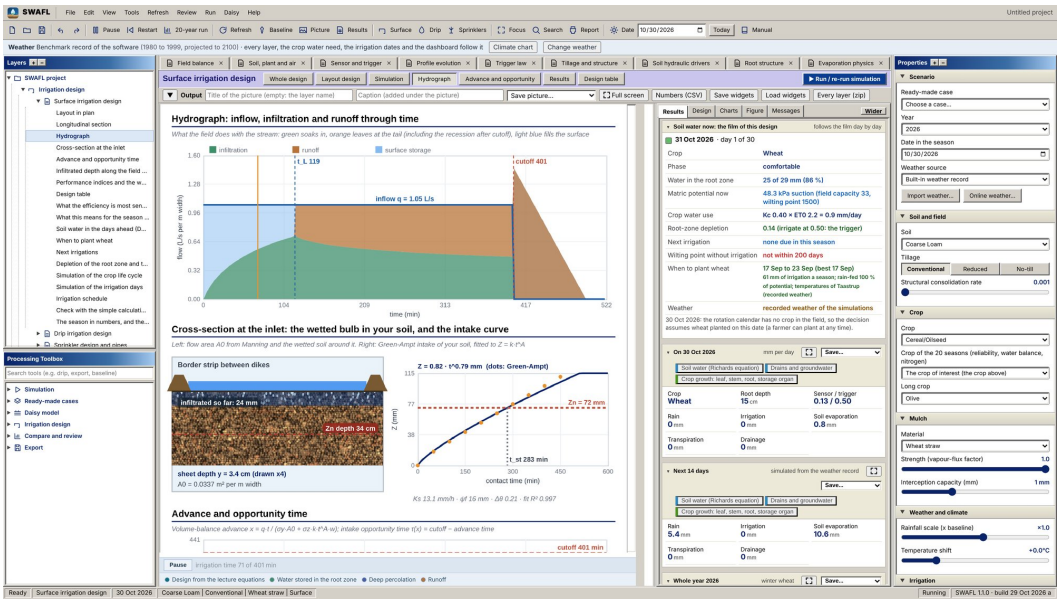


(a) Surface-irrigation layer: system choice, intake, advance and the design inputs carried from the soil and trigger (Section 4.4).

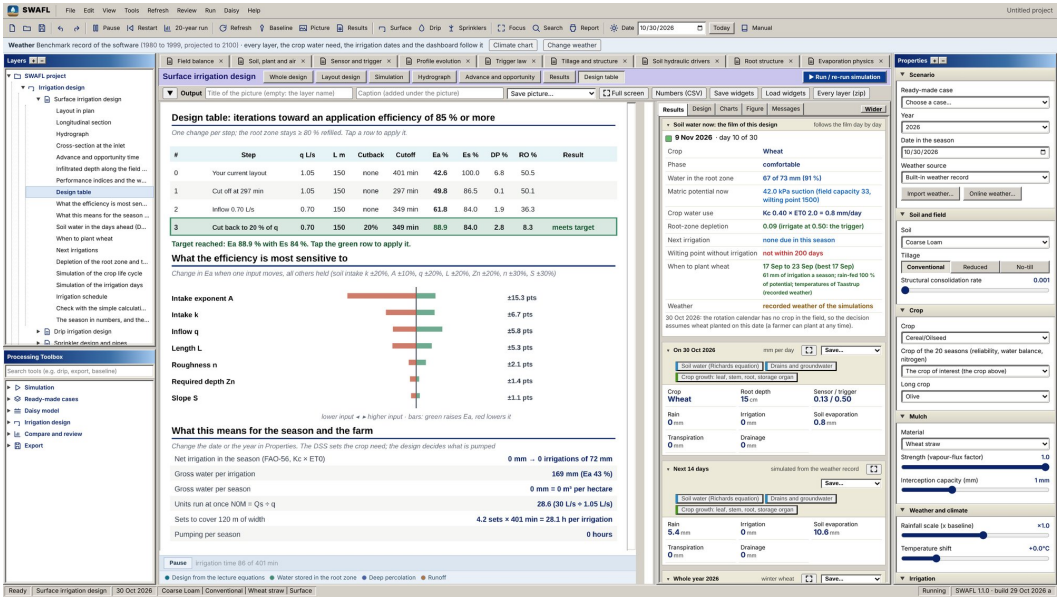


(b) Surface irrigation, longitudinal section: advance and infiltrated depth along the field (Eqs. (46) and (47)).

Figure 20: SWAFL Desktop: surface irrigation.

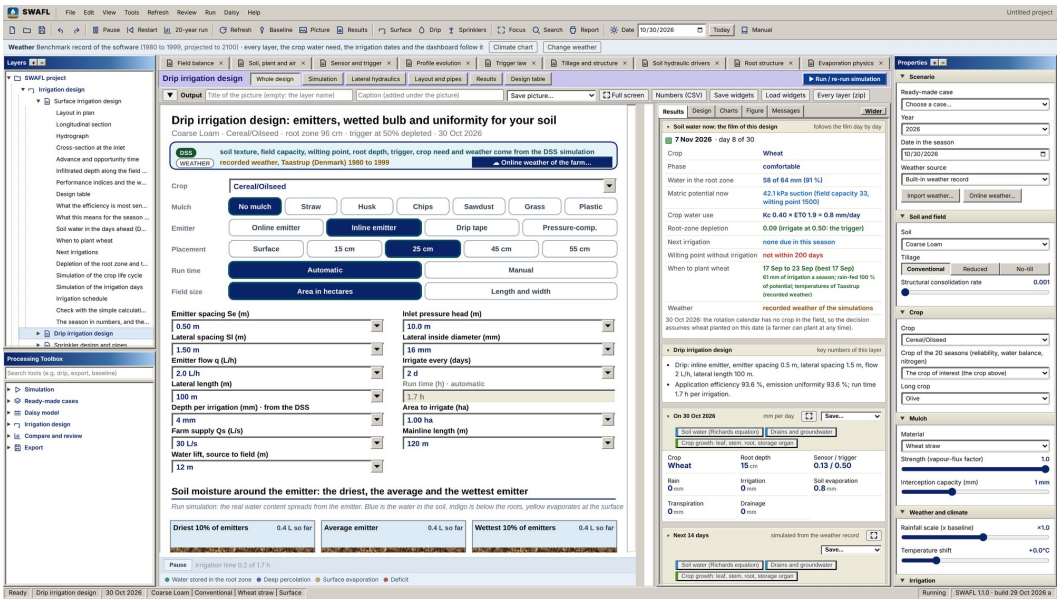


(a) Surface irrigation: hydrograph of inflow, infiltration and runoff through time.

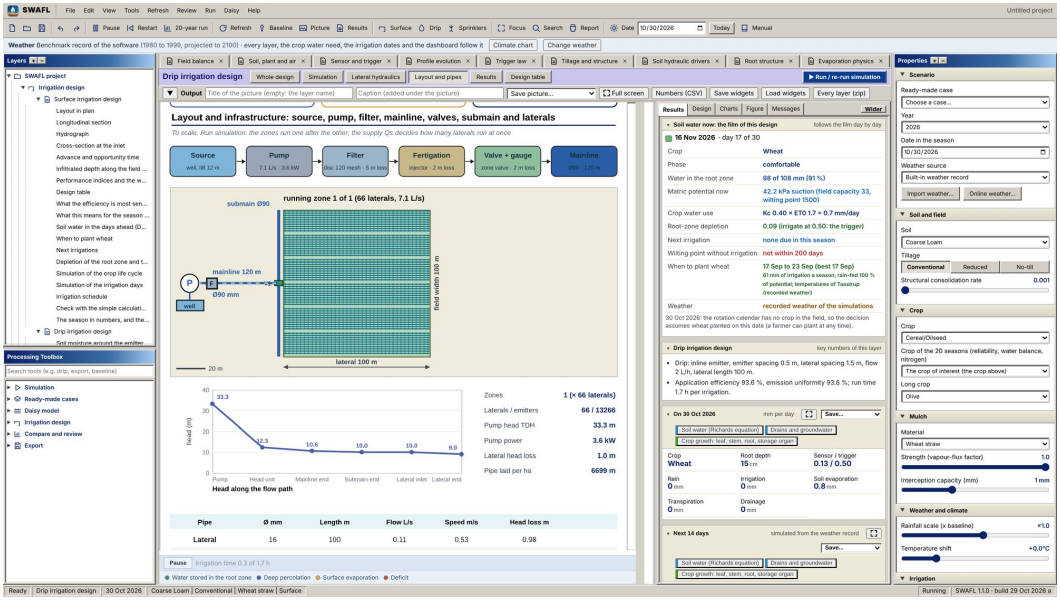


(b) Surface irrigation: design table of options with application and storage efficiency, deep percolation and runoff (Eq. (48)).

Figure 21: SWAFL Desktop: surface irrigation, schedule and design table.

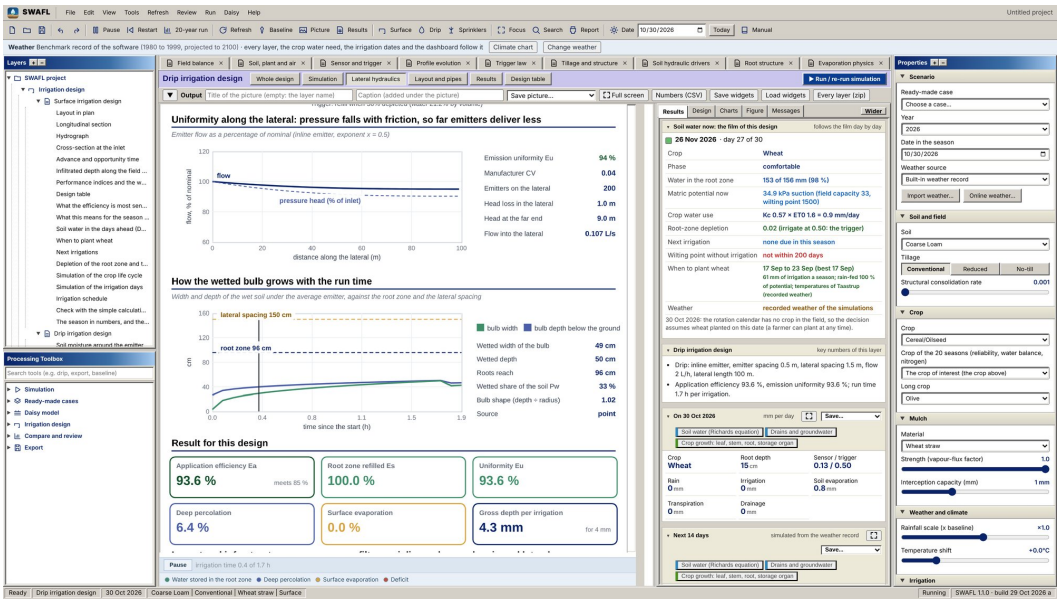


(a) Drip-irrigation layer: emitters, wetted bulb and uniformity for the chosen soil (Section 4.5).

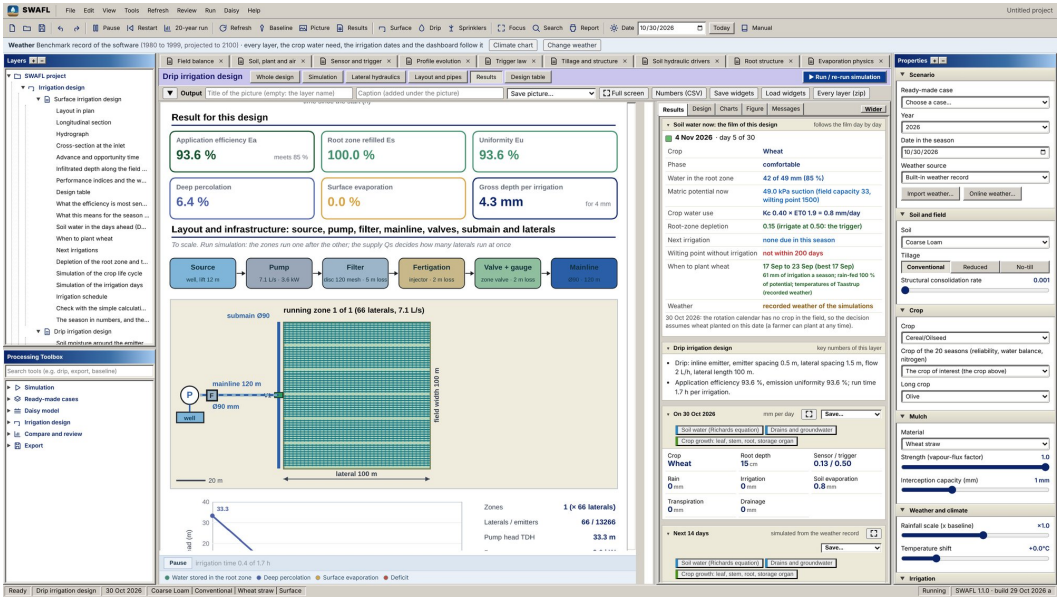


(b) Drip irrigation: field layout and infrastructure.

Figure 22: SWAFL Desktop: drip irrigation.

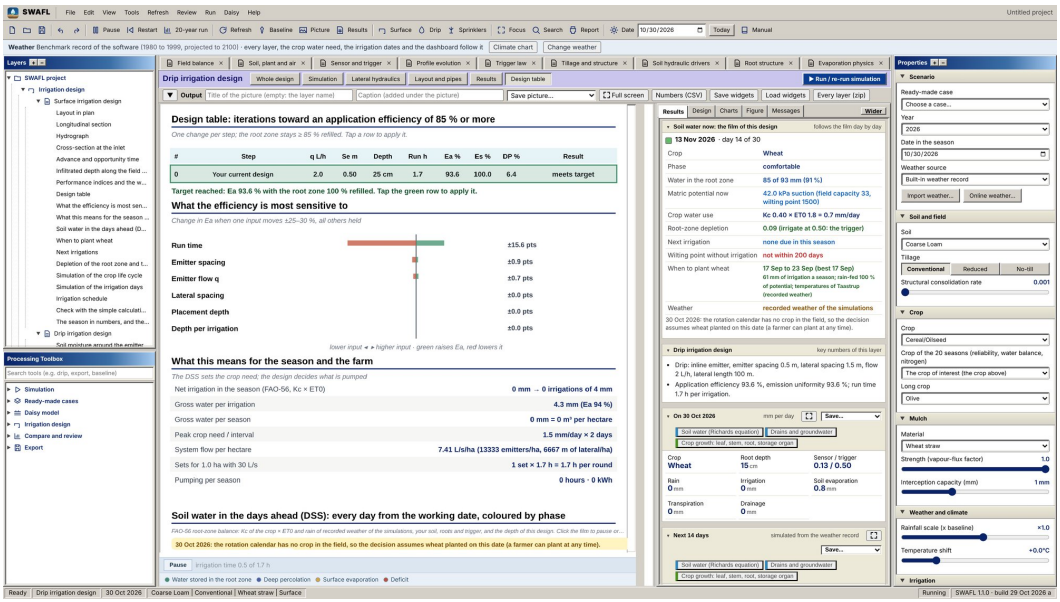


(a) Drip irrigation: head and emitter flow along the lateral (Eq. (49)) with the 20 % loss warning.



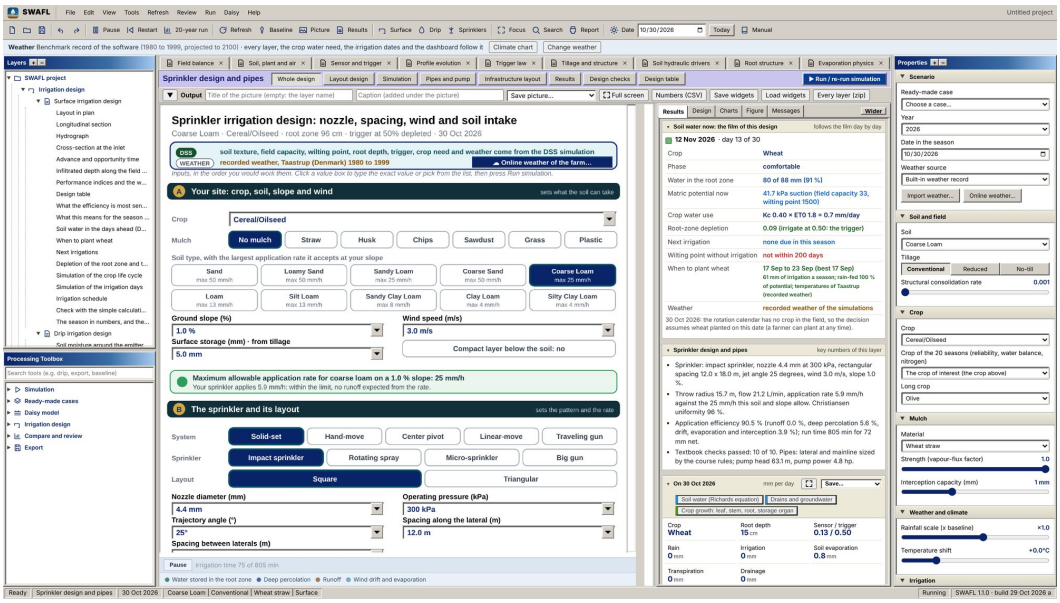
(b) Drip irrigation: result summary with emission uniformity (Eq. (50)) and efficiencies.

Figure 23: SWAFL Desktop: drip irrigation, lateral and results.

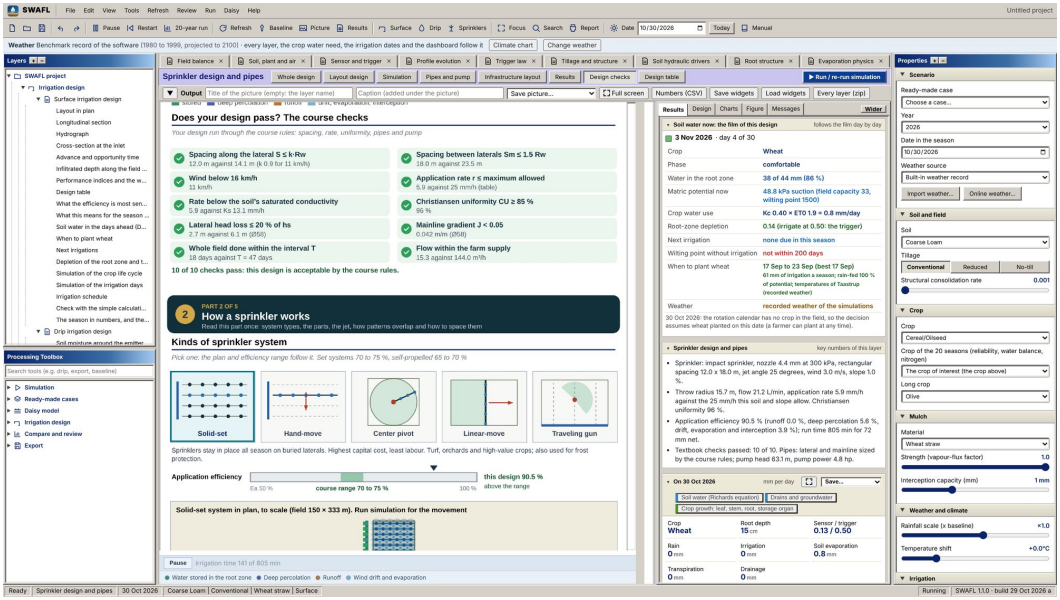


(a) Drip irrigation: design table, iterations toward an application efficiency of 85 % or more.

Figure 24: SWAFL Desktop: drip irrigation, design table.

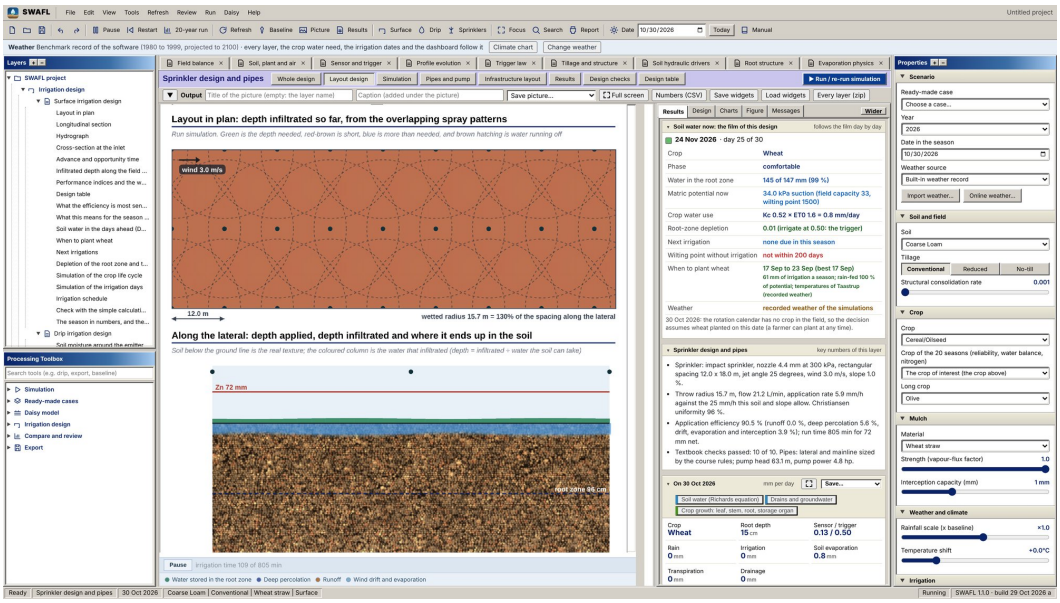


(a) Sprinkler layer: nozzle, wetted radius, pattern, overlap and uniformity (Eq. (51); Section 4.6).

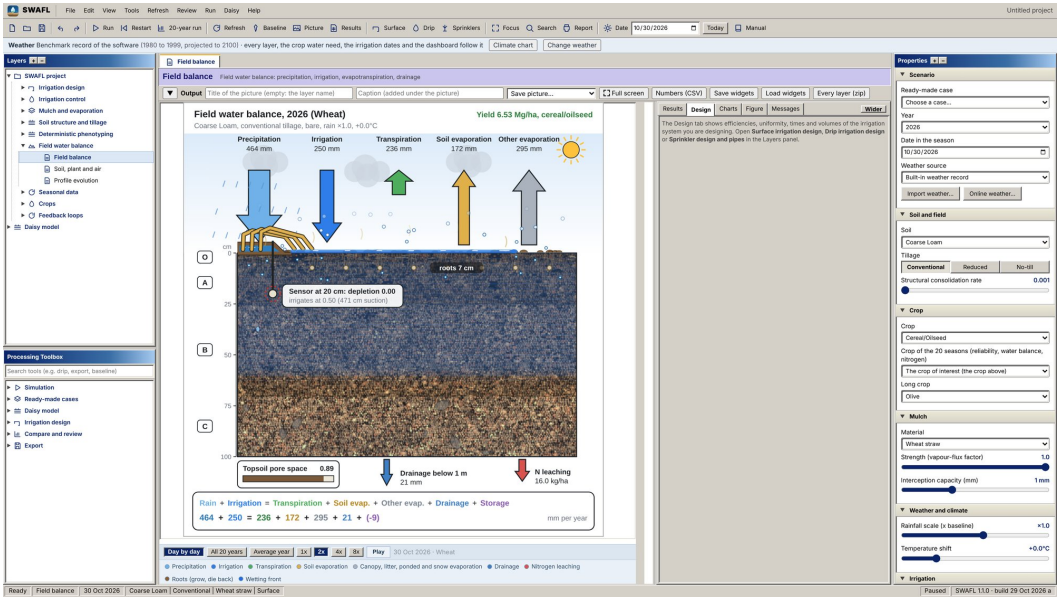


(b) Sprinkler design: textbook decision checklist (sprinkler class, spacing, application rate, sets per day).

Figure 25: SWAFL Desktop: sprinkler irrigation.

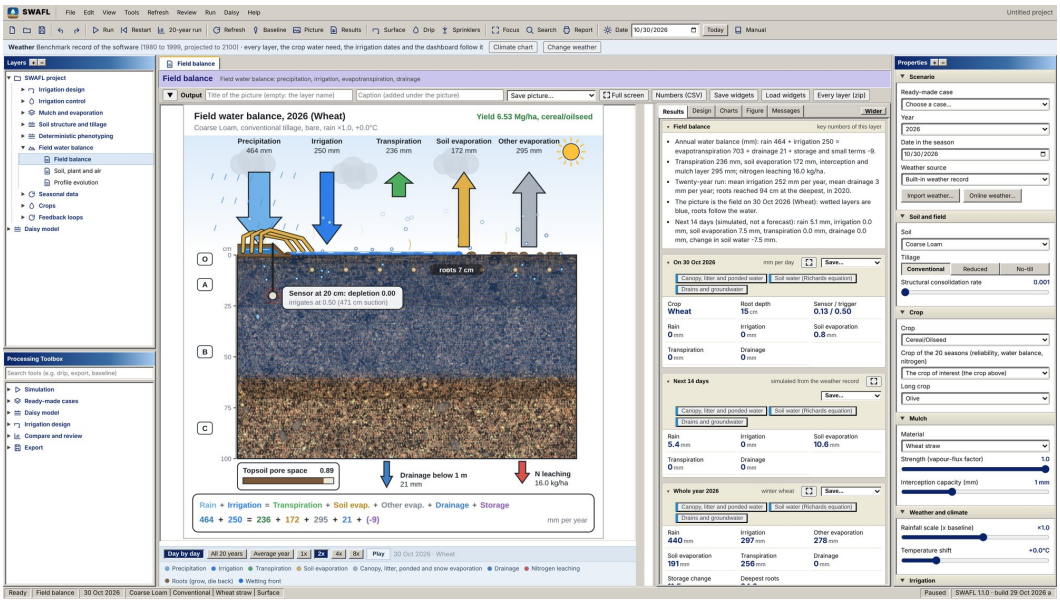


(a) Sprinkler: layout in plan, with the depth infiltrated from the overlapping spray patterns.

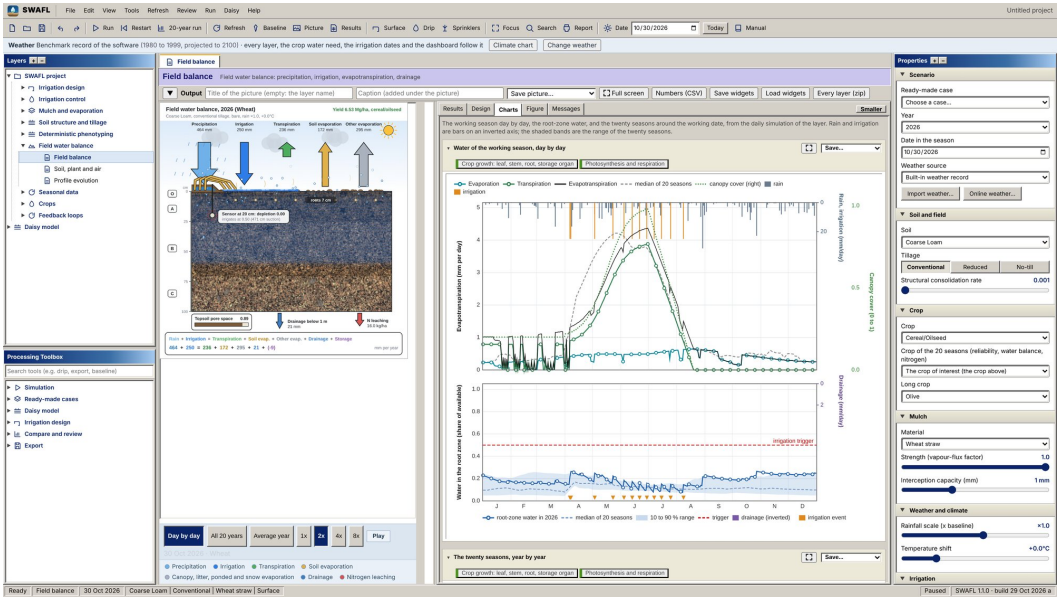


(b) Dashboard with the design tab open beside the field-balance layer.

Figure 26: SWAFL Desktop: sprinkler intake and the dashboard.

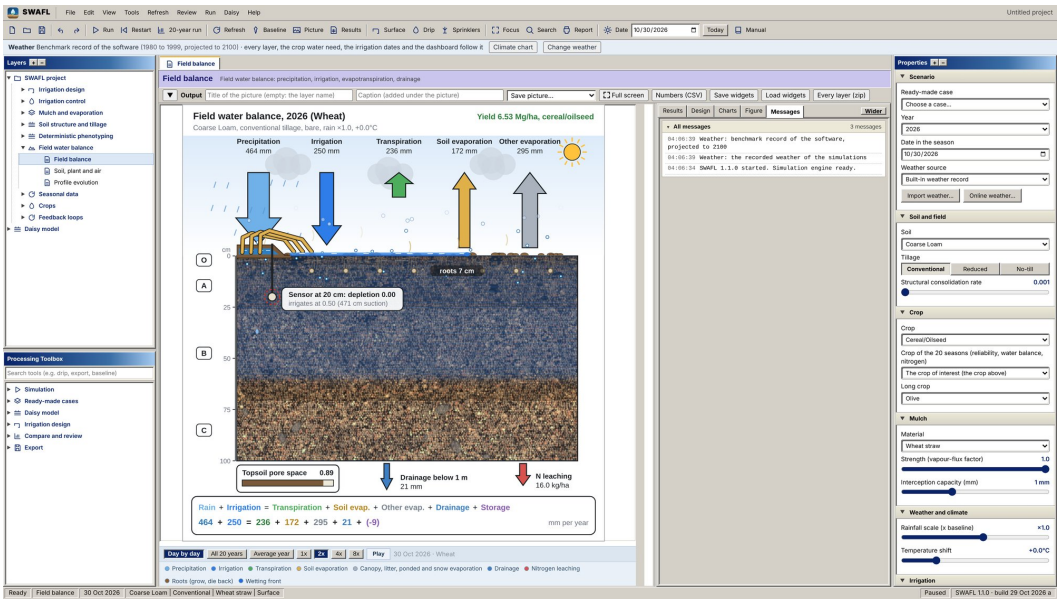


(a) Dashboard, results tab: annual and daily flux tiles of the field water balance.



(b) Dashboard, charts tab: seasonal rainfall, irrigation, drainage, evaporation and transpiration.

Figure 27: SWAFL Desktop: dashboard.



(a) Dashboard, messages tab: the decision messages and warnings of the current state.

Figure 28: SWAFL Desktop: dashboard messages.

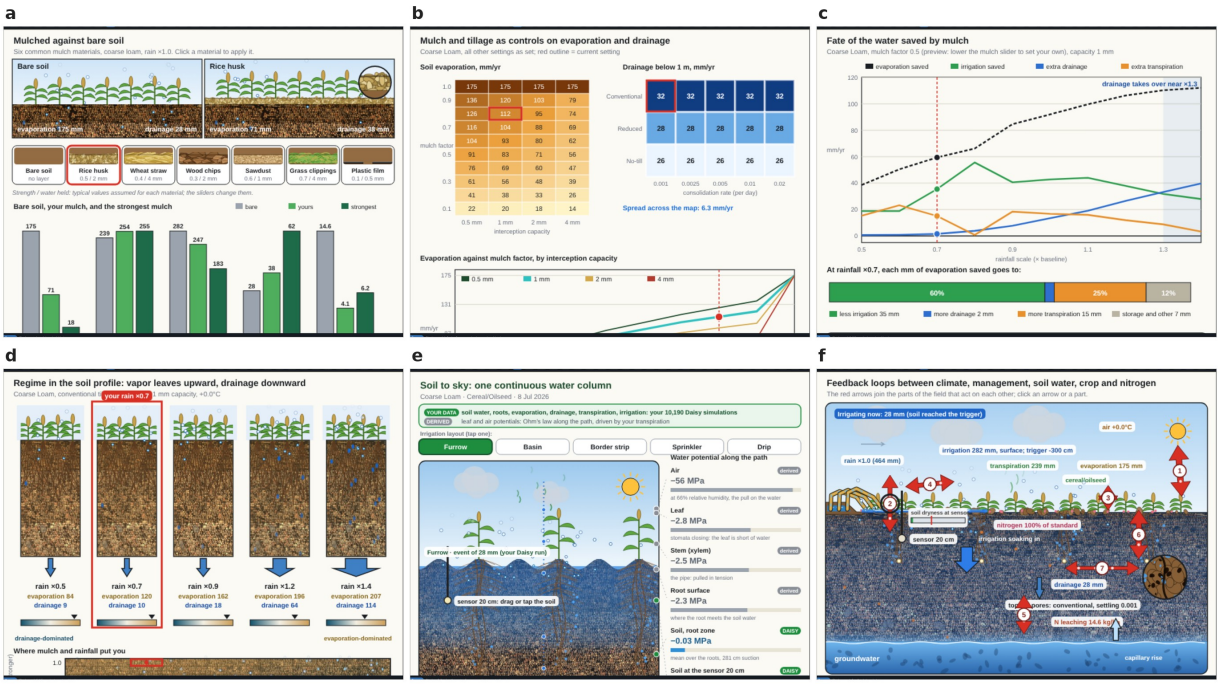


Figure 5: Result views for mulch and the whole system. (a) Mulch against bare soil; (b) mulch strength; (c) fate of the saved water; (d) regime map of where mulch pays; (e) soil to sky; (f) how the effects connect.